

1. ALL RESISTANCE VALUES ARE IN OHMS, 0.1 WATT +/- 5%.
2. ALL CAPACITANCE VALUES ARE IN MICROFARADS.
3. ALL CRYSTALS & OSCILLATOR VALUES ARE IN HERTZ.

J117 MLB

LAST MODIFIED=Thu Sep 18 13:37:48 2014
LAST MODIFICATION=Thu Sep 18 13:37:48 2014

REV	ECN	DESCRIPTION OF REVISION	CK APPD DATE
3	0003501996	ENGINEERING RELEASED	2014-12-09

<CSA>				
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[illegible]

DRAWING TITLE		SCHEM, MLB, J117	
 Apple Inc.	DRAWING NUMBER	051-00081	SIZE D
	REVISION	3.0.0	
	BRANCH		
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D	Schematic / PCB #'s <table><tr><th>PART#</th><th>QTY</th><th>DESCRIPTION</th><th>REFERENCE DESIGNATOR(S)</th><th>CRITICAL</th><th>BOM OPTION</th></tr><tr><td>051-00081</td><td>1</td><td>SCH_MLB,J117</td><td>SCH</td><td>CRITICAL</td><td></td></tr><tr><td>820-00034</td><td>1</td><td>PCBF_MLB,J117</td><td>PCB</td><td>CRITICAL</td><td></td></tr><tr><td>685-00013</td><td>1</td><td>PCBA_MLB,J117,COMMON PARTS</td><td>CMNPTS</td><td></td><td>MLB_CMNPTS</td></tr></table> Main BOM Variants <table><tr><th>BOM NUMBER</th><th>BOM NAME</th><th>BOM OPTIONS</th></tr><tr><td>985-00016</td><td>PCBA_MLB,DEV,J117</td><td>DEVELOPMENT,J117_DEVEL</td></tr><tr><td>939-00158</td><td>PCBA_MLB,DEV,J117,CPU_INT</td><td>MLB_CMNPTS,ALTERNATE,CPU:SOCKET,DDR3:HYNIX_8GB_1866,SSD:Y</td></tr><tr><td>639-00910</td><td>PCBA_MLB,J117,HY_8GB_29NM,HDD,1866</td><td>MLB_CMNPTS,ALTERNATE,CPU:ULT,DDR3:HYNIX_8GB_1866,SSD:N</td></tr><tr><td>639-00911</td><td>PCBA_MLB,J117,EL_8GB_25NM,HDD,1866</td><td>MLB_CMNPTS,ALTERNATE,CPU:ULT,DDR3:ELPIDA_8GB_1866,SSD:N</td></tr><tr><td>639-00912</td><td>PCBA_MLB,J117,SA_8GB_23NM,HDD,1866</td><td>MLB_CMNPTS,ALTERNATE,CPU:ULT,DDR3:SAMSUNG_8GB_1866,SSD:N</td></tr><tr><td>639-00913</td><td>PCBA_MLB,J117,EL_16GB_25NM,HDD,1866</td><td>MLB_CMNPTS,ALTERNATE,CPU:ULT,DDR3:ELPIDA_16GB_1866,SSD:N</td></tr><tr><td>639-00914</td><td>PCBA_MLB,J117,HY_16GB_25NM,HDD,1866</td><td>MLB_CMNPTS,ALTERNATE,CPU:ULT,DDR3:HYNIX_16GB_1866,SSD:N</td></tr><tr><td>639-00713</td><td>PCBA_MLB,J117,HY_8GB_29NM,SSD,1866</td><td>MLB_CMNPTS,ALTERNATE,CPU:ULT,DDR3:HYNIX_8GB_1866,SSD:Y</td></tr><tr><td>639-00714</td><td>PCBA_MLB,J117,EL_8GB_25NM,SSD,1866</td><td>MLB_CMNPTS,ALTERNATE,CPU:ULT,DDR3:ELPIDA_8GB_1866,SSD:Y</td></tr><tr><td>639-00715</td><td>PCBA_MLB,J117,SA_8GB_23NM,SSD,1866</td><td>MLB_CMNPTS,ALTERNATE,CPU:ULT,DDR3:SAMSUNG_8GB_1866,SSD:Y</td></tr><tr><td>639-00716</td><td>PCBA_MLB,J117,EL_16GB_25NM,SSD,1866</td><td>MLB_CMNPTS,ALTERNATE,CPU:ULT,DDR3:ELPIDA_16GB_1866,SSD:Y</td></tr><tr><td>639-00717</td><td>PCBA_MLB,J117,HY_16GB_25NM,SSD,1866</td><td>MLB_CMNPTS,ALTERNATE,CPU:ULT,DDR3:HYNIX_16GB_1866,SSD:Y</td></tr><tr><td>685-00013</td><td>PCBA_MLB,J117,COMMON PARTS</td><td>J117_COMMON</td></tr></table>								PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION	051-00081	1	SCH_MLB,J117	SCH	CRITICAL		820-00034	1	PCBF_MLB,J117	PCB	CRITICAL		685-00013	1	PCBA_MLB,J117,COMMON PARTS	CMNPTS		MLB_CMNPTS	BOM NUMBER	BOM NAME	BOM OPTIONS	985-00016	PCBA_MLB,DEV,J117	DEVELOPMENT,J117_DEVEL	939-00158	PCBA_MLB,DEV,J117,CPU_INT	MLB_CMNPTS,ALTERNATE,CPU:SOCKET,DDR3:HYNIX_8GB_1866,SSD:Y	639-00910	PCBA_MLB,J117,HY_8GB_29NM,HDD,1866	MLB_CMNPTS,ALTERNATE,CPU:ULT,DDR3:HYNIX_8GB_1866,SSD:N	639-00911	PCBA_MLB,J117,EL_8GB_25NM,HDD,1866	MLB_CMNPTS,ALTERNATE,CPU:ULT,DDR3:ELPIDA_8GB_1866,SSD:N	639-00912	PCBA_MLB,J117,SA_8GB_23NM,HDD,1866	MLB_CMNPTS,ALTERNATE,CPU:ULT,DDR3:SAMSUNG_8GB_1866,SSD:N	639-00913	PCBA_MLB,J117,EL_16GB_25NM,HDD,1866	MLB_CMNPTS,ALTERNATE,CPU:ULT,DDR3:ELPIDA_16GB_1866,SSD:N	639-00914	PCBA_MLB,J117,HY_16GB_25NM,HDD,1866	MLB_CMNPTS,ALTERNATE,CPU:ULT,DDR3:HYNIX_16GB_1866,SSD:N	639-00713	PCBA_MLB,J117,HY_8GB_29NM,SSD,1866	MLB_CMNPTS,ALTERNATE,CPU:ULT,DDR3:HYNIX_8GB_1866,SSD:Y	639-00714	PCBA_MLB,J117,EL_8GB_25NM,SSD,1866	MLB_CMNPTS,ALTERNATE,CPU:ULT,DDR3:ELPIDA_8GB_1866,SSD:Y	639-00715	PCBA_MLB,J117,SA_8GB_23NM,SSD,1866	MLB_CMNPTS,ALTERNATE,CPU:ULT,DDR3:SAMSUNG_8GB_1866,SSD:Y	639-00716	PCBA_MLB,J117,EL_16GB_25NM,SSD,1866	MLB_CMNPTS,ALTERNATE,CPU:ULT,DDR3:ELPIDA_16GB_1866,SSD:Y	639-00717	PCBA_MLB,J117,HY_16GB_25NM,SSD,1866	MLB_CMNPTS,ALTERNATE,CPU:ULT,DDR3:HYNIX_16GB_1866,SSD:Y	685-00013	PCBA_MLB,J117,COMMON PARTS	J117_COMMON																																													
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685-00013	PCBA_MLB,J117,COMMON PARTS	J117_COMMON																																																																																																																					
C	BOM Groups <table><tr><th>BOM GROUP</th><th>BOM OPTIONS</th></tr><tr><td>J117_COMMON</td><td>COMMON,ALTERNATE,J117_PROGPARTS,SMCREG:SUP,XDP,SMBUS1:ISOL,USB_OC_ISO:Y,RTCRST:Y,AUDIO_DP_SNS:N</td></tr><tr><td>J117_PROGPARTS</td><td>SMC:PROG,BOOTROM:PROG,CAMROM:PROG,TBTROM:PROG,ENETROM:PROG</td></tr><tr><td>J117_DEVEL</td><td>XDP_CONN,TEMPSNSDEV,SAMCONN</td></tr></table> CPUs <table><tr><th>PART#</th><th>QTY</th><th>DESCRIPTION</th><th>REFERENCE DESIGNATOR(S)</th><th>CRITICAL</th><th>BOM OPTION</th></tr><tr><td>337S00101</td><td>1</td><td>IC,CPU,BOM-02Z,Q8T1,B9,B9-B2,Z13,1,4,15W,,95,1168</td><td>U0500</td><td>CRITICAL</td><td>CPU:ULT</td></tr><tr><td>998-7866</td><td>1</td><td>INTERPOSER,RGA168,SINGLE SIDE</td><td>U0500</td><td>CRITICAL</td><td>CPU:SOCKET</td></tr></table> ASIC Parts <table><tr><th>PART#</th><th>QTY</th><th>DESCRIPTION</th><th>REFERENCE DESIGNATOR(S)</th><th>CRITICAL</th><th>BOM OPTION</th></tr><tr><td>338S1247</td><td>1</td><td>IC,TBT,F8-4C,A0,F9Q,C10,SR1JC,PCBGA288</td><td>U2800</td><td>CRITICAL</td><td></td></tr><tr><td>343S0616</td><td>1</td><td>IC,BCH57766A,C1V+,A0,8x8</td><td>U3900</td><td>CRITICAL</td><td></td></tr></table> Programmable Parts <table><tr><th>PART#</th><th>QTY</th><th>DESCRIPTION</th><th>REFERENCE DESIGNATOR(S)</th><th>CRITICAL</th><th>BOM OPTION</th></tr><tr><td>341S00158</td><td>1</td><td>IC,EFI,V0108,J117</td><td>U5210</td><td>CRITICAL</td><td>BOOTROM:PROG</td></tr><tr><td>335S00006</td><td>1</td><td>IC,SERIAL_FLASH,64MBIT,3V,B9,WS08,QB=1</td><td>U5210</td><td>CRITICAL</td><td>BOOTROM:BLANK</td></tr><tr><td>341S00110</td><td>1</td><td>IC,SMC-B1,EXTERNAL,V2.24A19,POC,J117</td><td>U5000</td><td>CRITICAL</td><td>SMC:PROG</td></tr><tr><td>338S1214</td><td>1</td><td>IC,DM12-B1,AMB2/50MIPS,M0,10780A</td><td>U5000</td><td>CRITICAL</td><td>SMC:BLANK</td></tr><tr><td>341S3778</td><td>1</td><td>IC,CAMERA,FLASH,V7229,J16</td><td>U4202</td><td>CRITICAL</td><td>CAMROM:PROG</td></tr><tr><td>335S0852</td><td>1</td><td>IC,FLASH,SPI,1MBIT,3V3</td><td>U4202</td><td>CRITICAL</td><td>CAMROM:BLANK</td></tr><tr><td>341S00154</td><td>1</td><td>IC,EPROM,T29,FALCON RIDGE,V27.1,J117</td><td>U2890</td><td>CRITICAL</td><td>TBTROM:PROG</td></tr><tr><td>335S0915</td><td>1</td><td>IC,FLASH,SPI,4MBIT,50MHZ</td><td>U2890</td><td>CRITICAL</td><td>TBTROM:BLANK</td></tr><tr><td>341S3912</td><td>1</td><td>IC,ENET SPI ROM,HYUNIX,V1.15,,J16/Z166/Z17</td><td>U3990</td><td>CRITICAL</td><td>ENETROM:PROG</td></tr><tr><td>335S1025</td><td>1</td><td>IC,SERIAL_FLASH,2MBIT,2.7V,REV F</td><td>U3990</td><td>CRITICAL</td><td>ENETROM:BLANK</td></tr></table>								BOM GROUP	BOM OPTIONS	J117_COMMON	COMMON,ALTERNATE,J117_PROGPARTS,SMCREG:SUP,XDP,SMBUS1:ISOL,USB_OC_ISO:Y,RTCRST:Y,AUDIO_DP_SNS:N	J117_PROGPARTS	SMC:PROG,BOOTROM:PROG,CAMROM:PROG,TBTROM:PROG,ENETROM:PROG	J117_DEVEL	XDP_CONN,TEMPSNSDEV,SAMCONN	PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION	337S00101	1	IC,CPU,BOM-02Z,Q8T1,B9,B9-B2,Z13,1,4,15W,,95,1168	U0500	CRITICAL	CPU:ULT	998-7866	1	INTERPOSER,RGA168,SINGLE SIDE	U0500	CRITICAL	CPU:SOCKET	PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION	338S1247	1	IC,TBT,F8-4C,A0,F9Q,C10,SR1JC,PCBGA288	U2800	CRITICAL		343S0616	1	IC,BCH57766A,C1V+,A0,8x8	U3900	CRITICAL		PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION	341S00158	1	IC,EFI,V0108,J117	U5210	CRITICAL	BOOTROM:PROG	335S00006	1	IC,SERIAL_FLASH,64MBIT,3V,B9,WS08,QB=1	U5210	CRITICAL	BOOTROM:BLANK	341S00110	1	IC,SMC-B1,EXTERNAL,V2.24A19,POC,J117	U5000	CRITICAL	SMC:PROG	338S1214	1	IC,DM12-B1,AMB2/50MIPS,M0,10780A	U5000	CRITICAL	SMC:BLANK	341S3778	1	IC,CAMERA,FLASH,V7229,J16	U4202	CRITICAL	CAMROM:PROG	335S0852	1	IC,FLASH,SPI,1MBIT,3V3	U4202	CRITICAL	CAMROM:BLANK	341S00154	1	IC,EPROM,T29,FALCON RIDGE,V27.1,J117	U2890	CRITICAL	TBTROM:PROG	335S0915	1	IC,FLASH,SPI,4MBIT,50MHZ	U2890	CRITICAL	TBTROM:BLANK	341S3912	1	IC,ENET SPI ROM,HYUNIX,V1.15,,J16/Z166/Z17	U3990	CRITICAL	ENETROM:PROG	335S1025	1	IC,SERIAL_FLASH,2MBIT,2.7V,REV F	U3990	CRITICAL	ENETROM:BLANK	
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J117_COMMON	COMMON,ALTERNATE,J117_PROGPARTS,SMCREG:SUP,XDP,SMBUS1:ISOL,USB_OC_ISO:Y,RTCRST:Y,AUDIO_DP_SNS:N																																																																																																																						
J117_PROGPARTS	SMC:PROG,BOOTROM:PROG,CAMROM:PROG,TBTROM:PROG,ENETROM:PROG																																																																																																																						
J117_DEVEL	XDP_CONN,TEMPSNSDEV,SAMCONN																																																																																																																						
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337S00101	1	IC,CPU,BOM-02Z,Q8T1,B9,B9-B2,Z13,1,4,15W,,95,1168	U0500	CRITICAL	CPU:ULT																																																																																																																		
998-7866	1	INTERPOSER,RGA168,SINGLE SIDE	U0500	CRITICAL	CPU:SOCKET																																																																																																																		
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338S1247	1	IC,TBT,F8-4C,A0,F9Q,C10,SR1JC,PCBGA288	U2800	CRITICAL																																																																																																																			
343S0616	1	IC,BCH57766A,C1V+,A0,8x8	U3900	CRITICAL																																																																																																																			
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335S00006	1	IC,SERIAL_FLASH,64MBIT,3V,B9,WS08,QB=1	U5210	CRITICAL	BOOTROM:BLANK																																																																																																																		
341S00110	1	IC,SMC-B1,EXTERNAL,V2.24A19,POC,J117	U5000	CRITICAL	SMC:PROG																																																																																																																		
338S1214	1	IC,DM12-B1,AMB2/50MIPS,M0,10780A	U5000	CRITICAL	SMC:BLANK																																																																																																																		
341S3778	1	IC,CAMERA,FLASH,V7229,J16	U4202	CRITICAL	CAMROM:PROG																																																																																																																		
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341S00154	1	IC,EPROM,T29,FALCON RIDGE,V27.1,J117	U2890	CRITICAL	TBTROM:PROG																																																																																																																		
335S0915	1	IC,FLASH,SPI,4MBIT,50MHZ	U2890	CRITICAL	TBTROM:BLANK																																																																																																																		
341S3912	1	IC,ENET SPI ROM,HYUNIX,V1.15,,J16/Z166/Z17	U3990	CRITICAL	ENETROM:PROG																																																																																																																		
335S1025	1	IC,SERIAL_FLASH,2MBIT,2.7V,REV F	U3990	CRITICAL	ENETROM:BLANK																																																																																																																		
B	CPU DRAM SPD Straps <table><tr><th>BOM GROUP</th><th>BOM OPTIONS</th></tr><tr><td>DDR3:HYNIX_8GB_1600</td><td>RAMCFG0:L,RAMCFG1:L,RAMCFG2:L,RAMCFG3:L,RAMCFG4:L,HYNIX_8GB_1600</td></tr><tr><td>DDR3:HYNIX_16GB_1600</td><td>RAMCFG0:L,RAMCFG1:L,RAMCFG2:H,RAMCFG3:L,RAMCFG4:L,HYNIX_16GB_1600</td></tr><tr><td>DDR3:HYNIX_8GB_1866</td><td>RAMCFG0:L,RAMCFG1:L,RAMCFG2:L,RAMCFG3:L,RAMCFG4:H,HYNIX_8GB_1866</td></tr><tr><td>DDR3:HYNIX_16GB_1866</td><td>RAMCFG0:L,RAMCFG1:L,RAMCFG2:H,RAMCFG3:L,RAMCFG4:H,HYNIX_16GB_1866</td></tr><tr><td>DDR3:ELPIDA_8GB_1600</td><td>RAMCFG0:H,RAMCFG1:H,RAMCFG2:L,RAMCFG3:L,RAMCFG4:L,ELPIDA_8GB_1600</td></tr><tr><td>DDR3:ELPIDA_16GB_1600</td><td>RAMCFG0:H,RAMCFG1:H,RAMCFG2:H,RAMCFG3:L,RAMCFG4:L,ELPIDA_16GB_1600</td></tr><tr><td>DDR3:ELPIDA_8GB_1866</td><td>RAMCFG0:H,RAMCFG1:H,RAMCFG2:L,RAMCFG3:L,RAMCFG4:H,ELPIDA_8GB_1866</td></tr><tr><td>DDR3:ELPIDA_16GB_1866</td><td>RAMCFG0:H,RAMCFG1:H,RAMCFG2:H,RAMCFG3:L,RAMCFG4:H,ELPIDA_16GB_1866</td></tr><tr><td>DDR3:SAMSUNG_8GB_1600</td><td>RAMCFG0:H,RAMCFG1:L,RAMCFG2:L,RAMCFG3:L,RAMCFG4:L,SAMSUNG_8GB_1600</td></tr><tr><td>DDR3:SAMSUNG_8GB_1866</td><td>RAMCFG0:H,RAMCFG1:L,RAMCFG2:L,RAMCFG3:L,RAMCFG4:H,SAMSUNG_8GB_1866</td></tr></table> DRAM Parts <table><tr><th>PART NUMBER</th><th>QTY</th><th>DESCRIPTION</th><th>REFERENCE DES</th><th>CRITICAL</th><th>BOM OPTION</th></tr><tr><td>333S0783</td><td>4</td><td>IC,SDRAM,25nm 32Gb,LPDDR3-1600,178P FBGA</td><td>U2300,U2400,U2500,U2600</td><td>CRITICAL</td><td>HYNIX_16GB_1600</td></tr><tr><td>333S0784</td><td>4</td><td>IC,SDRAM,25nm 32Gb,LPDDR3-1866,178P FBGA</td><td>U2300,U2400,U2500,U2600</td><td>CRITICAL</td><td>HYNIX_16GB_1866</td></tr><tr><td>333S0785</td><td>4</td><td>IC,SDRAM,29nm 16Gb,LPDDR3-1600,178P FBGA</td><td>U2300,U2400,U2500,U2600</td><td>CRITICAL</td><td>HYNIX_8GB_1600</td></tr><tr><td>333S0786</td><td>4</td><td>IC,SDRAM,29nm 16Gb,LPDDR3-1866,178P FBGA</td><td>U2300,U2400,U2500,U2600</td><td>CRITICAL</td><td>HYNIX_8GB_1866</td></tr><tr><td>333S0789</td><td>4</td><td>IC,SDRAM,25nm 32Gb,LPDDR3-1600,178P FBGA</td><td>U2300,U2400,U2500,U2600</td><td>CRITICAL</td><td>ELPIDA_16GB_1600</td></tr><tr><td>333S0790</td><td>4</td><td>IC,SDRAM,25nm 32Gb,LPDDR3-1866,178P FBGA</td><td>U2300,U2400,U2500,U2600</td><td>CRITICAL</td><td>ELPIDA_16GB_1866</td></tr><tr><td>333S0791</td><td>4</td><td>IC,SDRAM,25nm 16Gb,LPDDR3-1600,178P FBGA</td><td>U2300,U2400,U2500,U2600</td><td>CRITICAL</td><td>ELPIDA_8GB_1600</td></tr><tr><td>333S0792</td><td>4</td><td>IC,SDRAM,25nm 16Gb,LPDDR3-1866,178P FBGA</td><td>U2300,U2400,U2500,U2600</td><td>CRITICAL</td><td>ELPIDA_8GB_1866</td></tr><tr><td>333S00003</td><td>4</td><td>IC,SDRAM,23nm 16Gb,LPDDR3-1600,178P FBGA</td><td>U2300,U2400,U2500,U2600</td><td>CRITICAL</td><td>SAMSUNG_8GB_1600</td></tr><tr><td>333S00004</td><td>4</td><td>IC,SDRAM,23nm 16Gb,LPDDR3-1866,178P FBGA</td><td>U2300,U2400,U2500,U2600</td><td>CRITICAL</td><td>SAMSUNG_8GB_1866</td></tr></table>								BOM GROUP	BOM OPTIONS	DDR3:HYNIX_8GB_1600	RAMCFG0:L,RAMCFG1:L,RAMCFG2:L,RAMCFG3:L,RAMCFG4:L,HYNIX_8GB_1600	DDR3:HYNIX_16GB_1600	RAMCFG0:L,RAMCFG1:L,RAMCFG2:H,RAMCFG3:L,RAMCFG4:L,HYNIX_16GB_1600	DDR3:HYNIX_8GB_1866	RAMCFG0:L,RAMCFG1:L,RAMCFG2:L,RAMCFG3:L,RAMCFG4:H,HYNIX_8GB_1866	DDR3:HYNIX_16GB_1866	RAMCFG0:L,RAMCFG1:L,RAMCFG2:H,RAMCFG3:L,RAMCFG4:H,HYNIX_16GB_1866	DDR3:ELPIDA_8GB_1600	RAMCFG0:H,RAMCFG1:H,RAMCFG2:L,RAMCFG3:L,RAMCFG4:L,ELPIDA_8GB_1600	DDR3:ELPIDA_16GB_1600	RAMCFG0:H,RAMCFG1:H,RAMCFG2:H,RAMCFG3:L,RAMCFG4:L,ELPIDA_16GB_1600	DDR3:ELPIDA_8GB_1866	RAMCFG0:H,RAMCFG1:H,RAMCFG2:L,RAMCFG3:L,RAMCFG4:H,ELPIDA_8GB_1866	DDR3:ELPIDA_16GB_1866	RAMCFG0:H,RAMCFG1:H,RAMCFG2:H,RAMCFG3:L,RAMCFG4:H,ELPIDA_16GB_1866	DDR3:SAMSUNG_8GB_1600	RAMCFG0:H,RAMCFG1:L,RAMCFG2:L,RAMCFG3:L,RAMCFG4:L,SAMSUNG_8GB_1600	DDR3:SAMSUNG_8GB_1866	RAMCFG0:H,RAMCFG1:L,RAMCFG2:L,RAMCFG3:L,RAMCFG4:H,SAMSUNG_8GB_1866	PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION	333S0783	4	IC,SDRAM,25nm 32Gb,LPDDR3-1600,178P FBGA	U2300,U2400,U2500,U2600	CRITICAL	HYNIX_16GB_1600	333S0784	4	IC,SDRAM,25nm 32Gb,LPDDR3-1866,178P FBGA	U2300,U2400,U2500,U2600	CRITICAL	HYNIX_16GB_1866	333S0785	4	IC,SDRAM,29nm 16Gb,LPDDR3-1600,178P FBGA	U2300,U2400,U2500,U2600	CRITICAL	HYNIX_8GB_1600	333S0786	4	IC,SDRAM,29nm 16Gb,LPDDR3-1866,178P FBGA	U2300,U2400,U2500,U2600	CRITICAL	HYNIX_8GB_1866	333S0789	4	IC,SDRAM,25nm 32Gb,LPDDR3-1600,178P FBGA	U2300,U2400,U2500,U2600	CRITICAL	ELPIDA_16GB_1600	333S0790	4	IC,SDRAM,25nm 32Gb,LPDDR3-1866,178P FBGA	U2300,U2400,U2500,U2600	CRITICAL	ELPIDA_16GB_1866	333S0791	4	IC,SDRAM,25nm 16Gb,LPDDR3-1600,178P FBGA	U2300,U2400,U2500,U2600	CRITICAL	ELPIDA_8GB_1600	333S0792	4	IC,SDRAM,25nm 16Gb,LPDDR3-1866,178P FBGA	U2300,U2400,U2500,U2600	CRITICAL	ELPIDA_8GB_1866	333S00003	4	IC,SDRAM,23nm 16Gb,LPDDR3-1600,178P FBGA	U2300,U2400,U2500,U2600	CRITICAL	SAMSUNG_8GB_1600	333S00004	4	IC,SDRAM,23nm 16Gb,LPDDR3-1866,178P FBGA	U2300,U2400,U2500,U2600	CRITICAL	SAMSUNG_8GB_1866																							
BOM GROUP	BOM OPTIONS																																																																																																																						
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333S0784	4	IC,SDRAM,25nm 32Gb,LPDDR3-1866,178P FBGA	U2300,U2400,U2500,U2600	CRITICAL	HYNIX_16GB_1866																																																																																																																		
333S0785	4	IC,SDRAM,29nm 16Gb,LPDDR3-1600,178P FBGA	U2300,U2400,U2500,U2600	CRITICAL	HYNIX_8GB_1600																																																																																																																		
333S0786	4	IC,SDRAM,29nm 16Gb,LPDDR3-1866,178P FBGA	U2300,U2400,U2500,U2600	CRITICAL	HYNIX_8GB_1866																																																																																																																		
333S0789	4	IC,SDRAM,25nm 32Gb,LPDDR3-1600,178P FBGA	U2300,U2400,U2500,U2600	CRITICAL	ELPIDA_16GB_1600																																																																																																																		
333S0790	4	IC,SDRAM,25nm 32Gb,LPDDR3-1866,178P FBGA	U2300,U2400,U2500,U2600	CRITICAL	ELPIDA_16GB_1866																																																																																																																		
333S0791	4	IC,SDRAM,25nm 16Gb,LPDDR3-1600,178P FBGA	U2300,U2400,U2500,U2600	CRITICAL	ELPIDA_8GB_1600																																																																																																																		
333S0792	4	IC,SDRAM,25nm 16Gb,LPDDR3-1866,178P FBGA	U2300,U2400,U2500,U2600	CRITICAL	ELPIDA_8GB_1866																																																																																																																		
333S00003	4	IC,SDRAM,23nm 16Gb,LPDDR3-1600,178P FBGA	U2300,U2400,U2500,U2600	CRITICAL	SAMSUNG_8GB_1600																																																																																																																		
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A	8	7	6	5	4	3	2	1	Strategic Silicon <table><tr><th>PART#</th><th>STRATEGIC VALUE</th><th>COMMENT</th></tr><tr><td>337S00065</td><td>08</td><td>CPU,BDW-ULT,2+GT3</td></tr><tr><td>338S1247</td><td>02</td><td>TBT,Falcon Ridge-4c</td></tr><tr><td>335S00006</td><td>07</td><td>IC,SERIAL_FLASH,Quad-10</td></tr><tr><td>333S0784</td><td>07</td><td>32Gb,25nm LPDDR3-1866</td></tr><tr><td>333S0786</td><td>07</td><td>16Gb,29nm LPDDR3-1866</td></tr><tr><td>333S0790</td><td>07</td><td>32Gb,25nm LPDDR3-1866</td></tr><tr><td>333S0792</td><td>07</td><td>16Gb,25nm LPDDR3-1866</td></tr><tr><td>333S00004</td><td>07</td><td>16Gb,23nm LPDDR3-1866</td></tr></table> CPU DRAM CFG Chart <table><tr><th></th><th>VENDOR</th><th>CFG 1</th><th>CFG 0</th></tr><tr><td></td><td>HYNIX</td><td>0</td><td>0</td></tr><tr><td></td><td>SAMSUNG</td><td>0</td><td>1</td></tr><tr><td></td><td>N/A</td><td>1</td><td>0</td></tr><tr><td></td><td>ELPIDA</td><td>1</td><td>1</td></tr></table><table><tr><th></th><th>SIZE</th><th>CFG 2</th></tr><tr><td></td><td>8GB</td><td>0</td></tr><tr><td></td><td>16GB</td><td>1</td></tr></table> SYNC_MASTER=J117 TONY SYNC_DATE=05/05/2014 PAGE TITLE Apple Inc. DRAWING NUMBER 051-00081 REVISION 3.0.0 NOTICE OF PROPRIETARY PROPERTY: THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE INC. THE POSSESSOR AGREES TO THE FOLLOWING: I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE II NOT TO REPRODUCE OR COPY IT III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART IV ALL RIGHTS RESERVED BRANCH PAGE 2 OF 123 SHEET 2 OF 81	PART#	STRATEGIC VALUE	COMMENT	337S00065	08	CPU,BDW-ULT,2+GT3	338S1247	02	TBT,Falcon Ridge-4c	335S00006	07	IC,SERIAL_FLASH,Quad-10	333S0784	07	32Gb,25nm LPDDR3-1866	333S0786	07	16Gb,29nm LPDDR3-1866	333S0790	07	32Gb,25nm LPDDR3-1866	333S0792	07	16Gb,25nm LPDDR3-1866	333S00004	07	16Gb,23nm LPDDR3-1866		VENDOR	CFG 1	CFG 0		HYNIX	0	0		SAMSUNG	0	1		N/A	1	0		ELPIDA	1	1		SIZE	CFG 2		8GB	0		16GB	1																																																						
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	16GB	1																																																																																																																					

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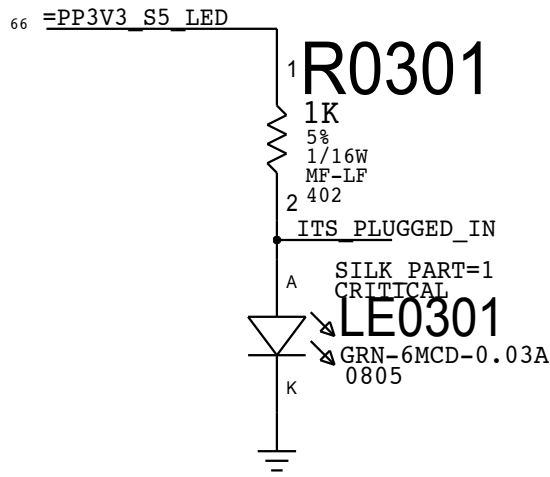
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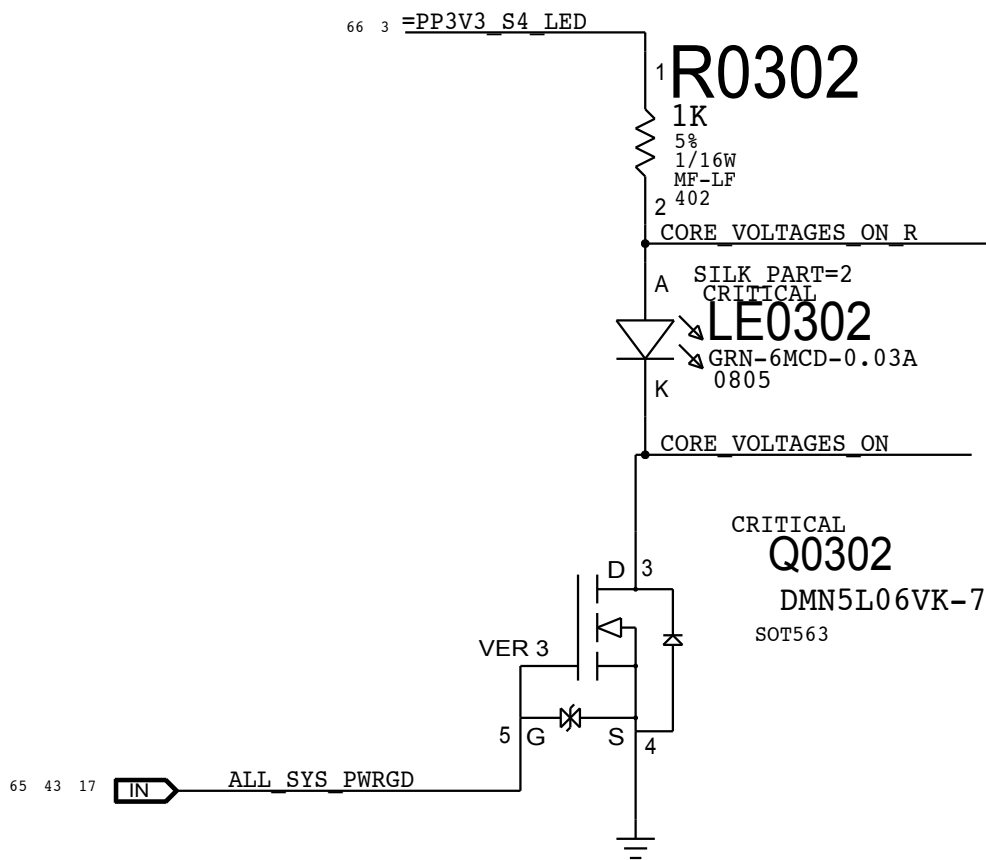
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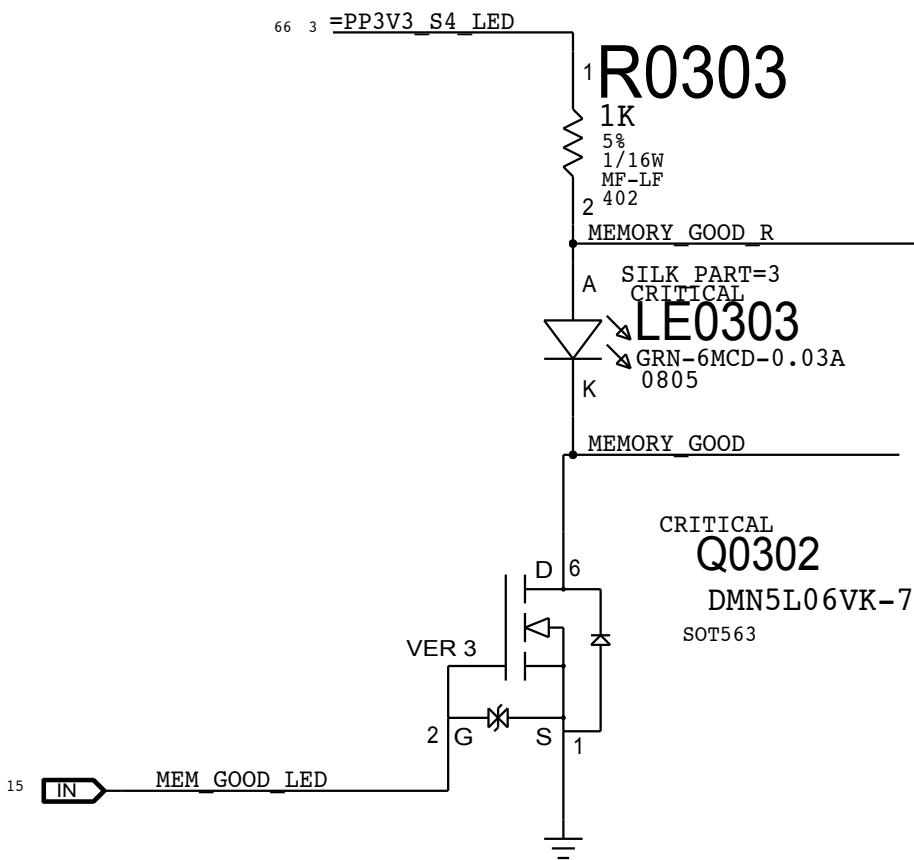
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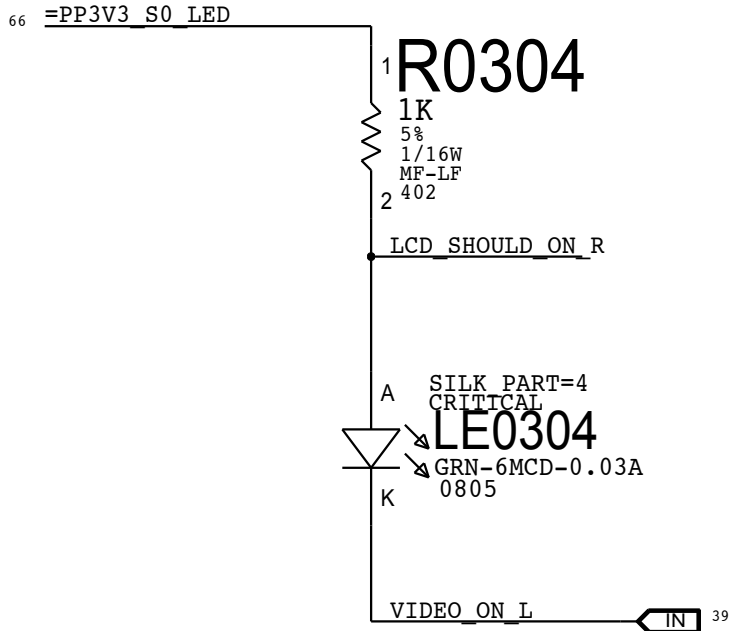
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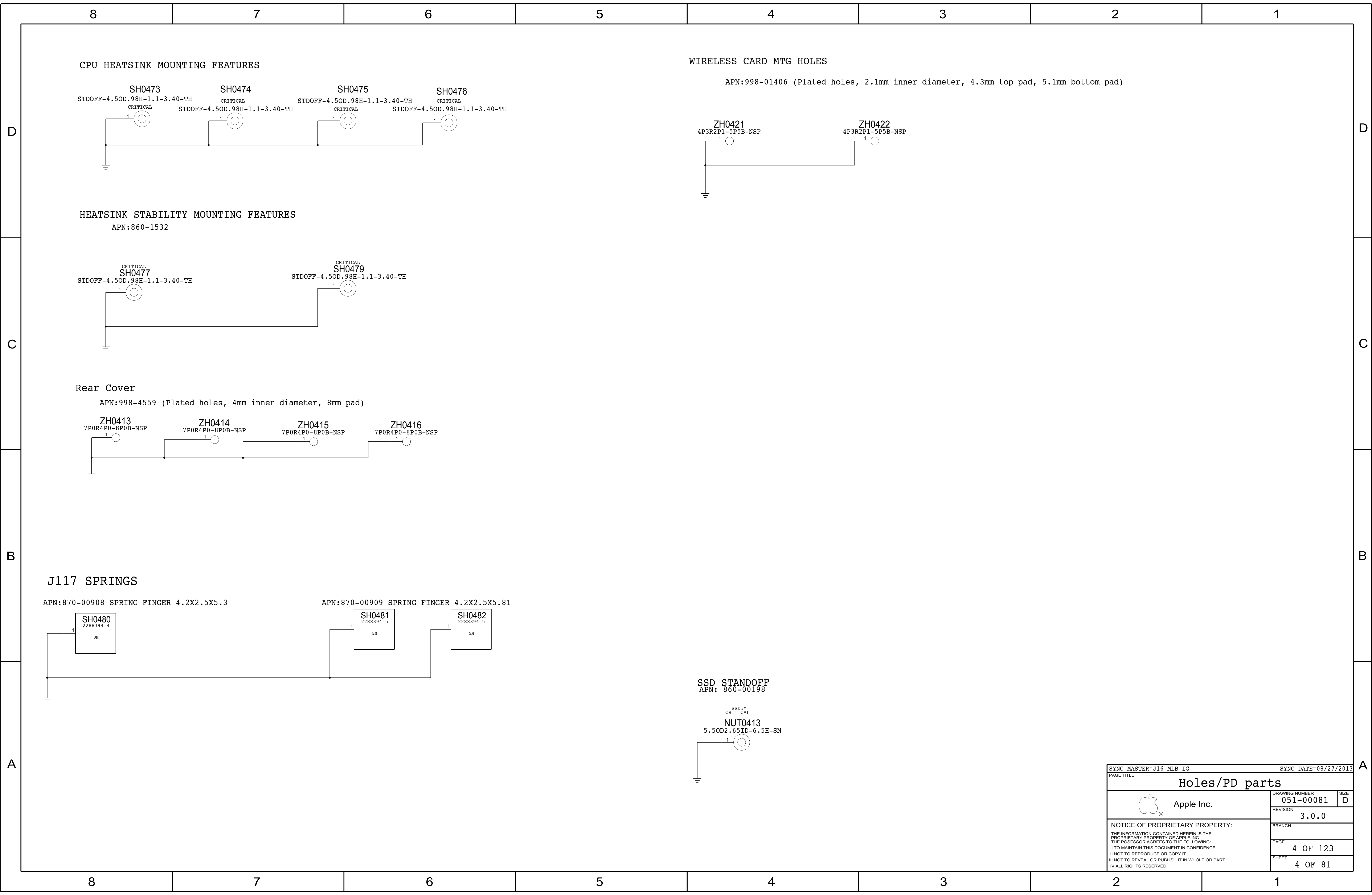


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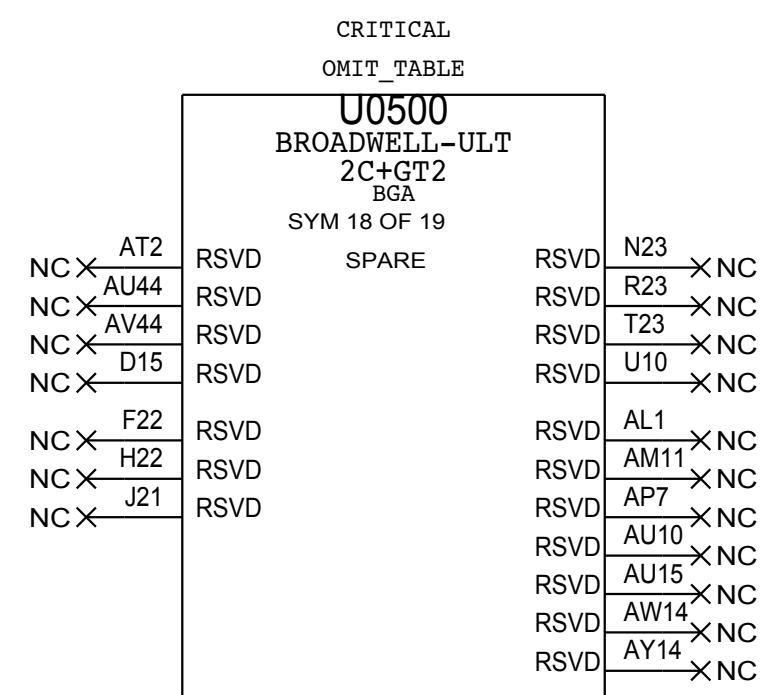
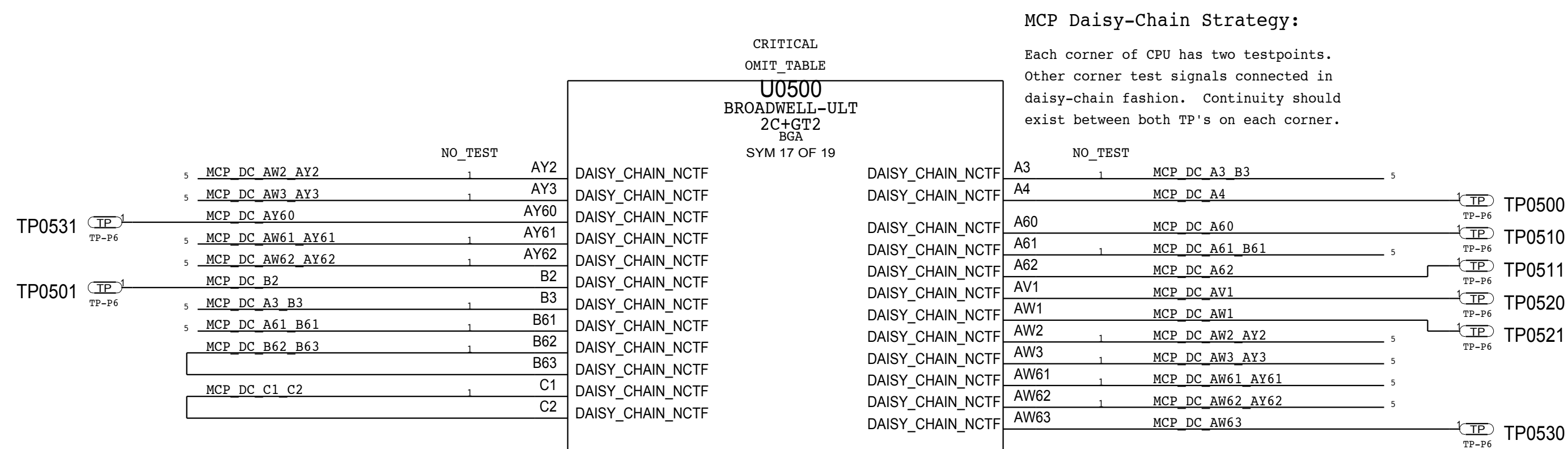
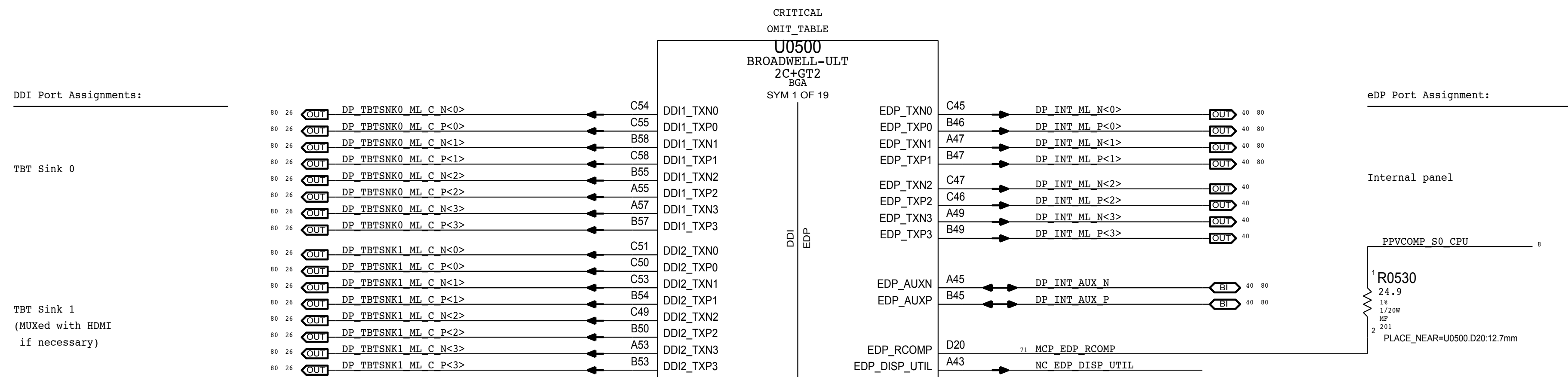


VIDEO ON Led





SYNC MASTER=J16 MLB IG		SYNC DATE=08/27/2013	
PAGE TITLE			
Holes/PD parts			
 Apple Inc.	DRAWING NUMBER		SIZE
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	BRANCH		
	PAGE		
	4 OF 123		
	SHEET		
	4 OF 81		



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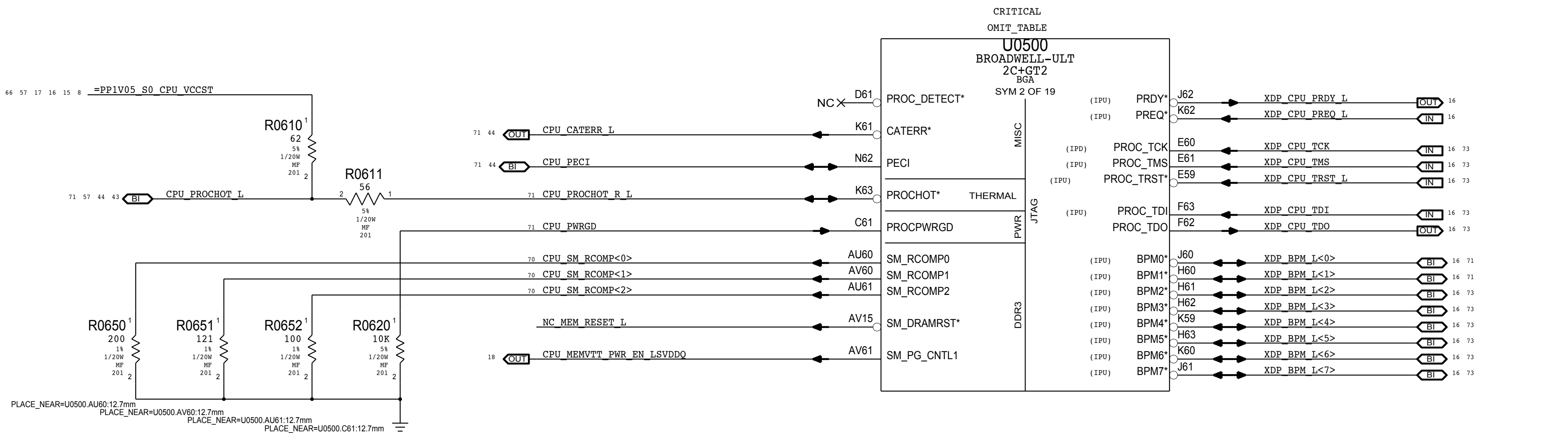
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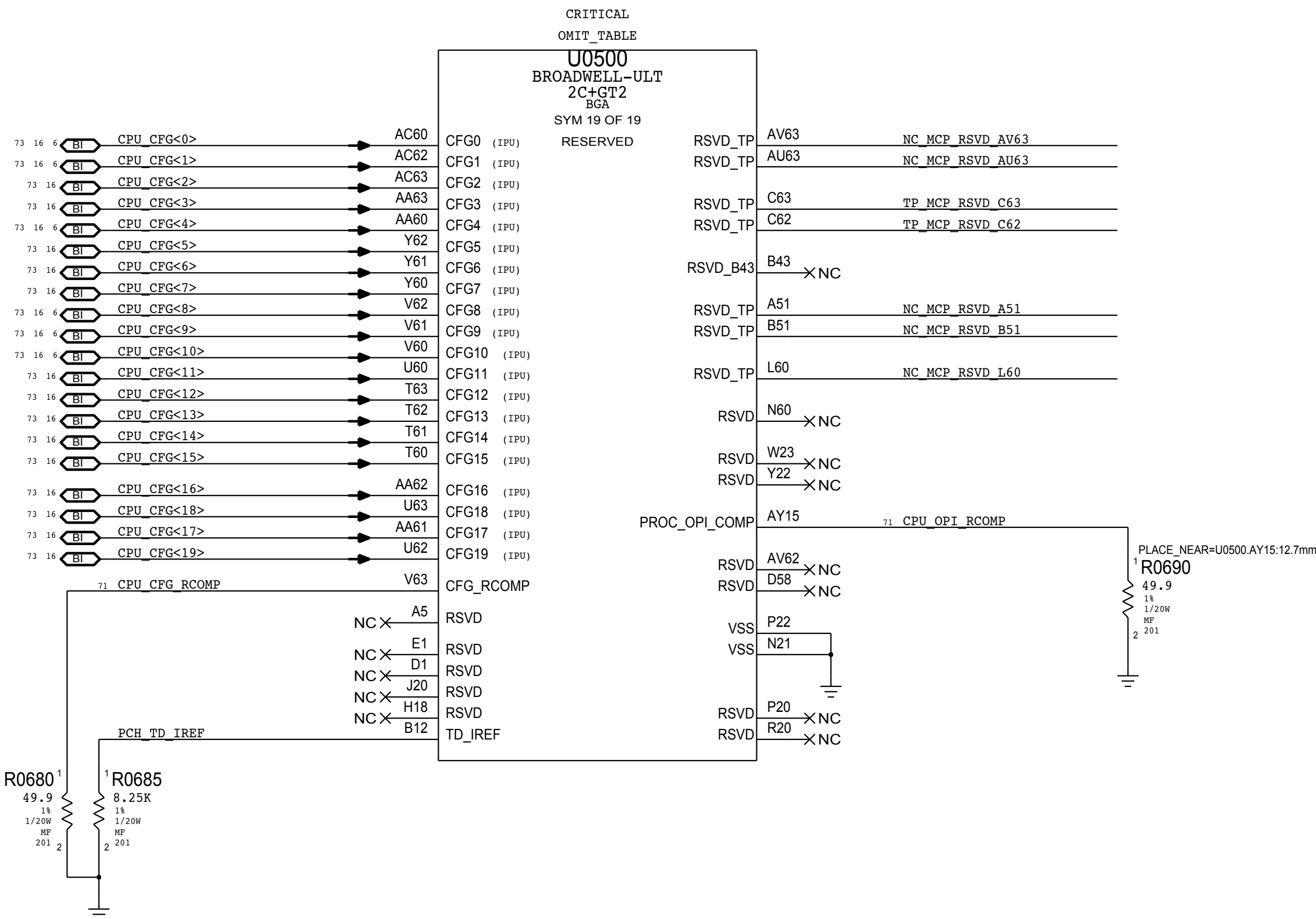
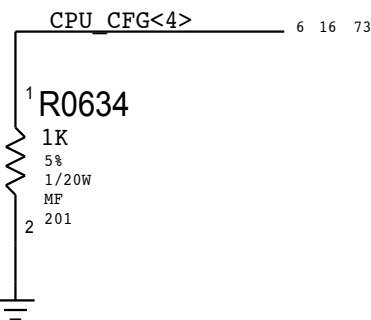
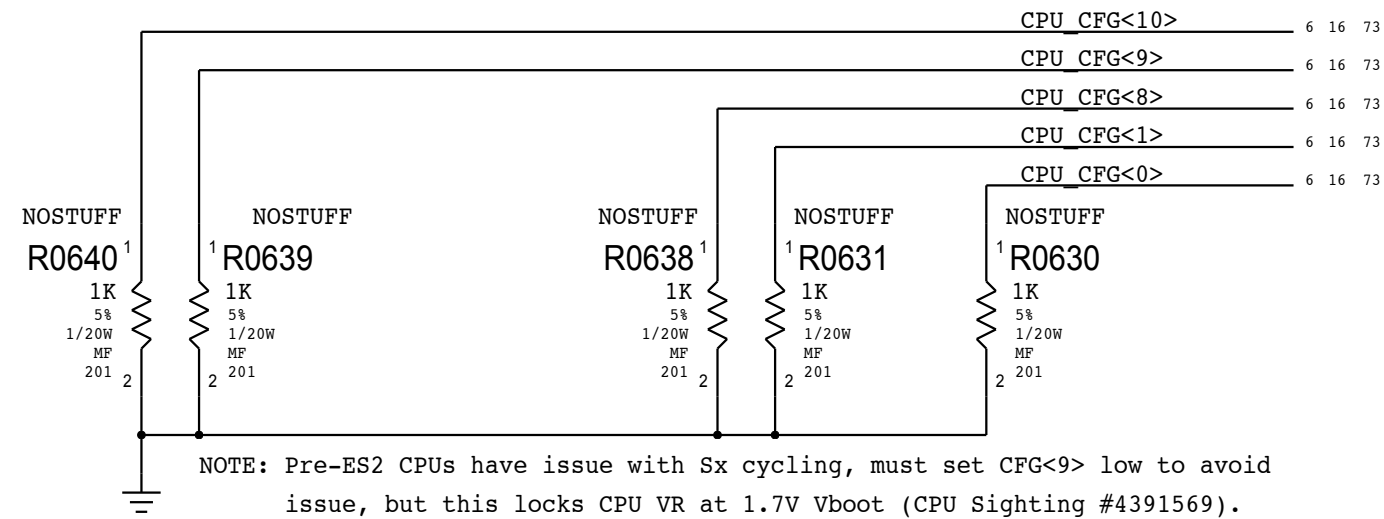
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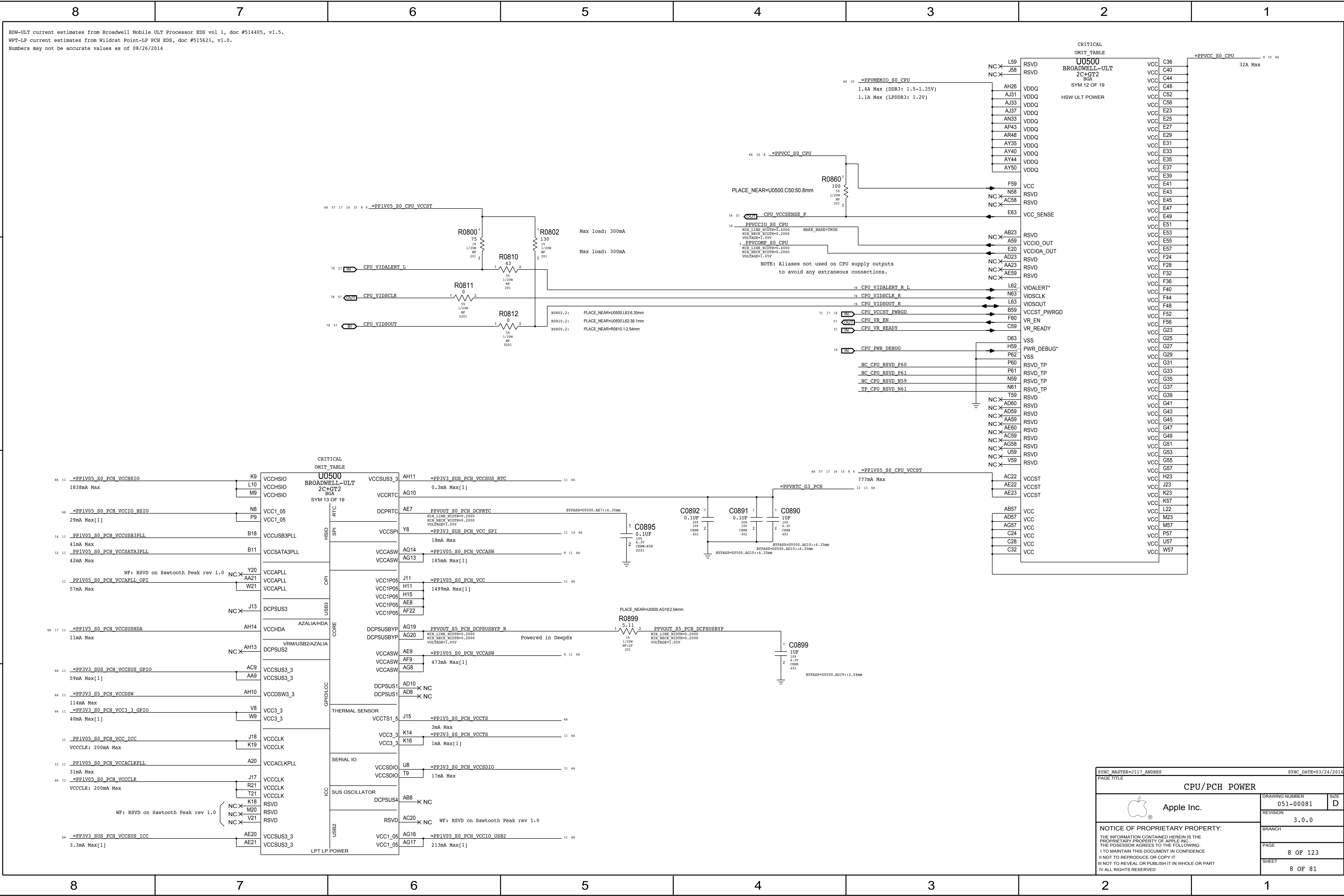
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CFG<9> :NO SVID-CAPABLE VR	1 = VR SUPPORTS SVID	0 = VR DOES NOT SUPPORT SVID
CFG<8> :ALLOW NOA ON LOCKED UNITS	1 = NORMAL OPERATION	0 = NOA ALWAYS UNLOCKED
CFG<4> :eDP ENABLE/DISABLE	1 = DISABLED	0 = ENABLED
CFG<1> :PCH-LESS MODE	1 = NORMAL OPERATION	0 = PCH-LESS MODE
CFG<0> :RESET SEQUENCE STALL	1 = NORMAL OPERATION	0 = STALL AFTER PCU PLL LOCK

These can be placed close to J1800
and are only for debug access



SYNC_MASTER=J117_ANDRES		SYNC_DATE=03/24/2014	
PAGE TITLE			
CPU Misc/JTAG/CFG/RSVD			
	DRAWING NUMBER		SIZE
	051-00081		D
REVISION		3.0.0	
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		PAGE	
		6 OF 123	
		SHEET	
		6 OF 81	





All Intel recommendations from Intel doc #503160 Shark Bay Ultrabook Platform Power Delivery Design Guide Rev 1.0 unless stated otherwise

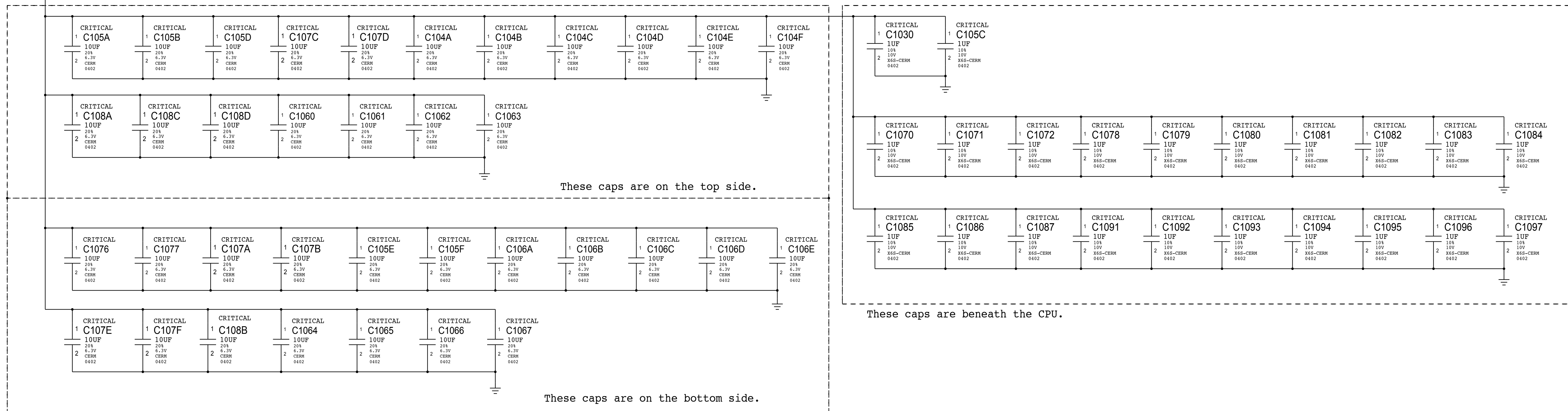
CPU VCC Decoupling

Intel recommendation (Table 5-1): 23x 22uF 0805 stuff, 7x 22uF 0805 nostuff

Apple implementation : 18x 10uF 0402 mirrored stuff, 1x 470uF stuff, 50x 10uF mirrored no stuff, 50x 10uF single sided no stuff

J70 implementation : 18x 10uF 0402 mirrored stuff, 32x 10uF single sided no stuff

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66 8  =PPVCC S0 CPU
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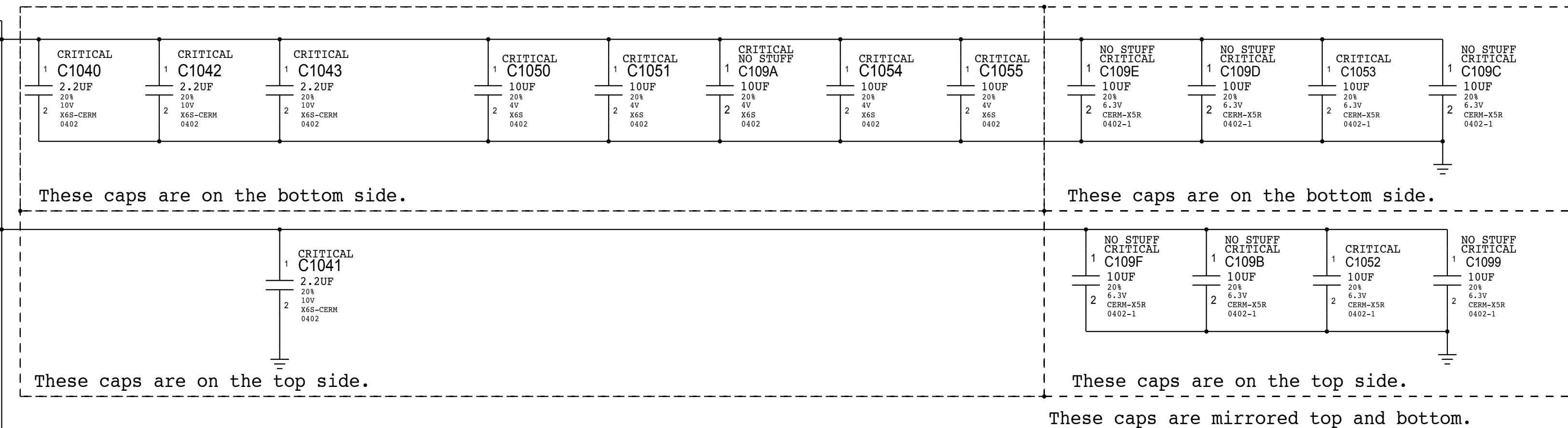


CPU VDDQ DECOUPLING

Intel recommendation (Table 5-4): 4x 2.2uF 0402, 6x 10uF 0603

Apple implementation : 4x 2.2uF 0402, 6x 10uF 0402, 6x 10uF no stuff, 1x 270 uF Bulk

```
66 8 =PPVMEMIO S0 CPU
```



CRITICAL
C1056
270UF-0.0060HM
20%
2V
TANT
CASE-D2

D

C

B

A

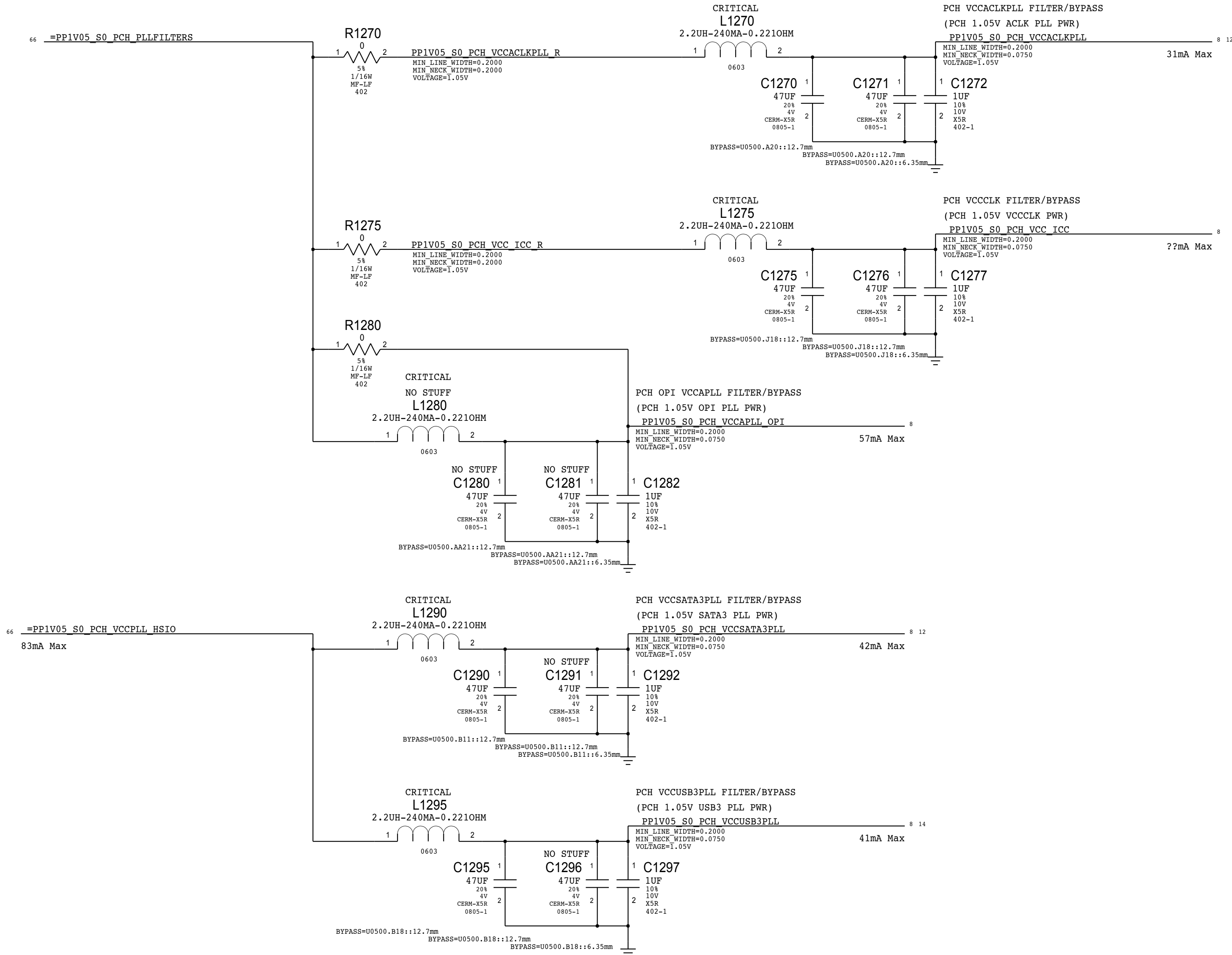
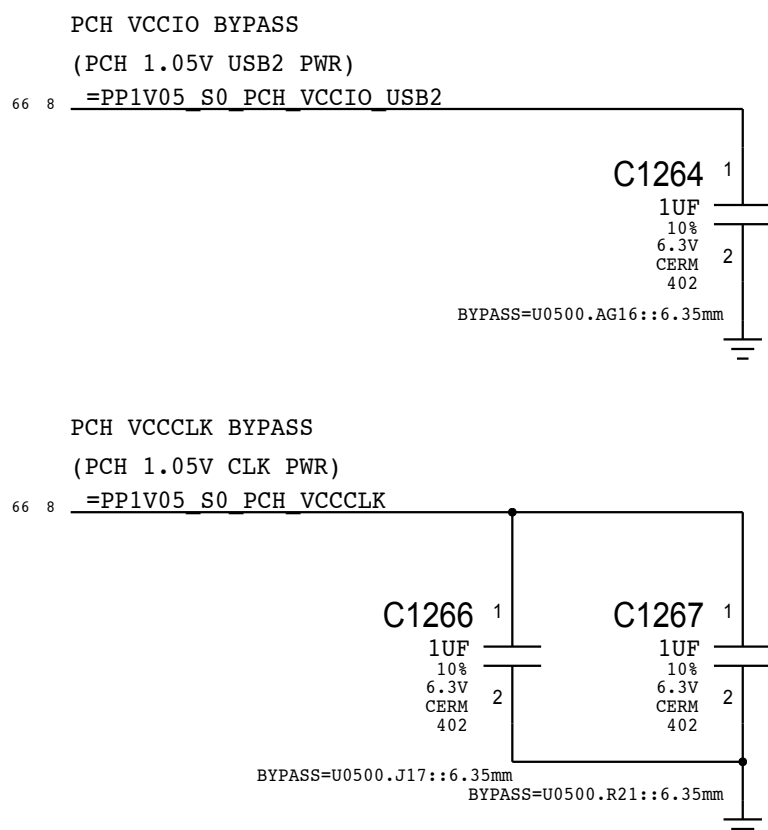
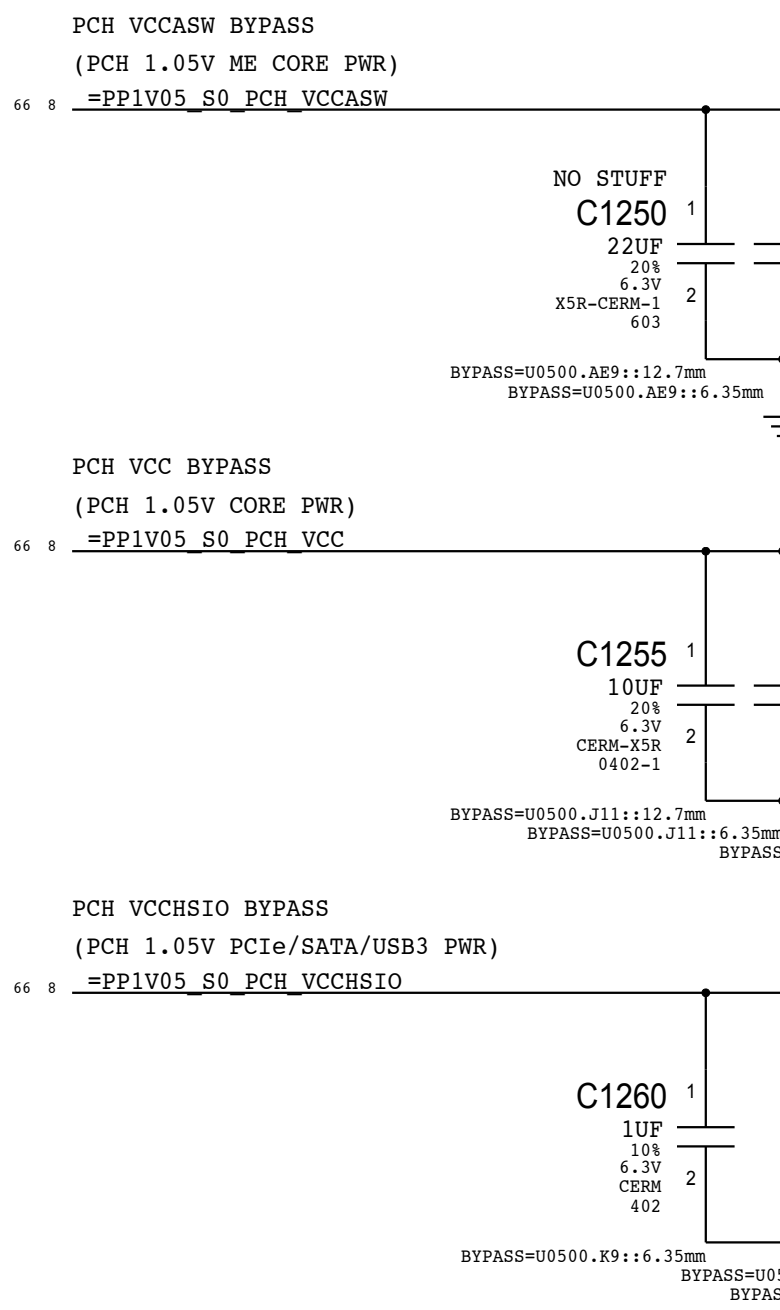
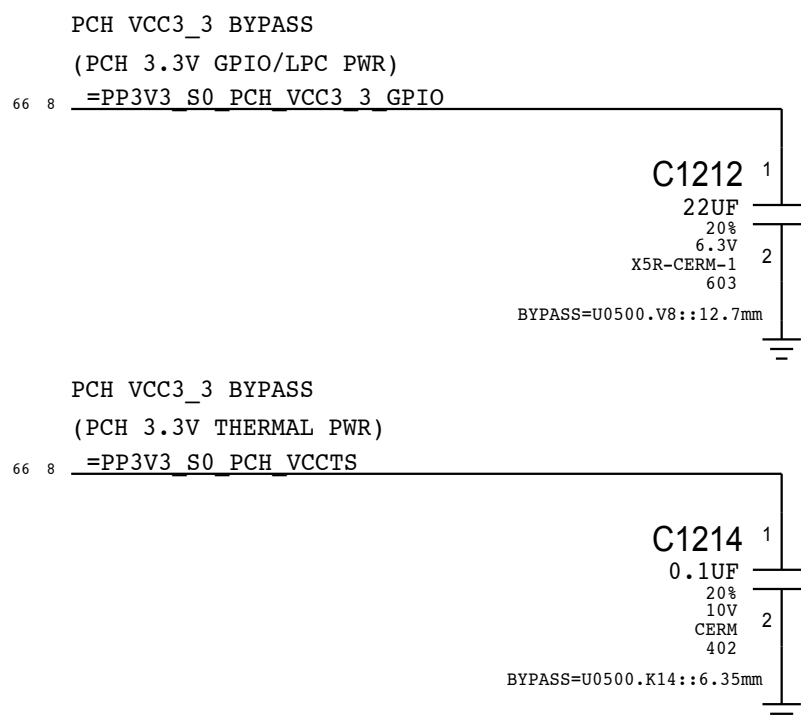
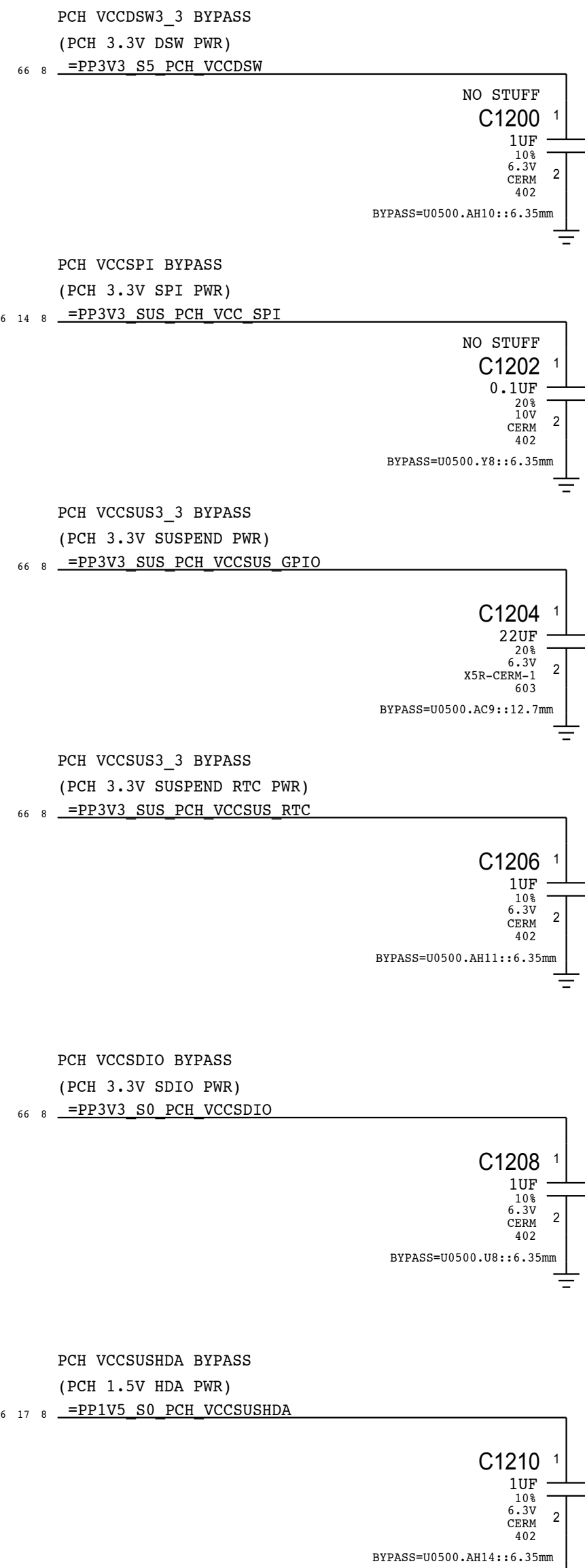
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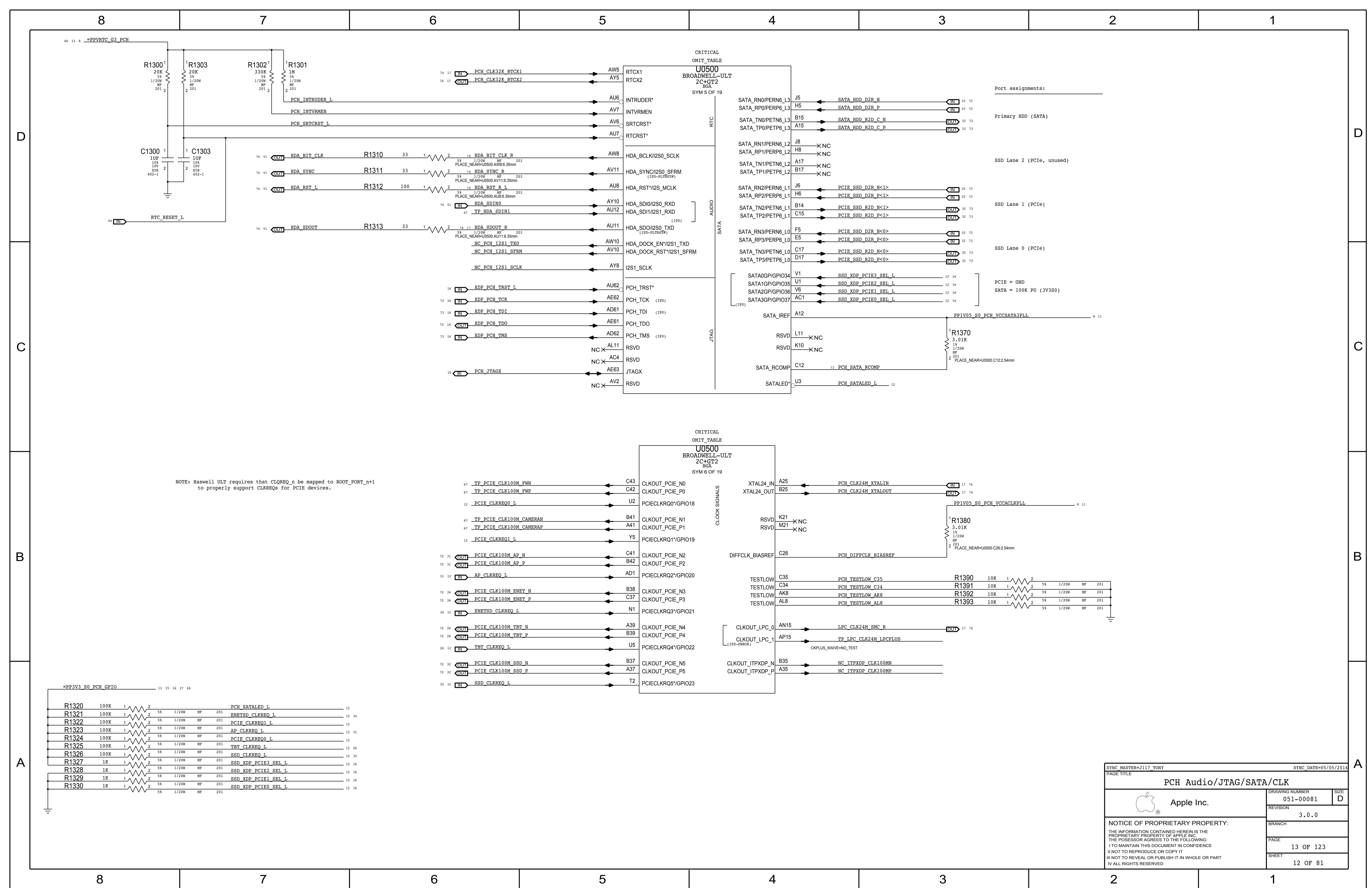
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 Apple Inc.	DRAWING NUMBER		SIZE
	051-00081		D
	REVISION		
	3.0.0		
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PAGE		12 OF 123	
SHEET		11 OF 81	





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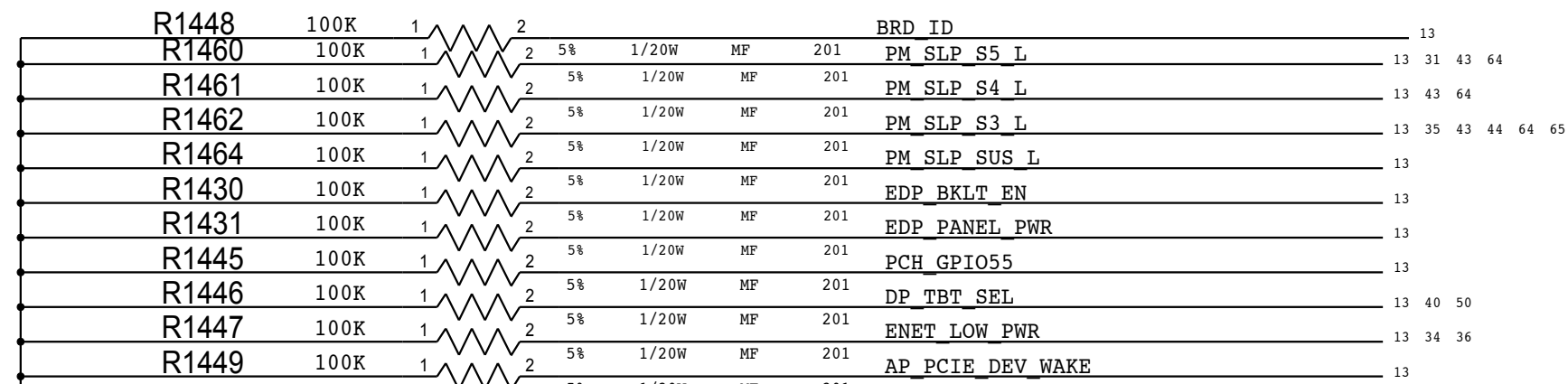
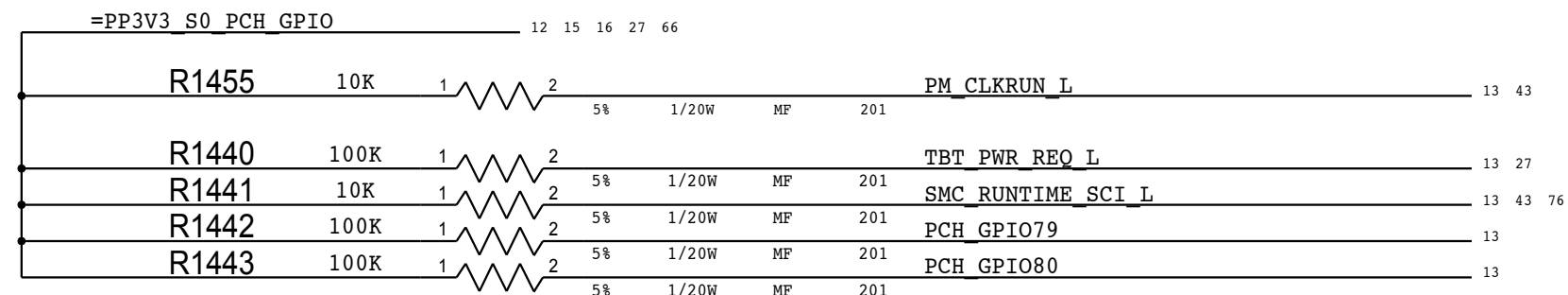
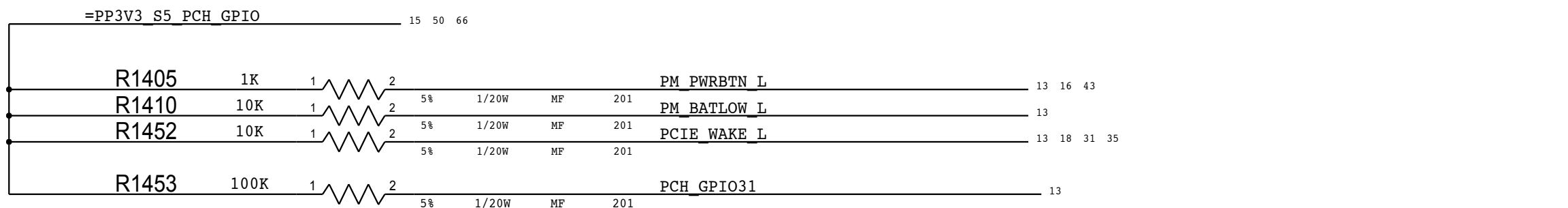
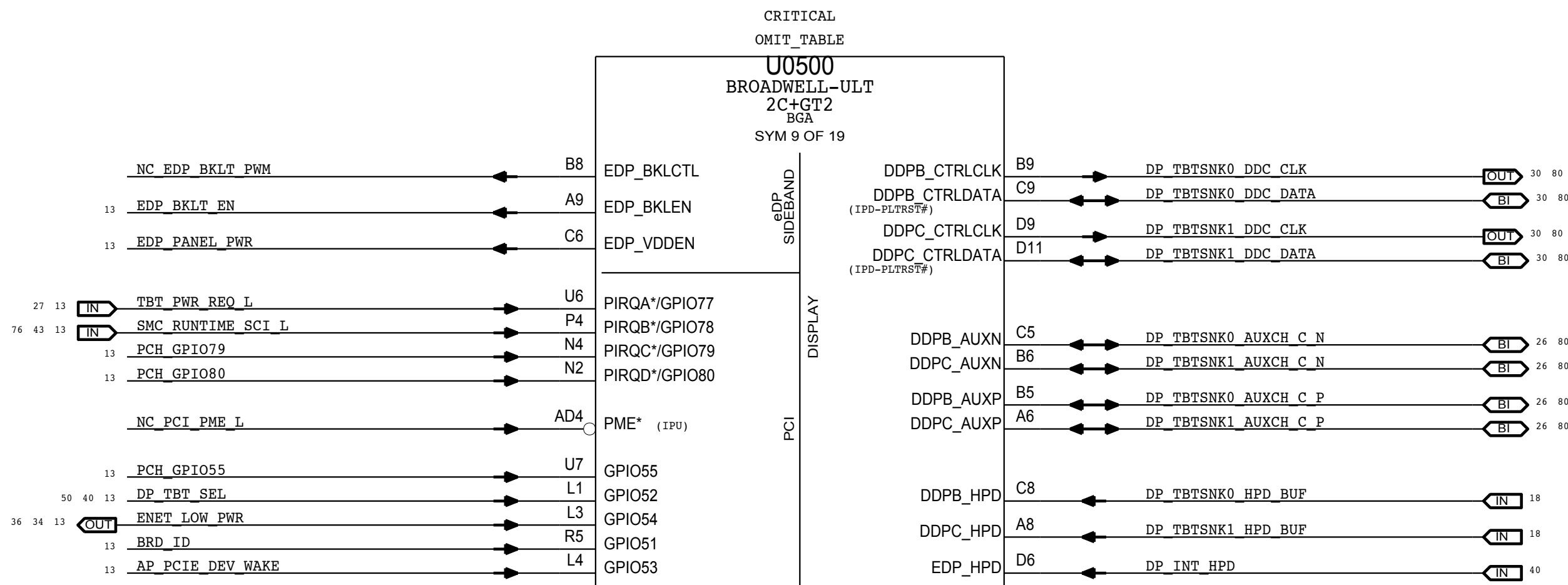
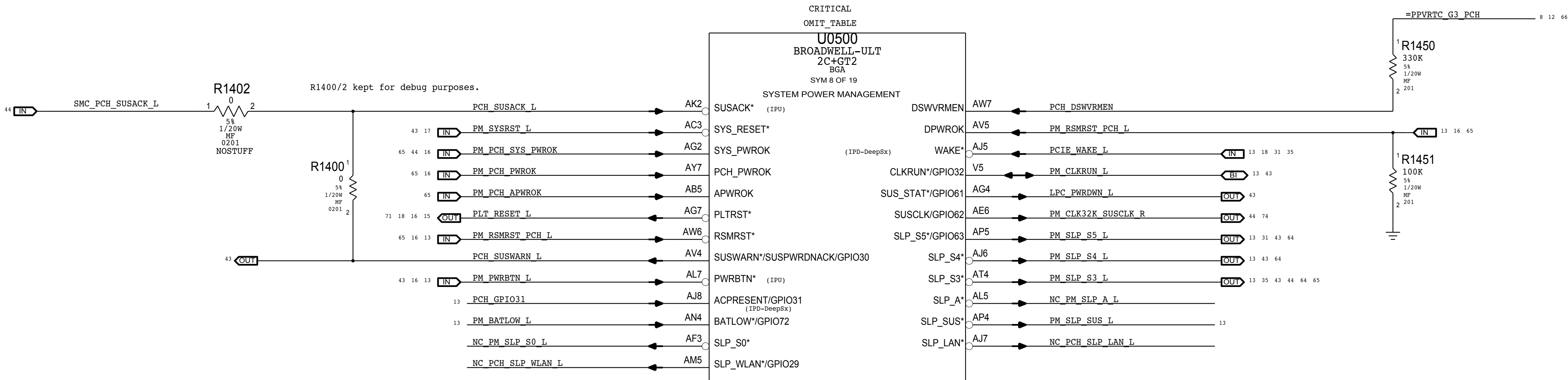
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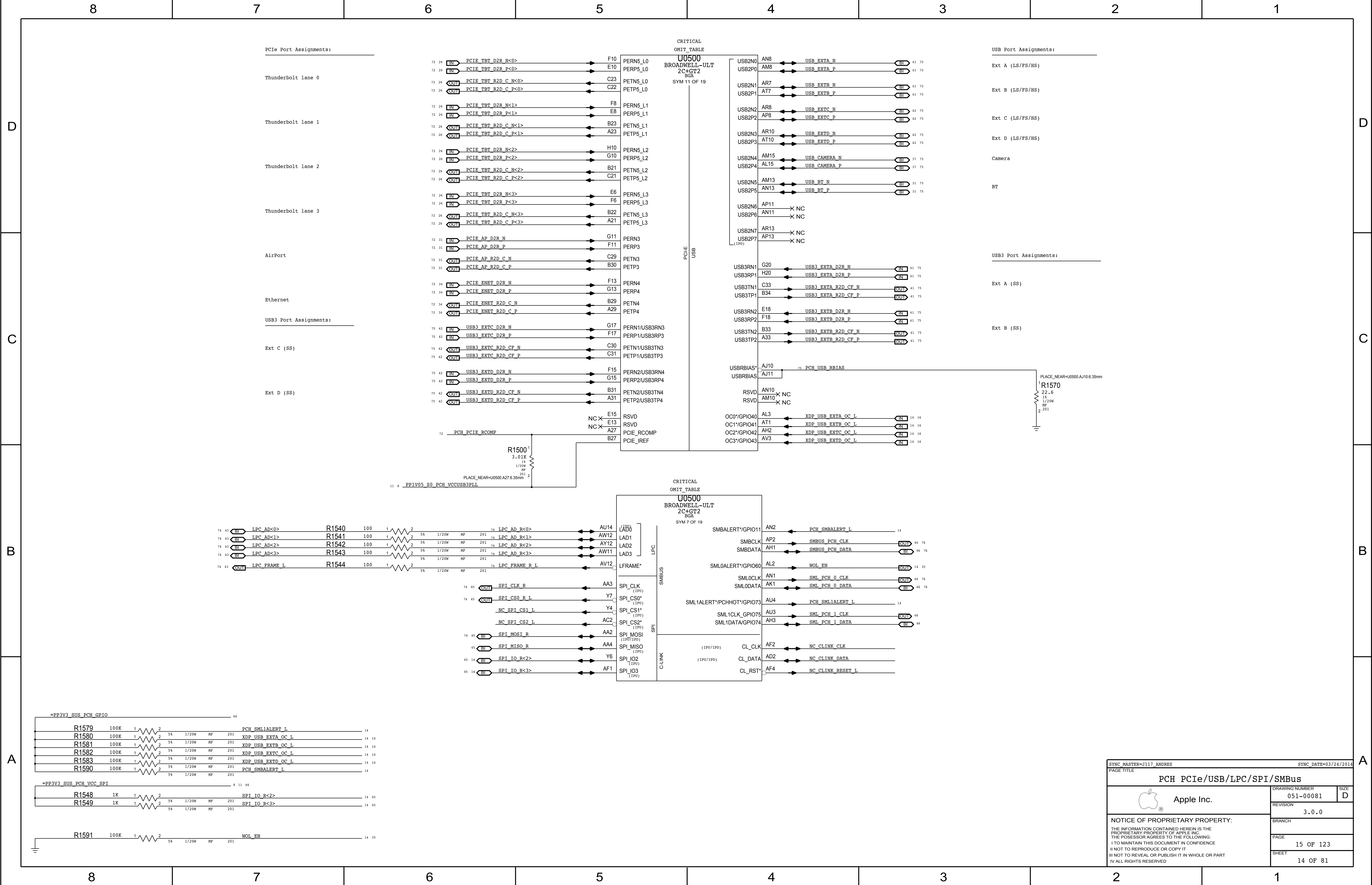
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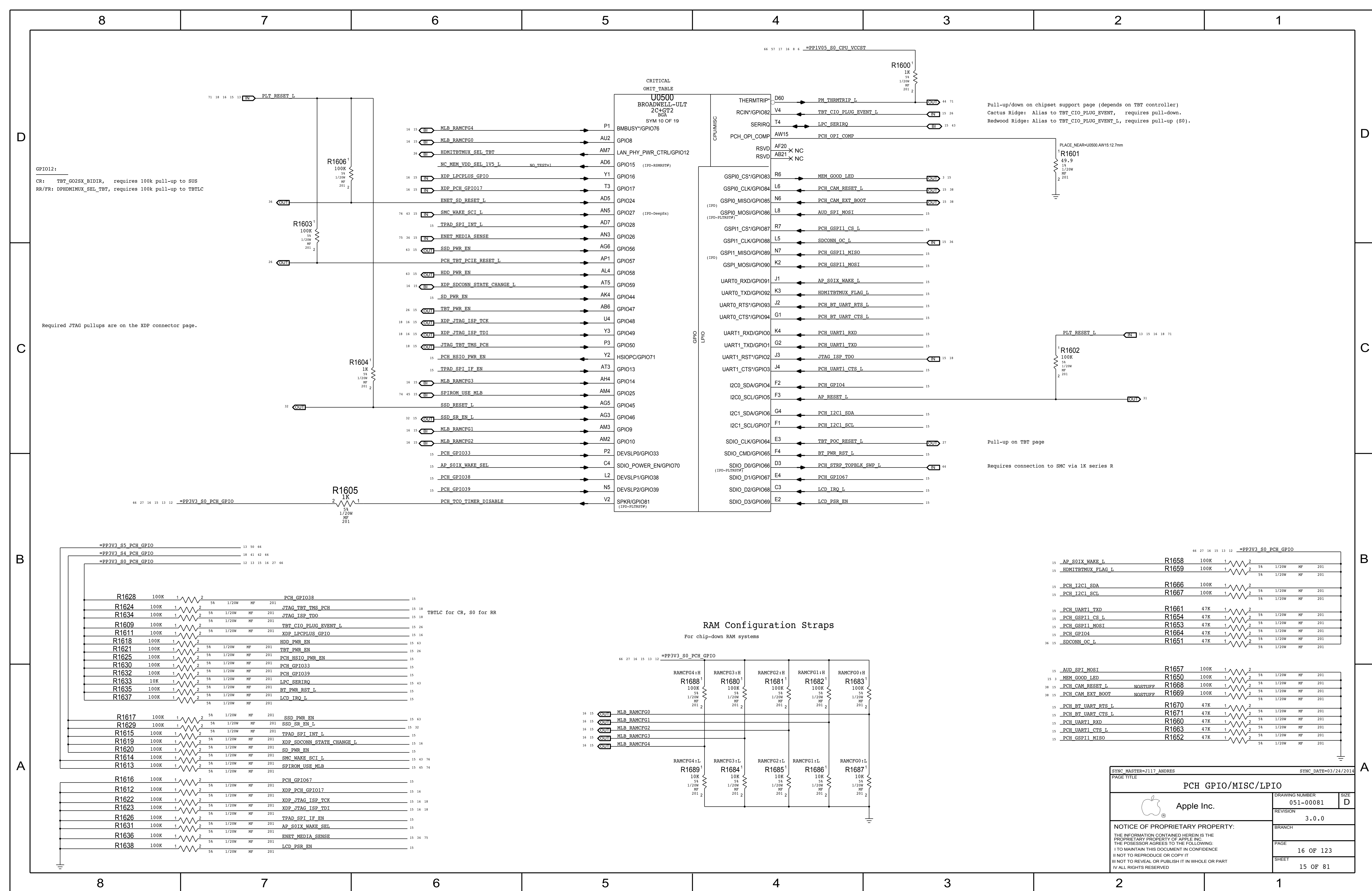
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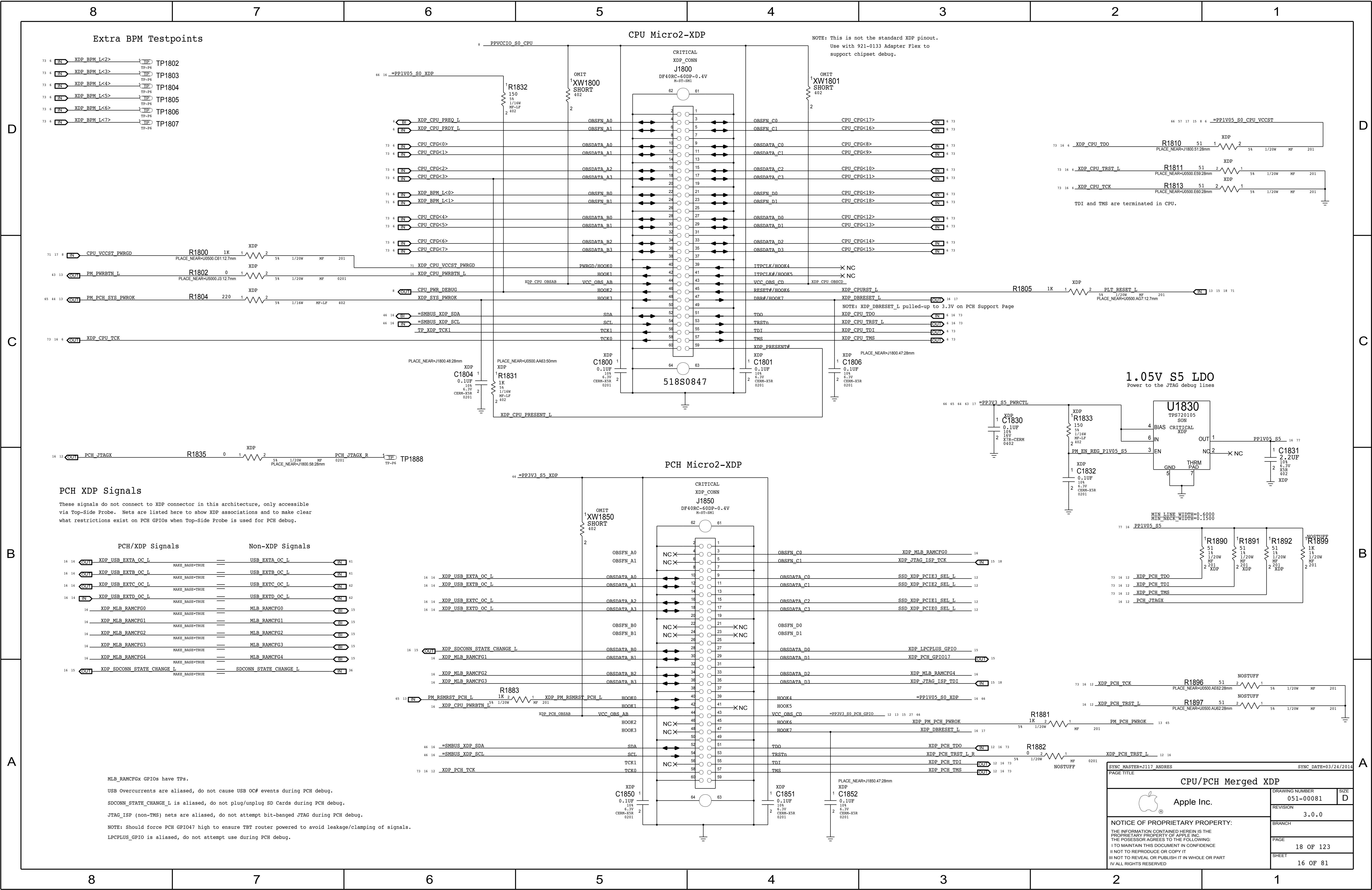
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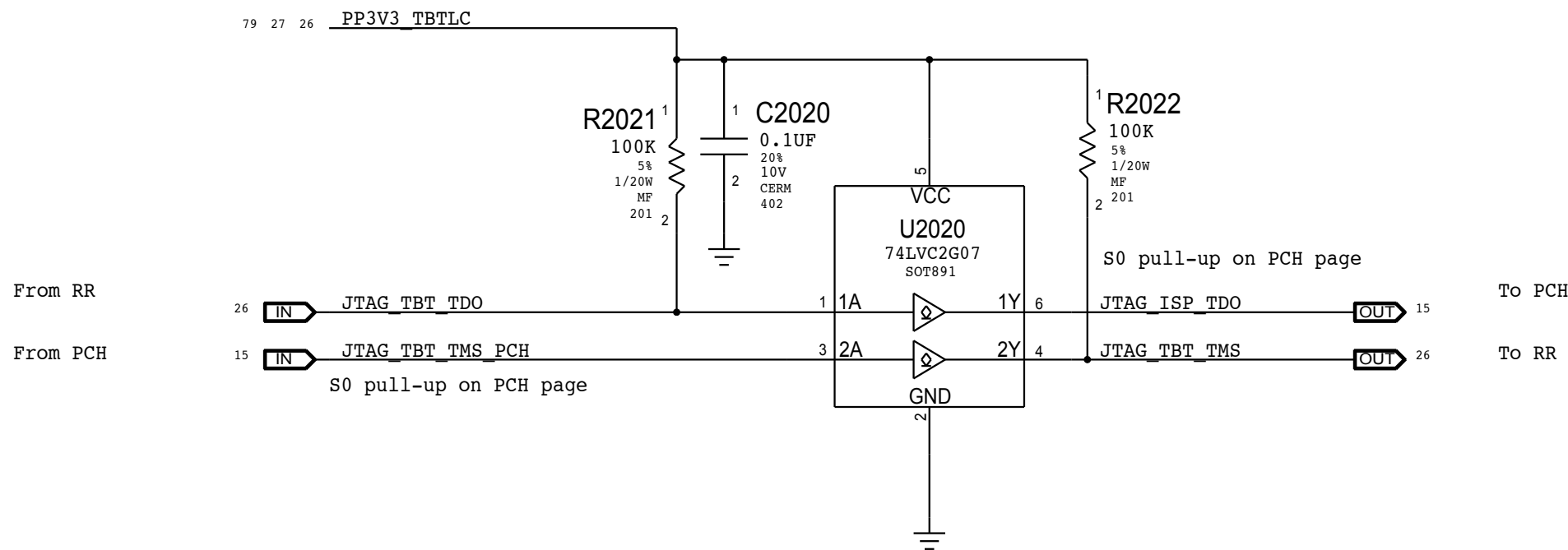






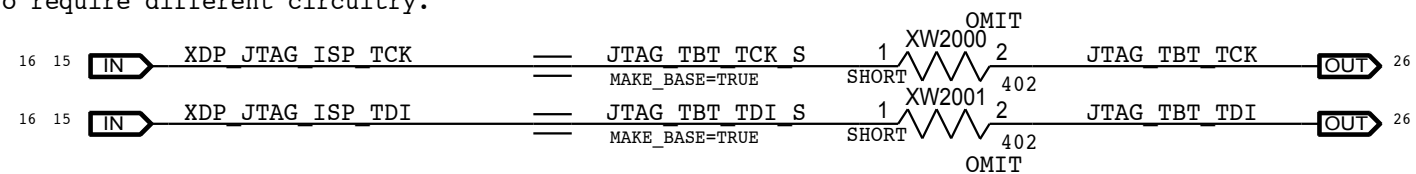
Falcon Ridge JTAG Isolation

TBTLC can be on when S0 is off, and vice-versa
Isolation ensures no leakage to RR or PCH



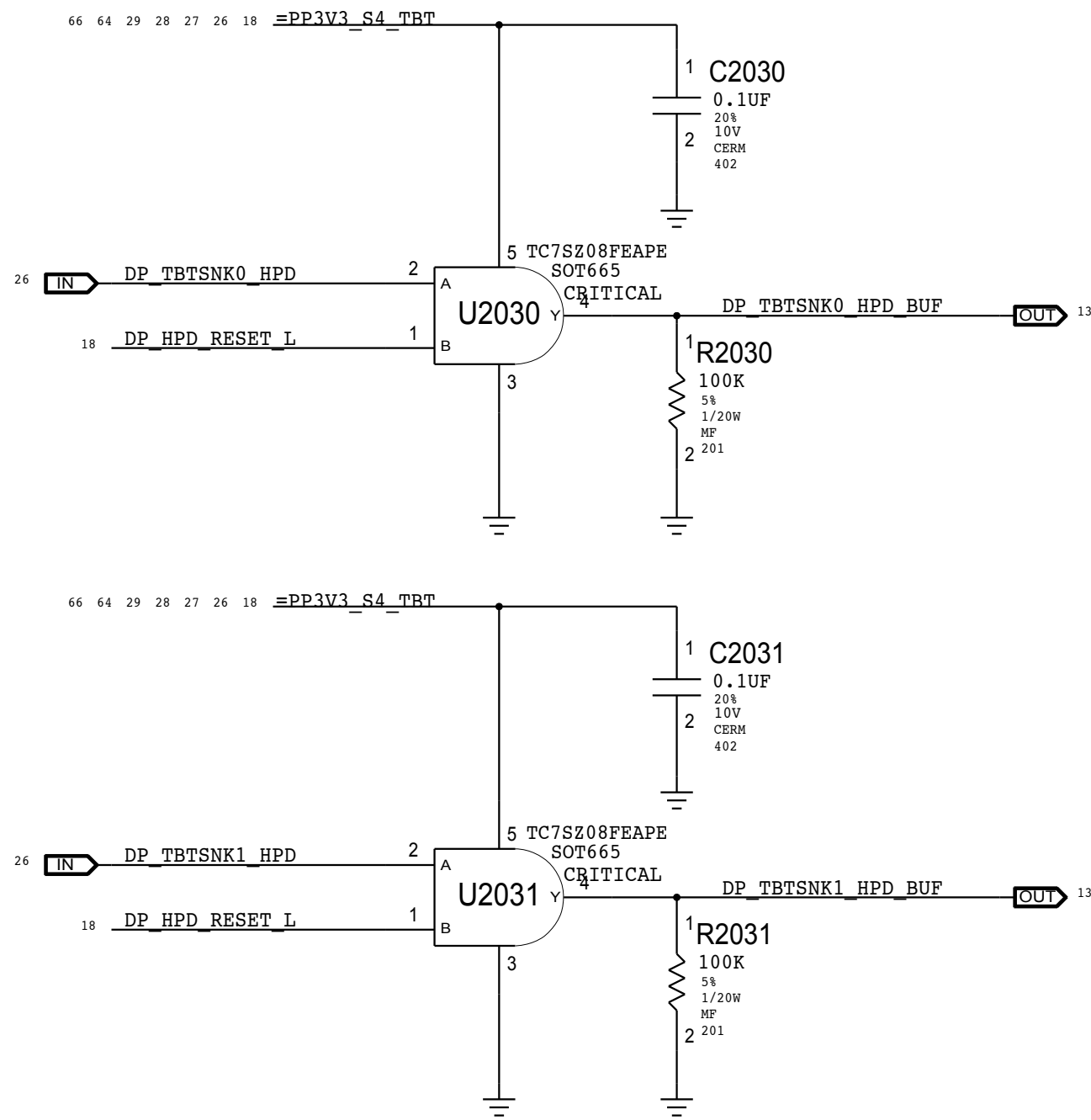
NOTE: Solution shown is for WPT-LP. Other PCH's may require isolation on TCK and TDI as well for PCH glitch-prevention.

NOTE: This reference schematic assumes PCH JTAG GPIOs are only used for Thunderbolt. If other ASIC JTAG signals are wired into these GPIOs different isolation techniques will likely be necessary. Multi-router designs also require different circuitry.

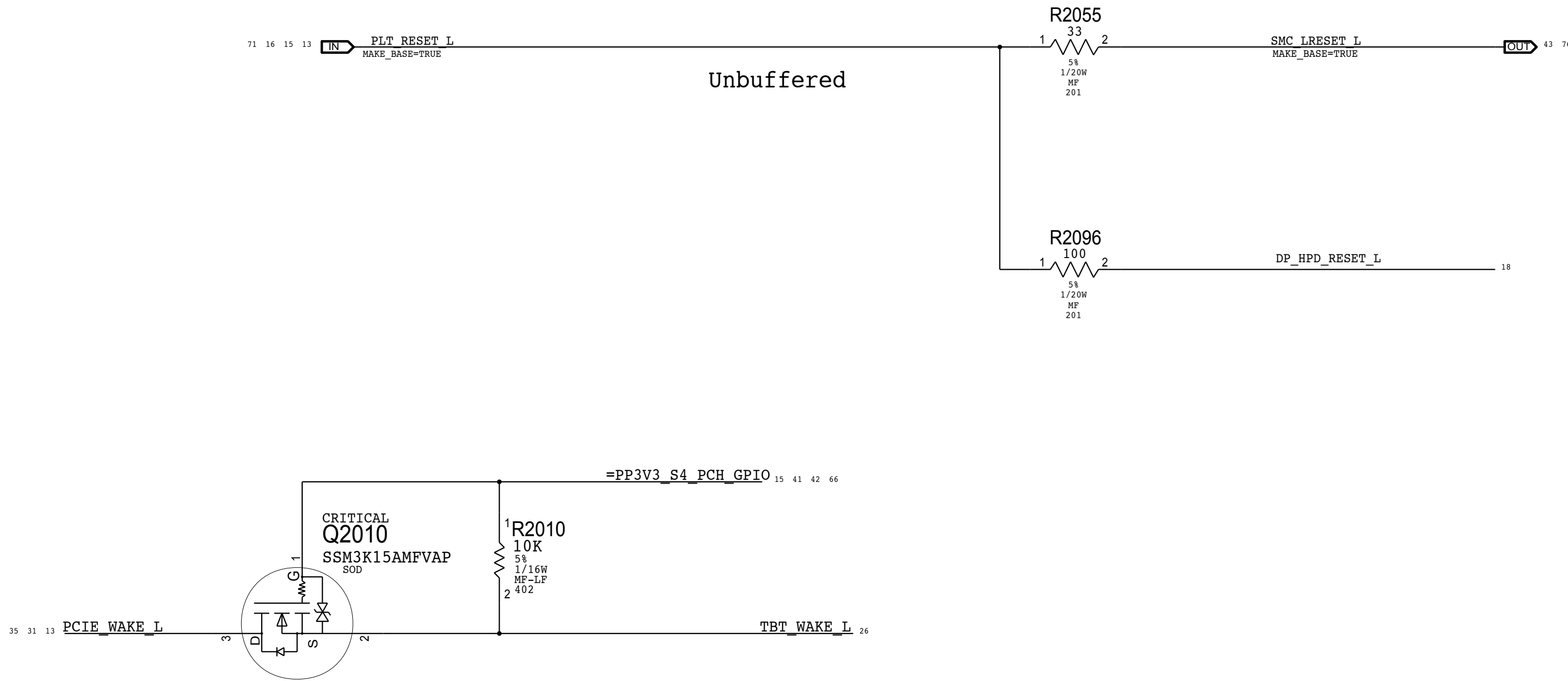


Falcon Ridge HPD Isolation

NOTE: PLT_RESET_L used as the other input to the AND gate so that HPD is only driven high to the PCH in S0.

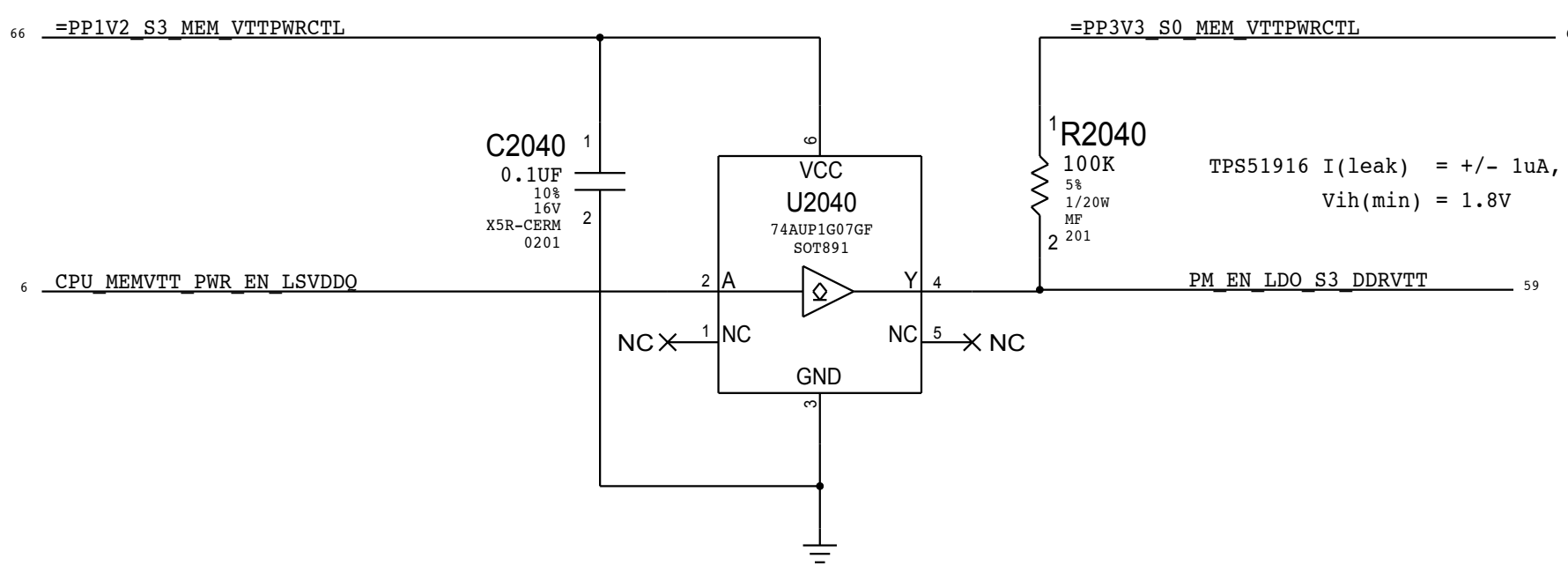


Platform Reset Connections



Memory VTT Enable Level-Shifter

CPU output is on VDDQ rail (1.2V), TP851916 has 1.8V Vih(min).



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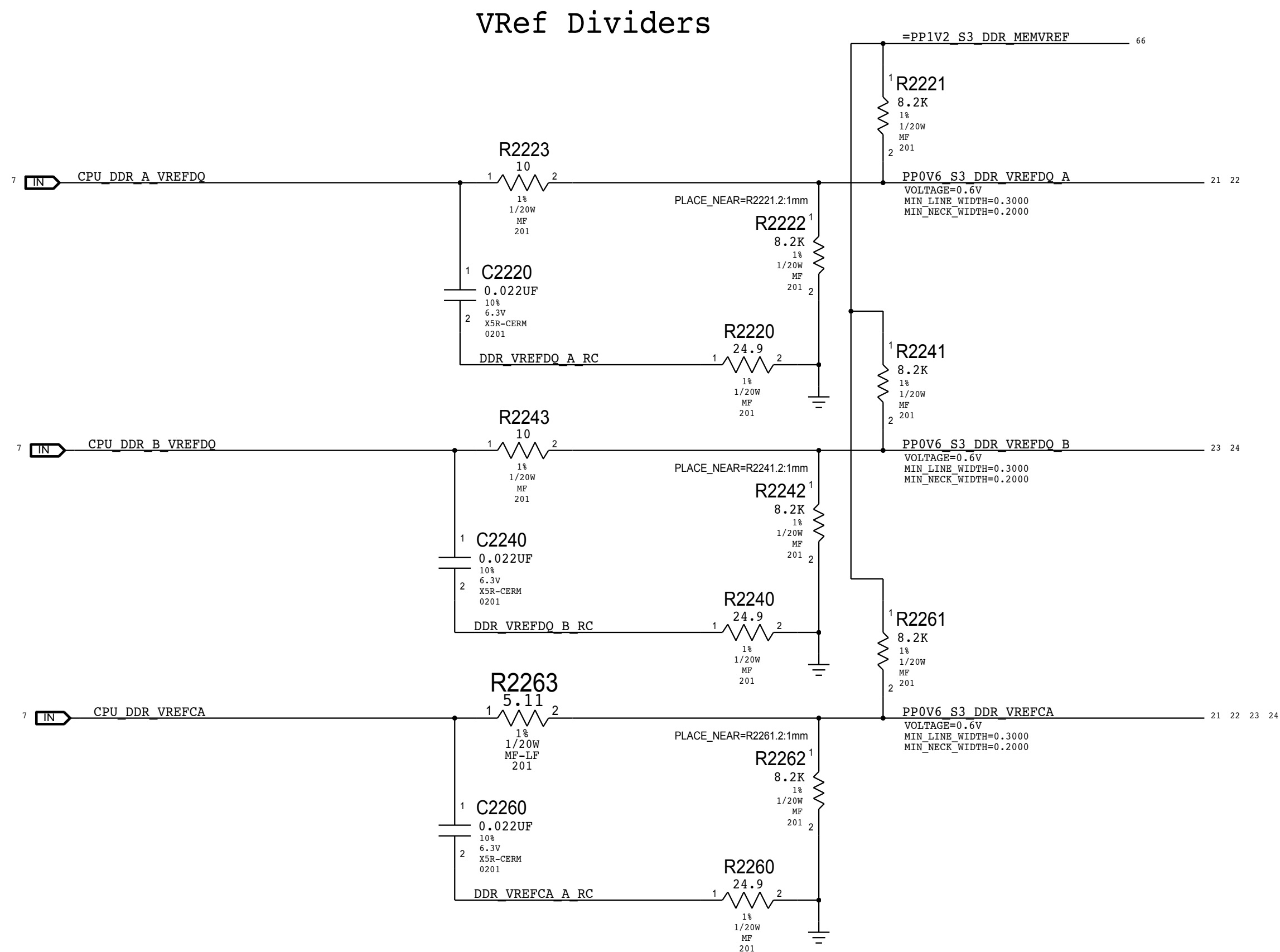
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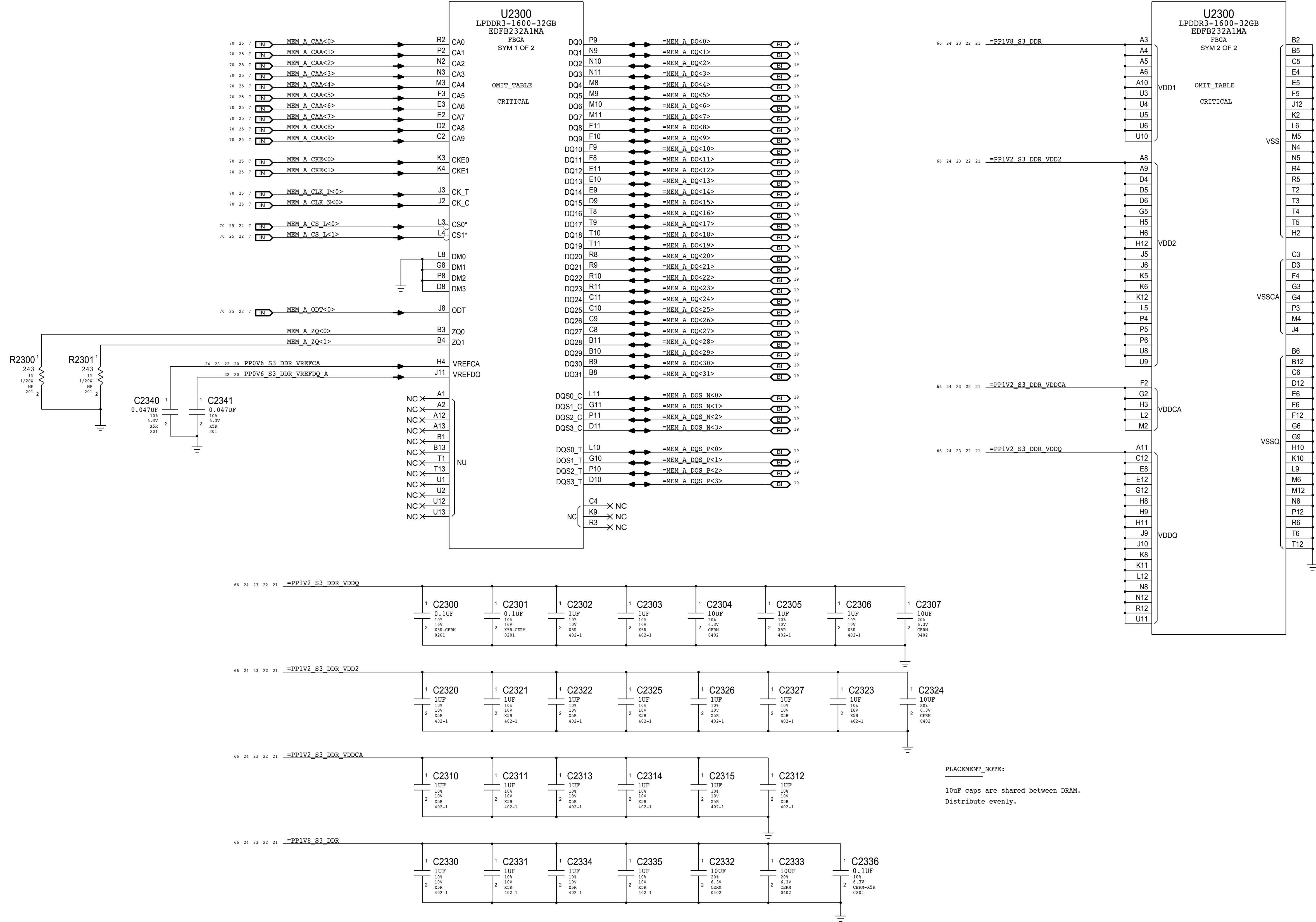
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	BRANCH		
	PAGE		22 OF 123
	SHEET		20 OF 81

LPDDR3 CHANNEL A (0-31)



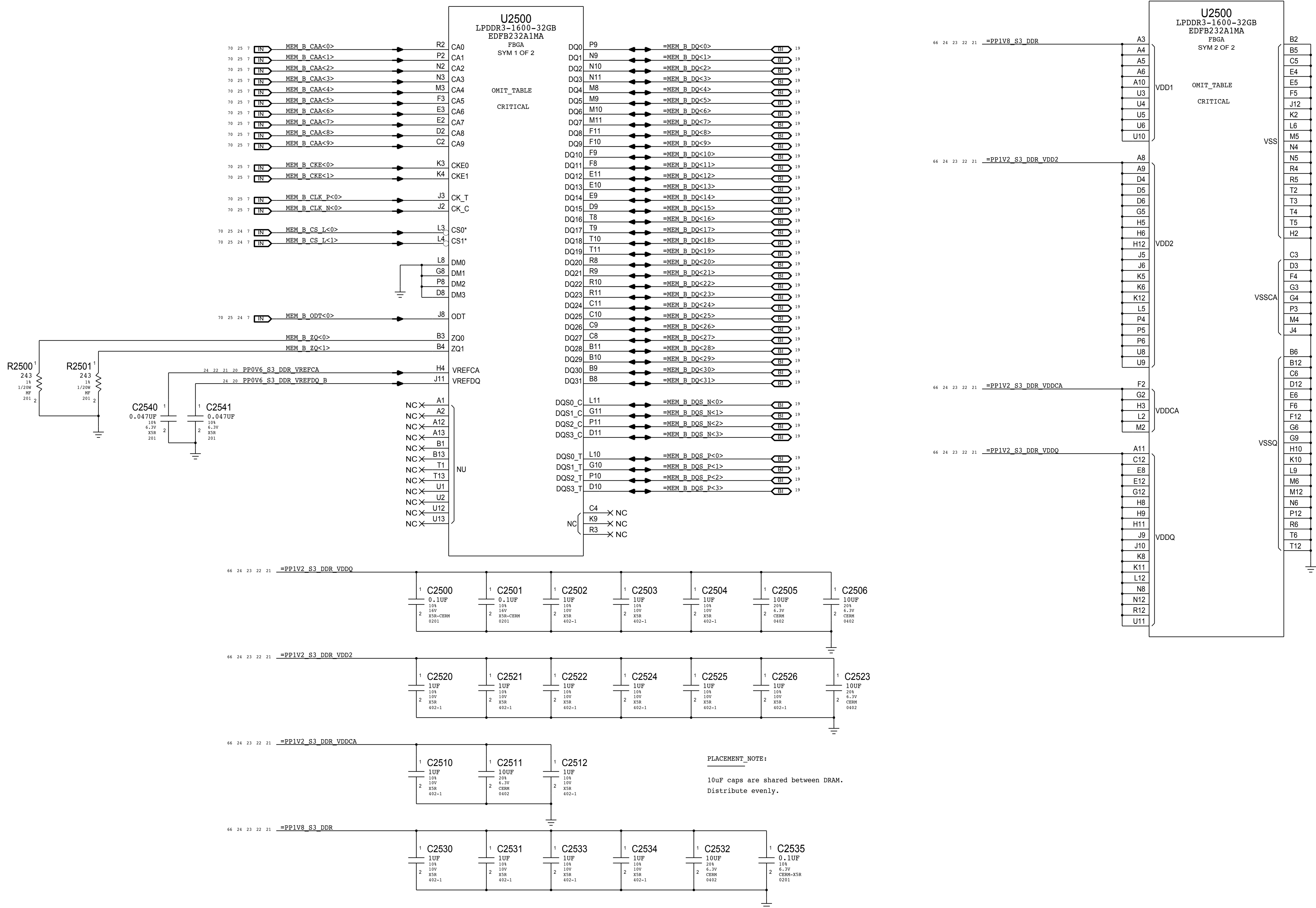
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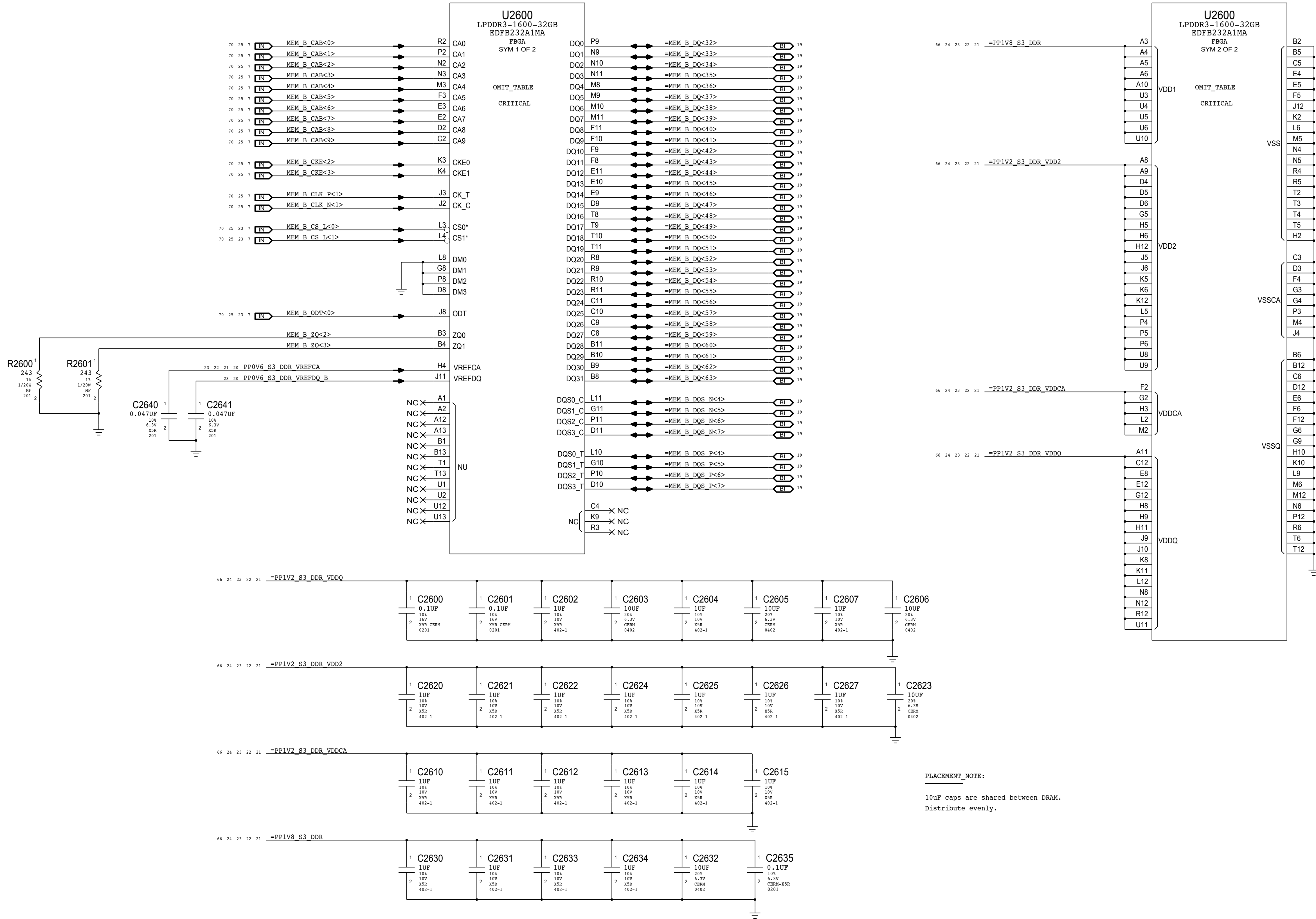
10uF caps are shared between DRAM.
Distribute evenly.

A

LPDDR3 CHANNEL B (0-31)



LPDDR3 CHANNEL B (32-63)



PLACEMENT NOTE:

10uF caps are shared between DRAM.
Distribute evenly.

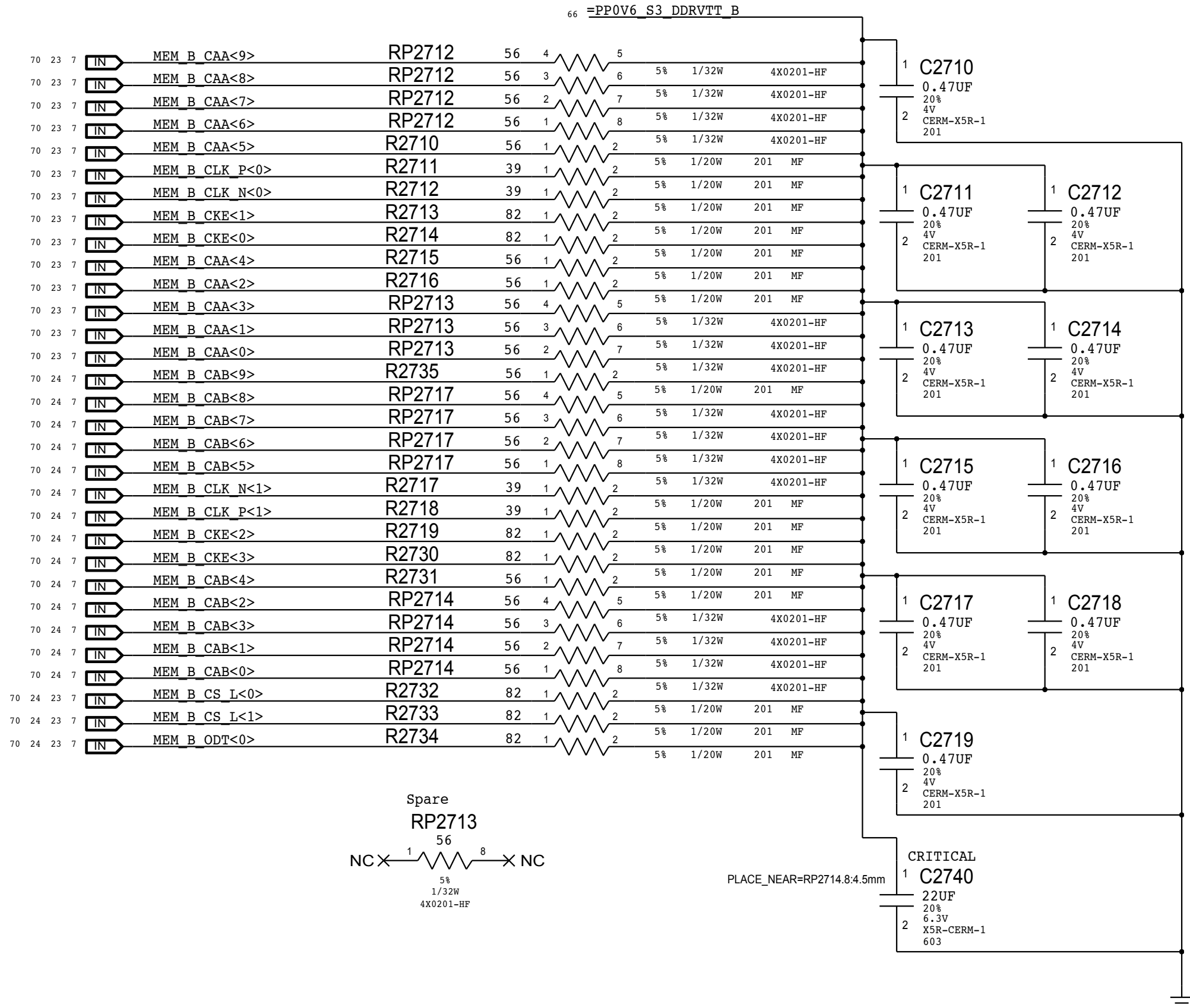
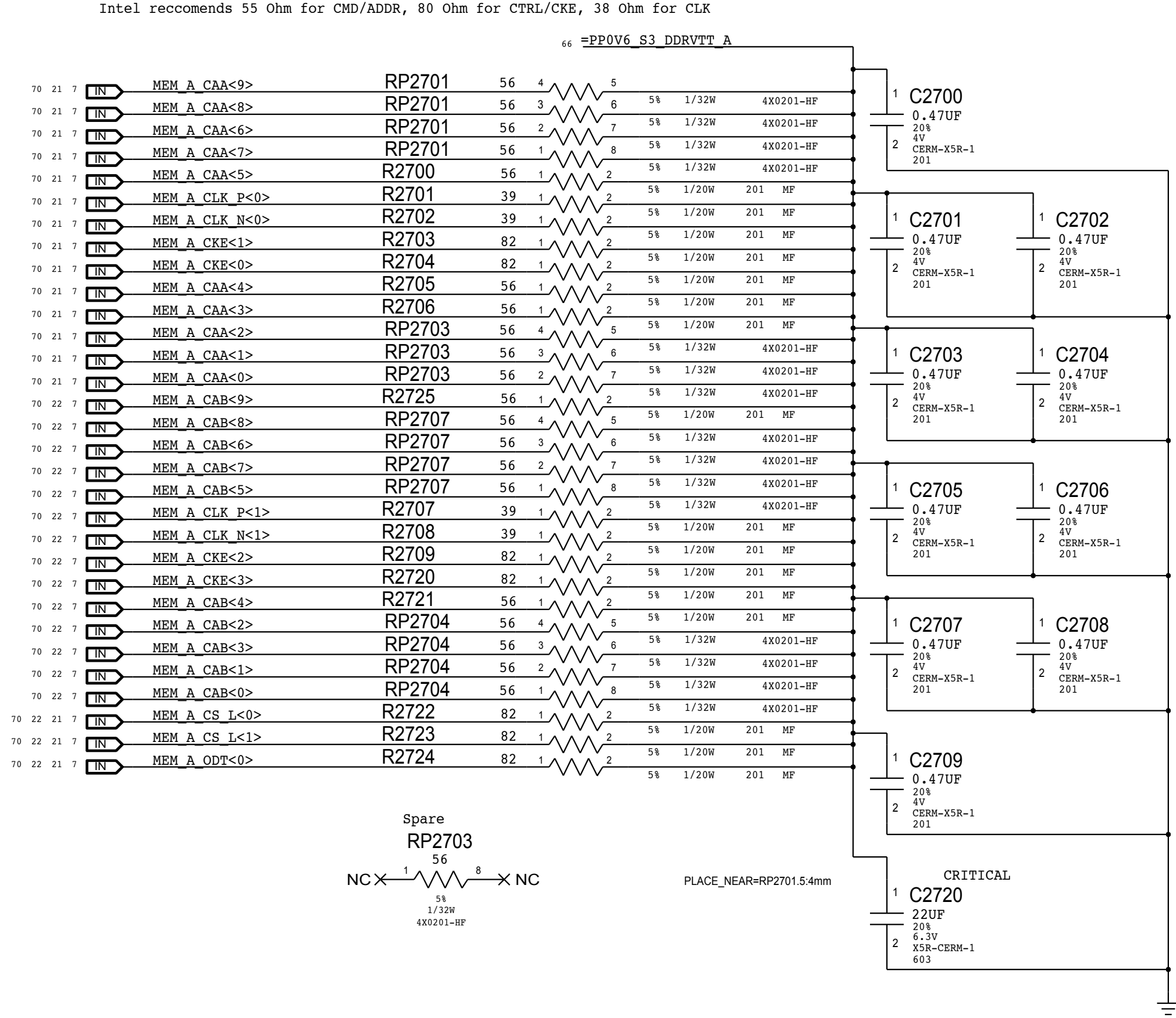
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LPDDR3 DRAM Channel B (32-63)			
 Apple Inc.	DRAWING NUMBER		SIZE
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		PAGE	26 OF 123
		SHEET	24 OF 81

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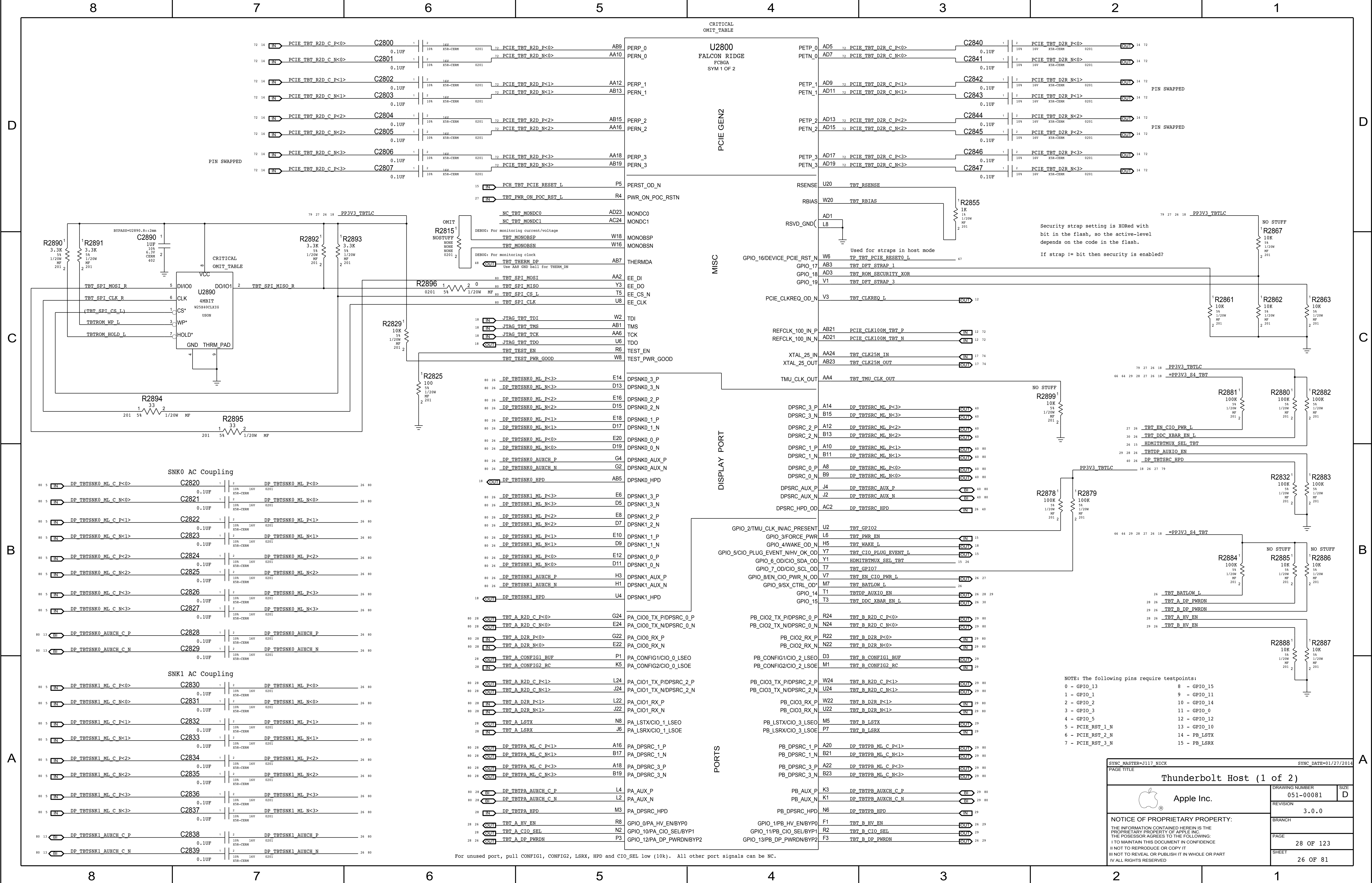


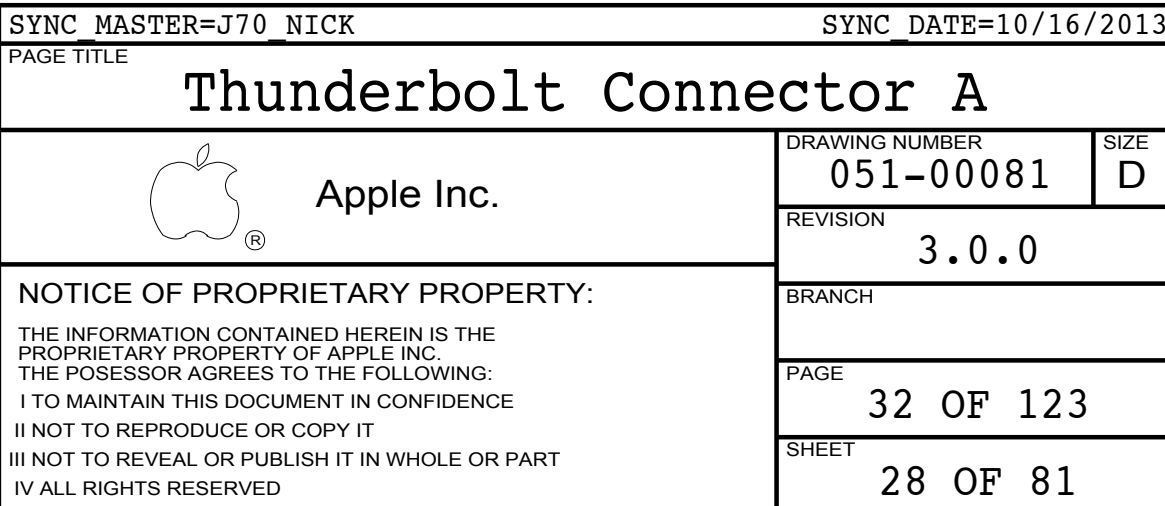
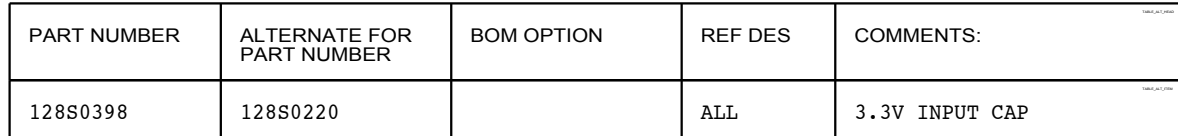
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3.3V/HV Power MUX

V3P3 must be S4 to support wake from Thunderbolt devices.

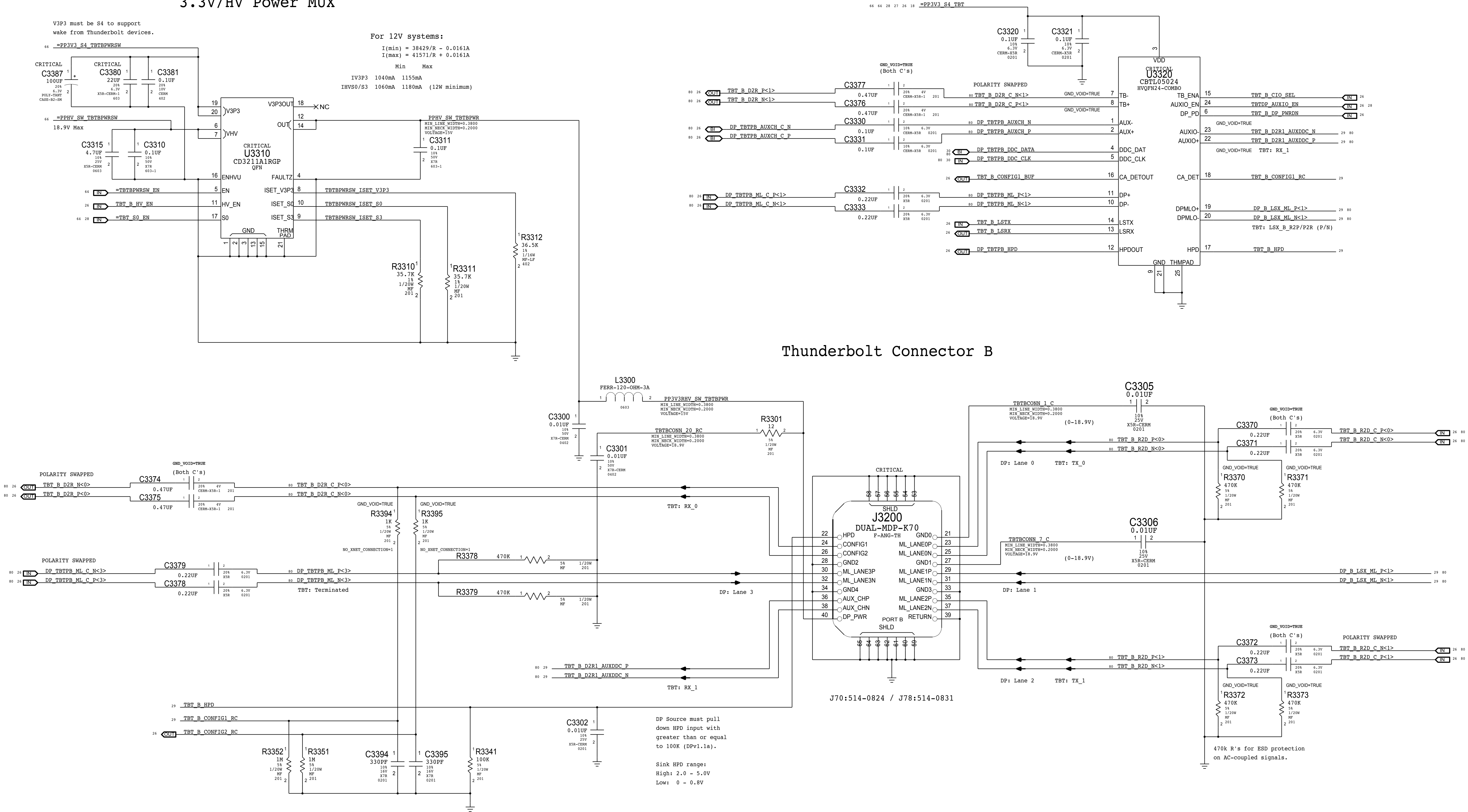
For 12V systems:

$I(\min) = 38429/R - 0.0161A$
 $I(\max) = 41571/R + 0.0161A$

Min Max

IV3P3 1040mA 1155mA

IHVS0/S3 1060mA 1180mA (12W minimum)



Thunderbolt Connector B

J70:514-0824 / J78:514-0831

SYNC_MASTER=J70_NICK SYNC_DATE=10/16/2013

PAGE TITLE

Thunderbolt Connector B



Apple Inc.

DRAWING NUMBER 051-00081

SIZE D

REVISION 3.0.0

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PAGE 33 OF 123

SHEET 29 OF 81

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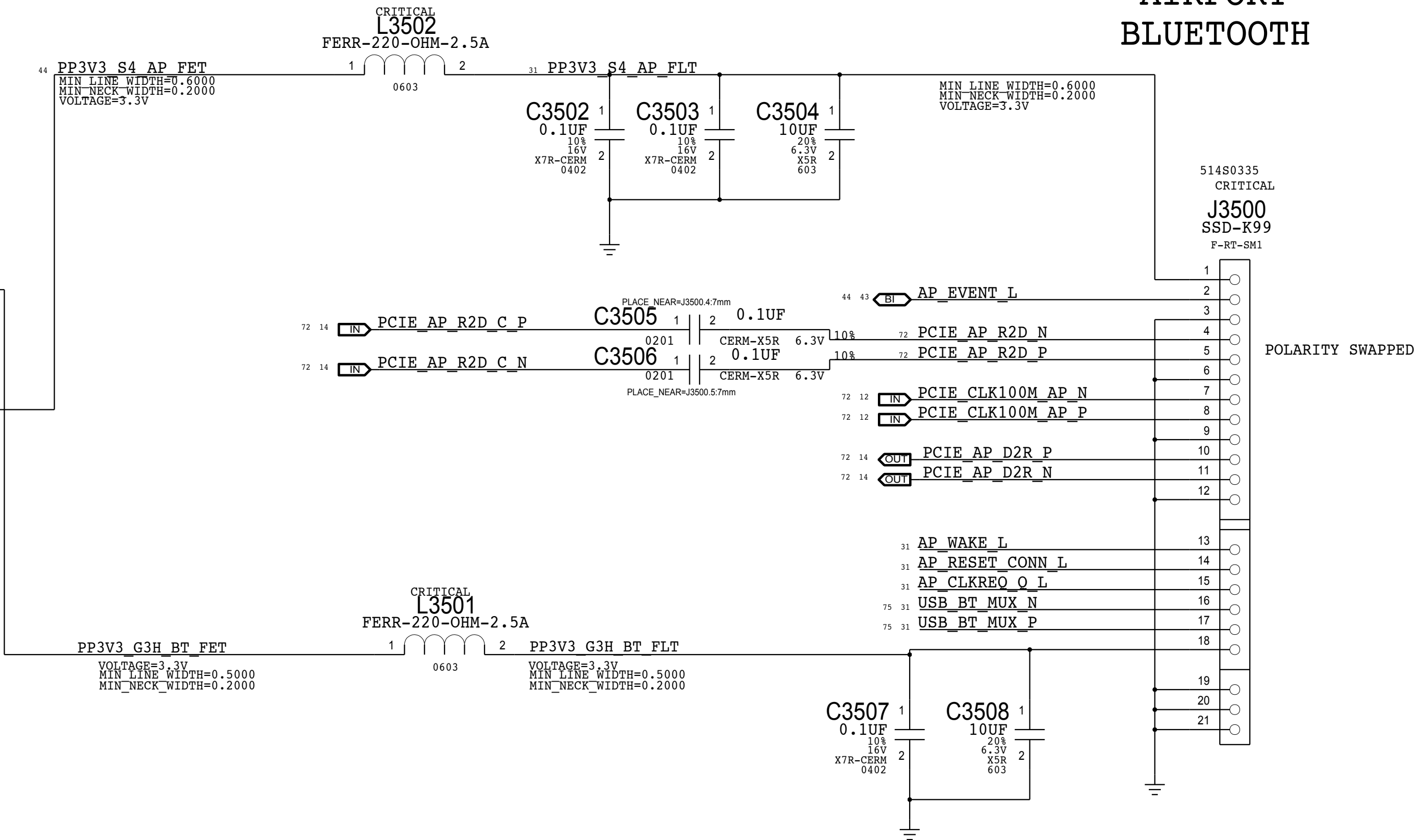
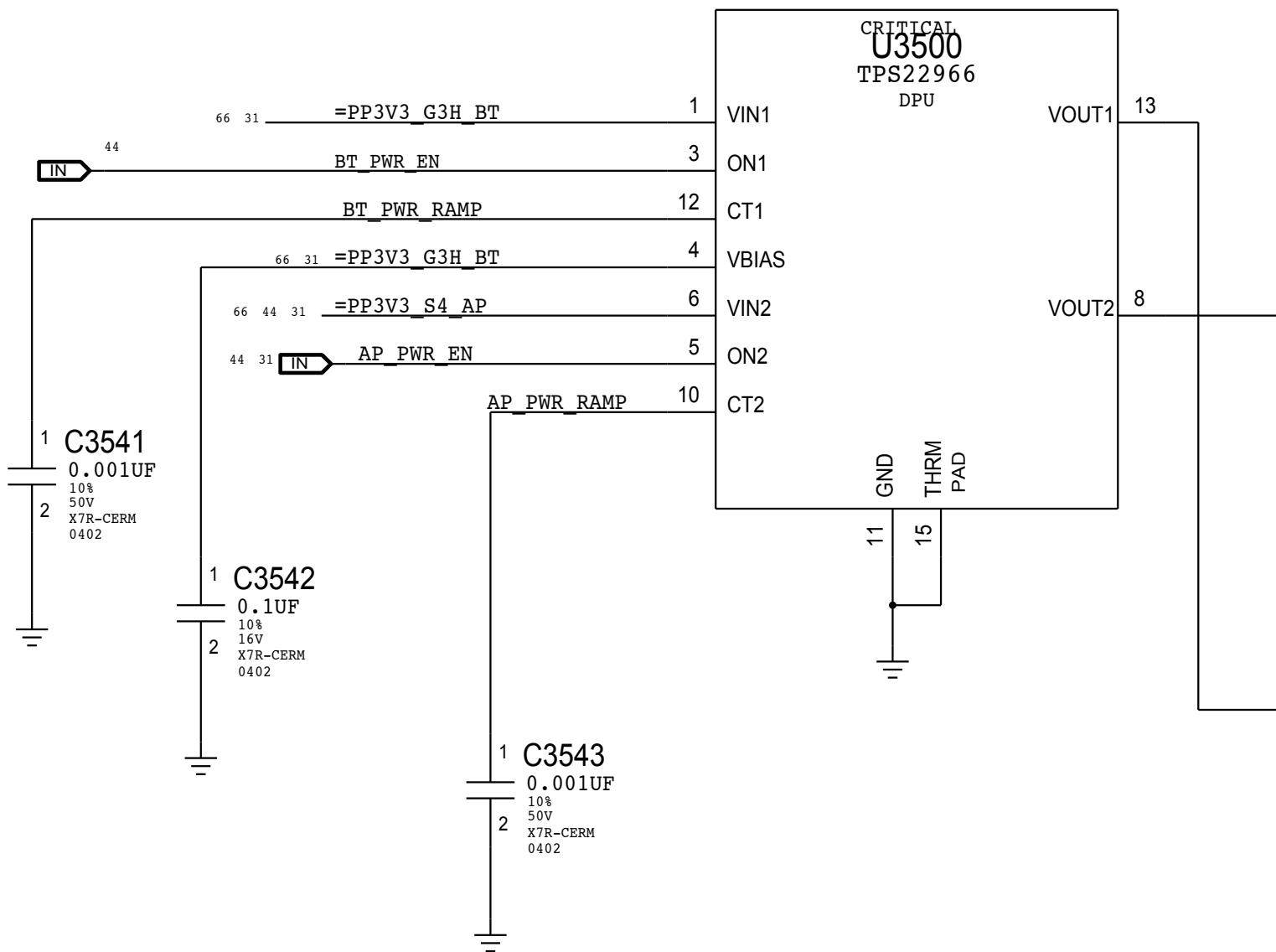
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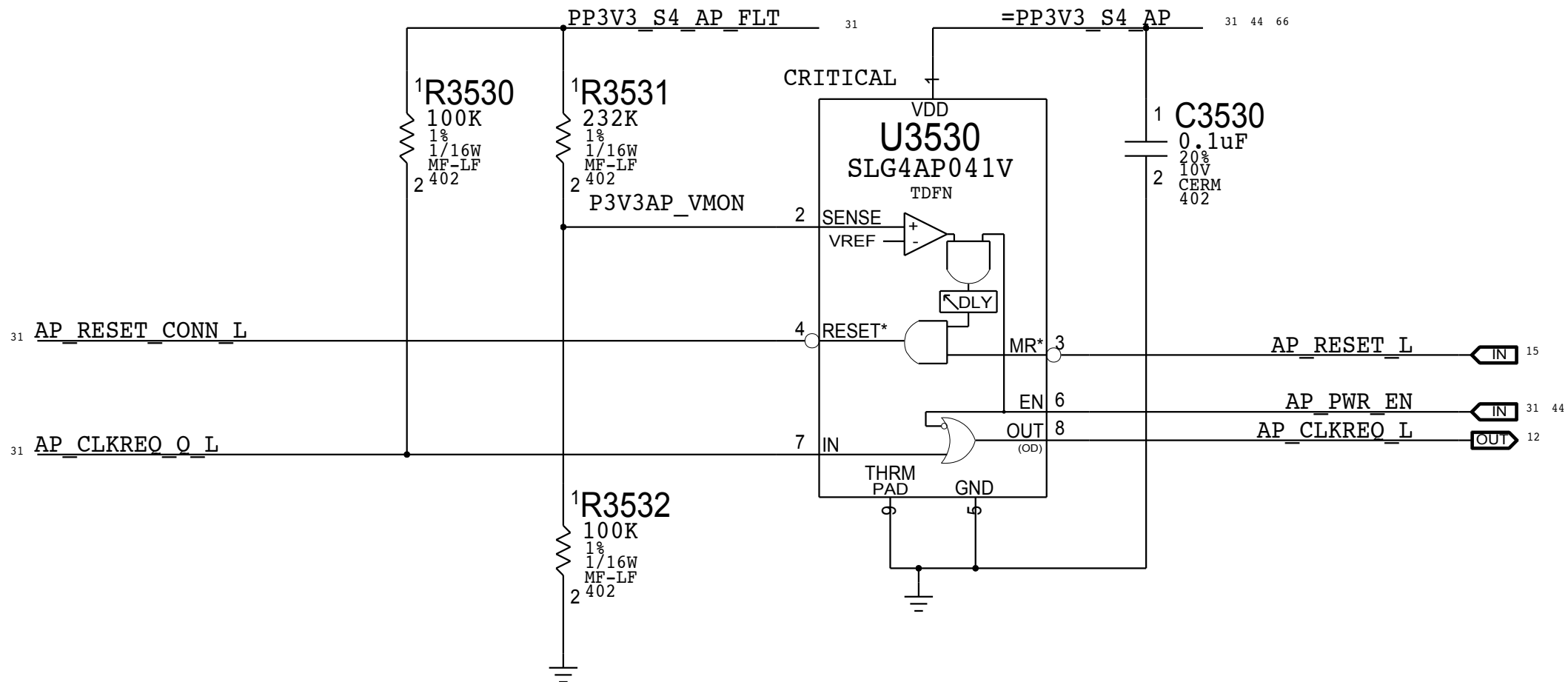
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AP & BT Load Switch

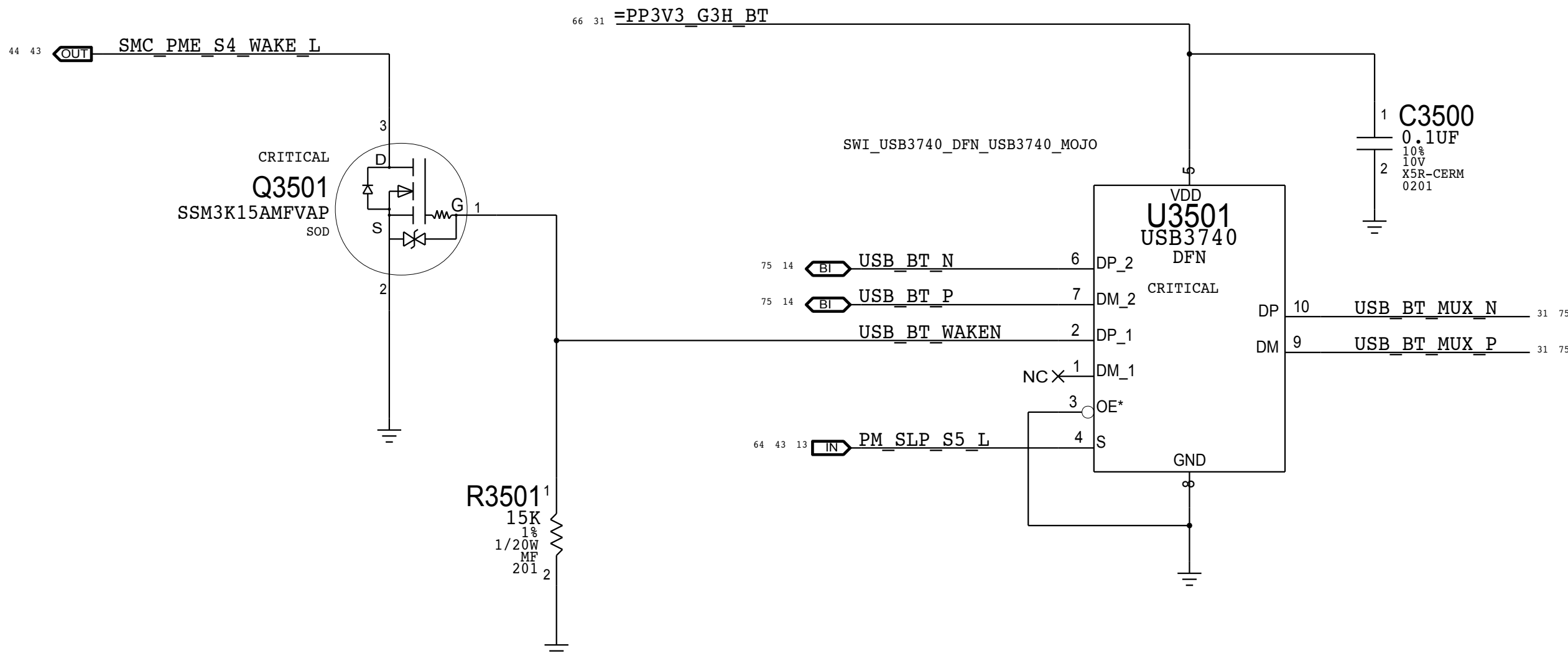
SWITCH	TPS22966
AP SLEW RATE	1185 us
BT SLEW RATE	1185 us
Equation	$0.32 * C_t + 13.7$



Supervisor & CLKREQ# Isolation
Delay = 130 ms +/- 20%



Wake from BT in G3H circuit



SYNC_MASTER=J70_DINI SYNC_DATE=10/08/2013

PAGE TITLE

AIRPORT/BT



Apple Inc.

DRAWING NUMBER
051-00081

REVISION
3.0.0

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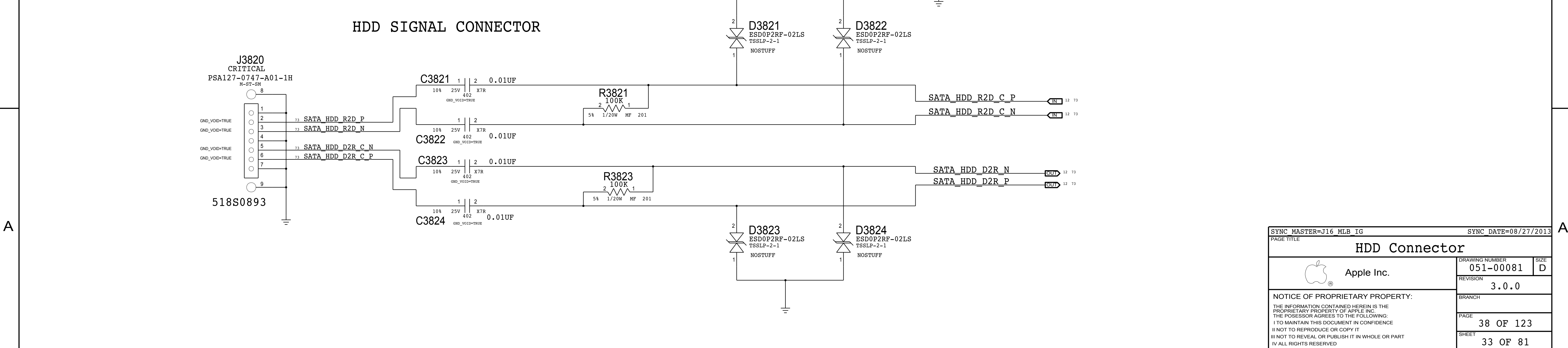
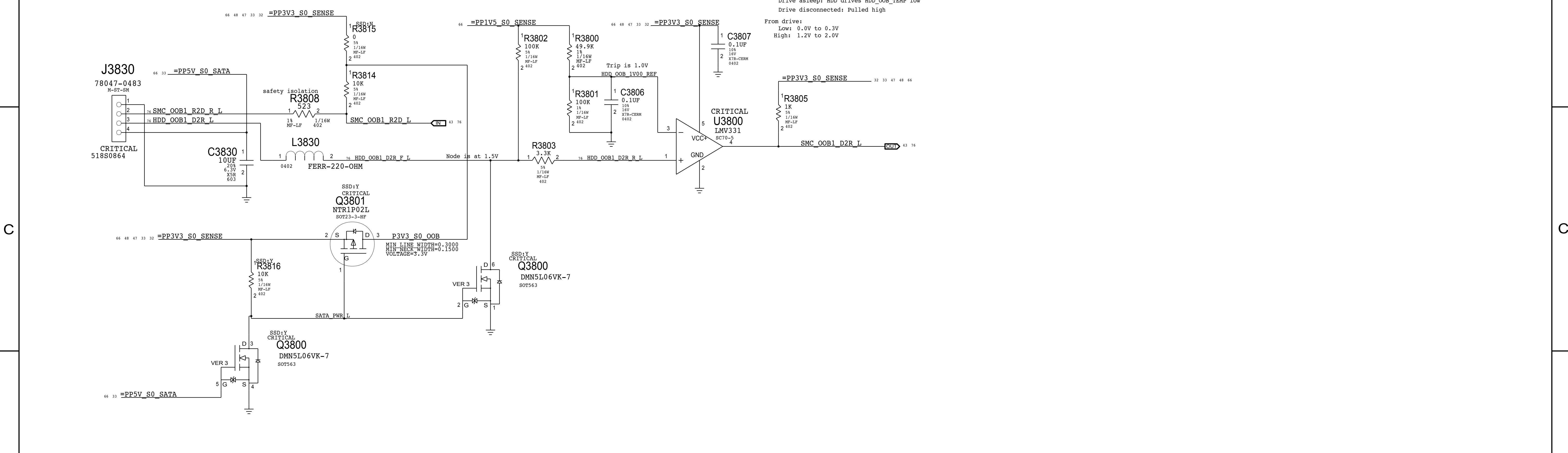
PAGE
35 OF 123

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31 OF 81

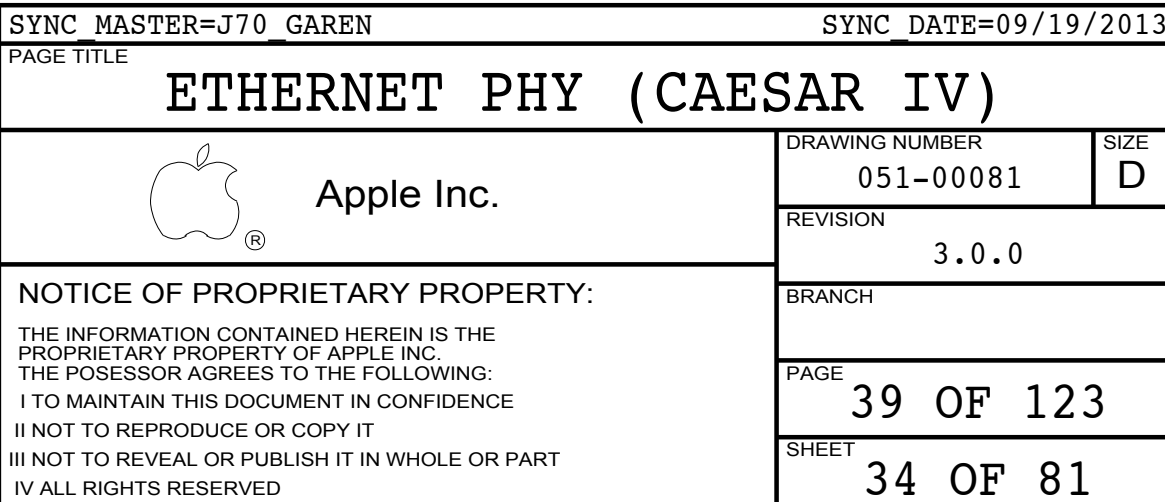
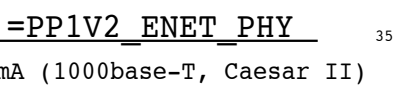
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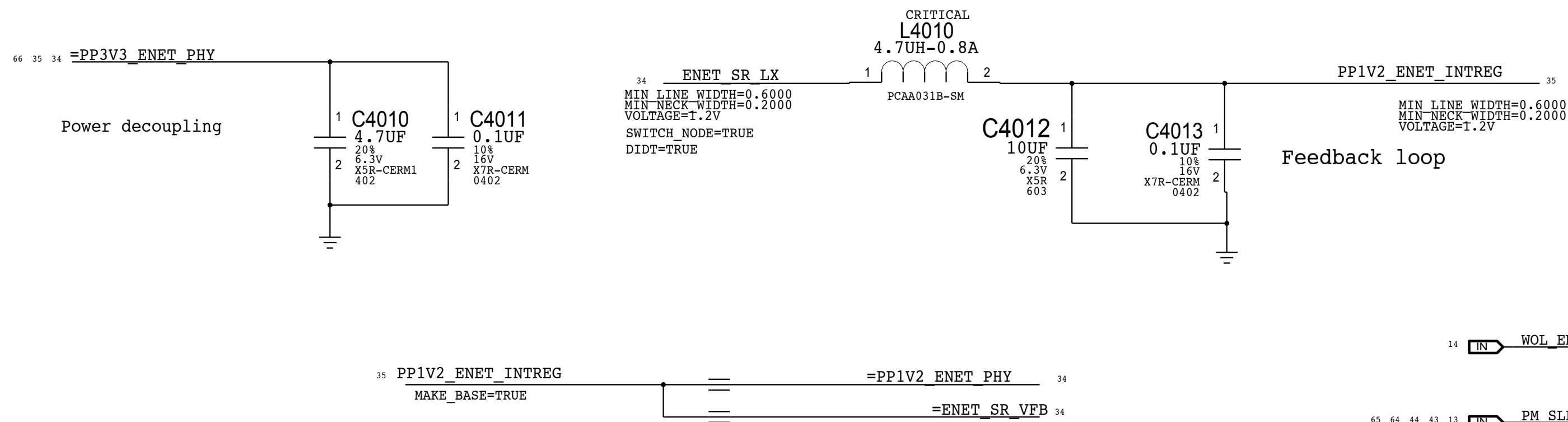
D	Notes: Drive active: Valid signal protocol	D
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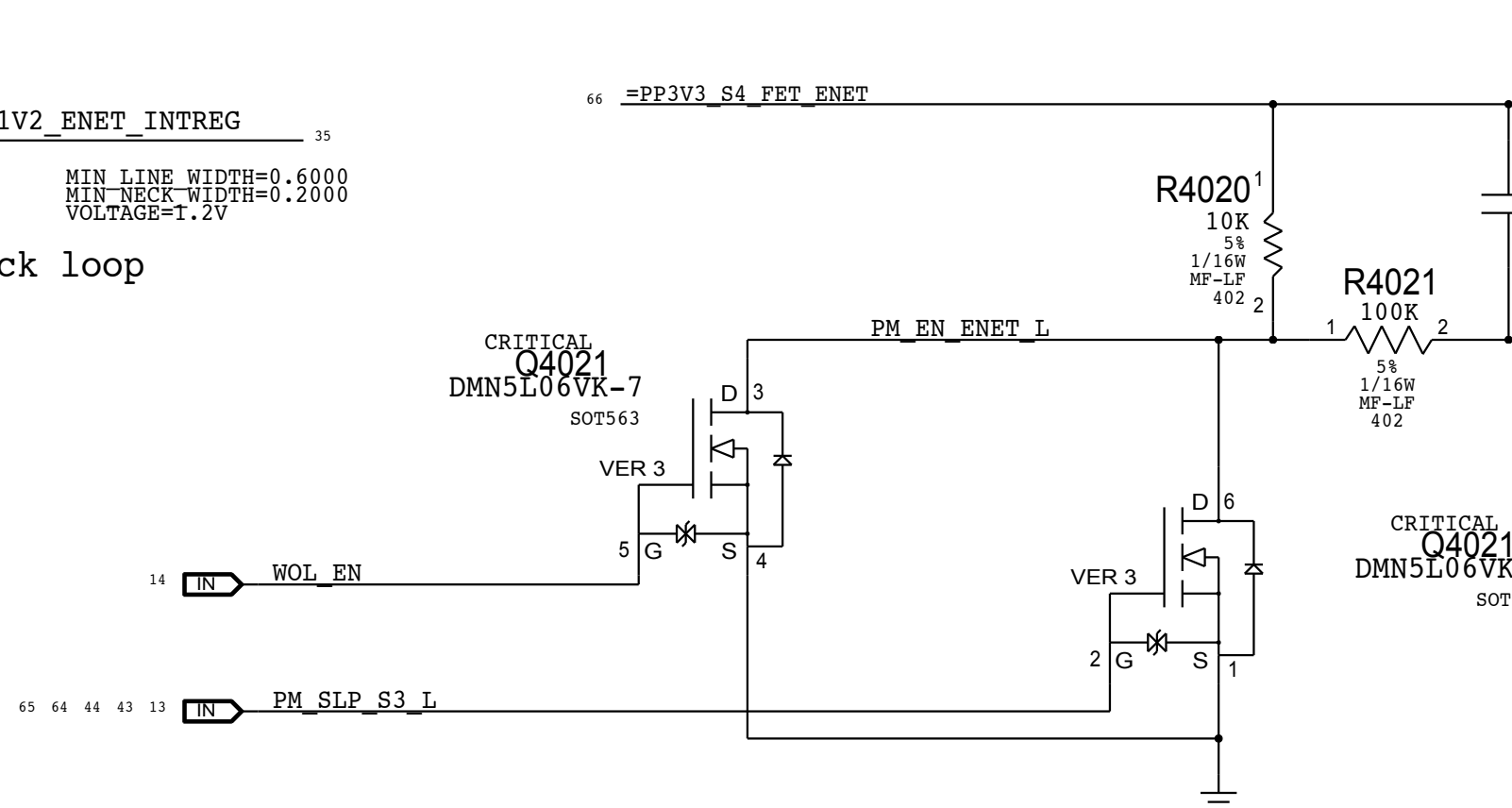
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CAESAR IV 1.2V INT.VR CMPTS

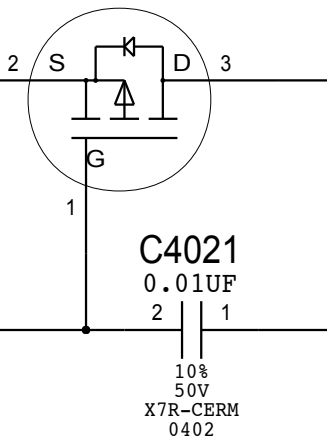


ENET Enable Generation

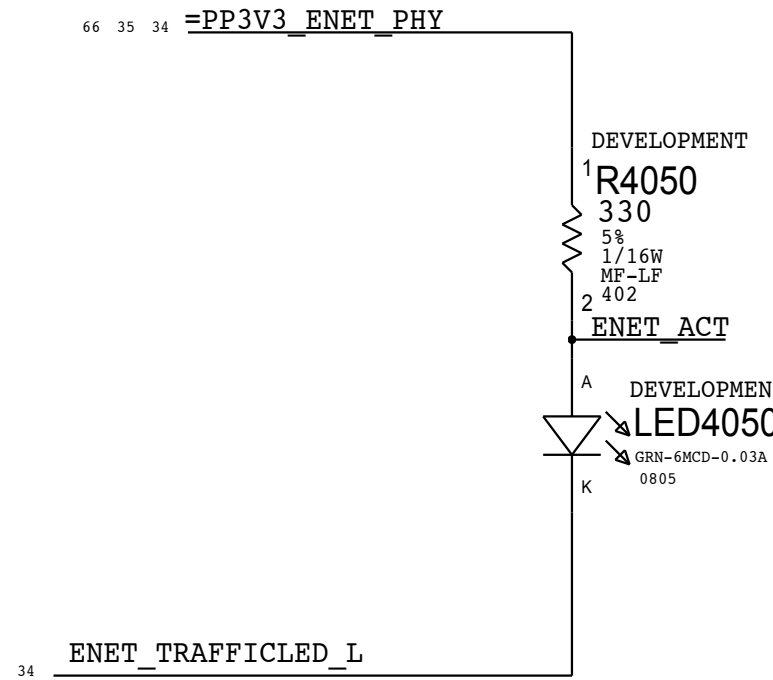


3.3V ENET FET

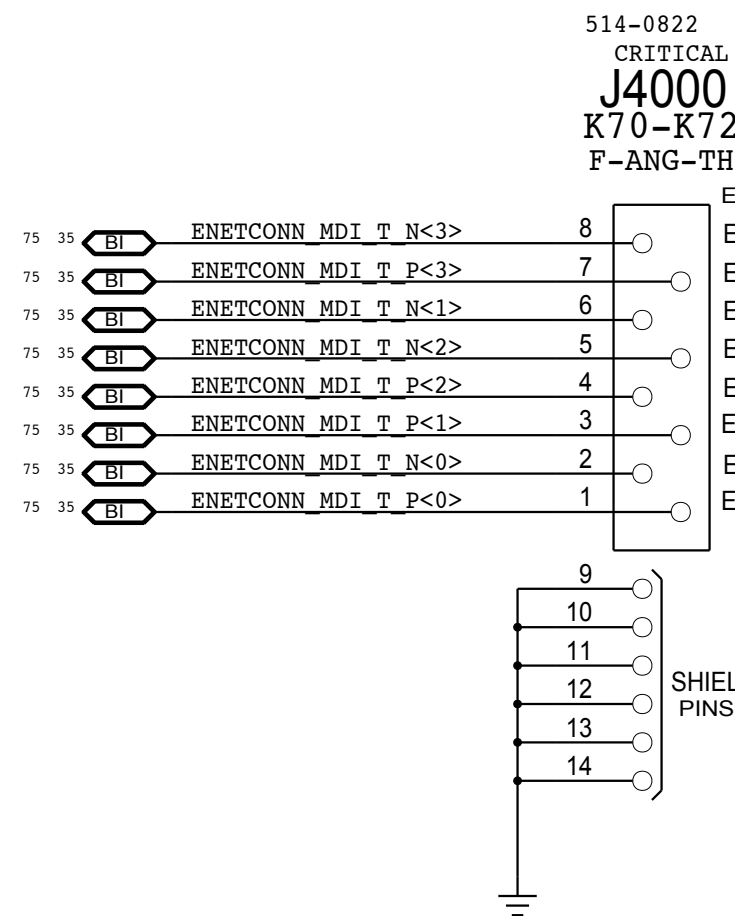
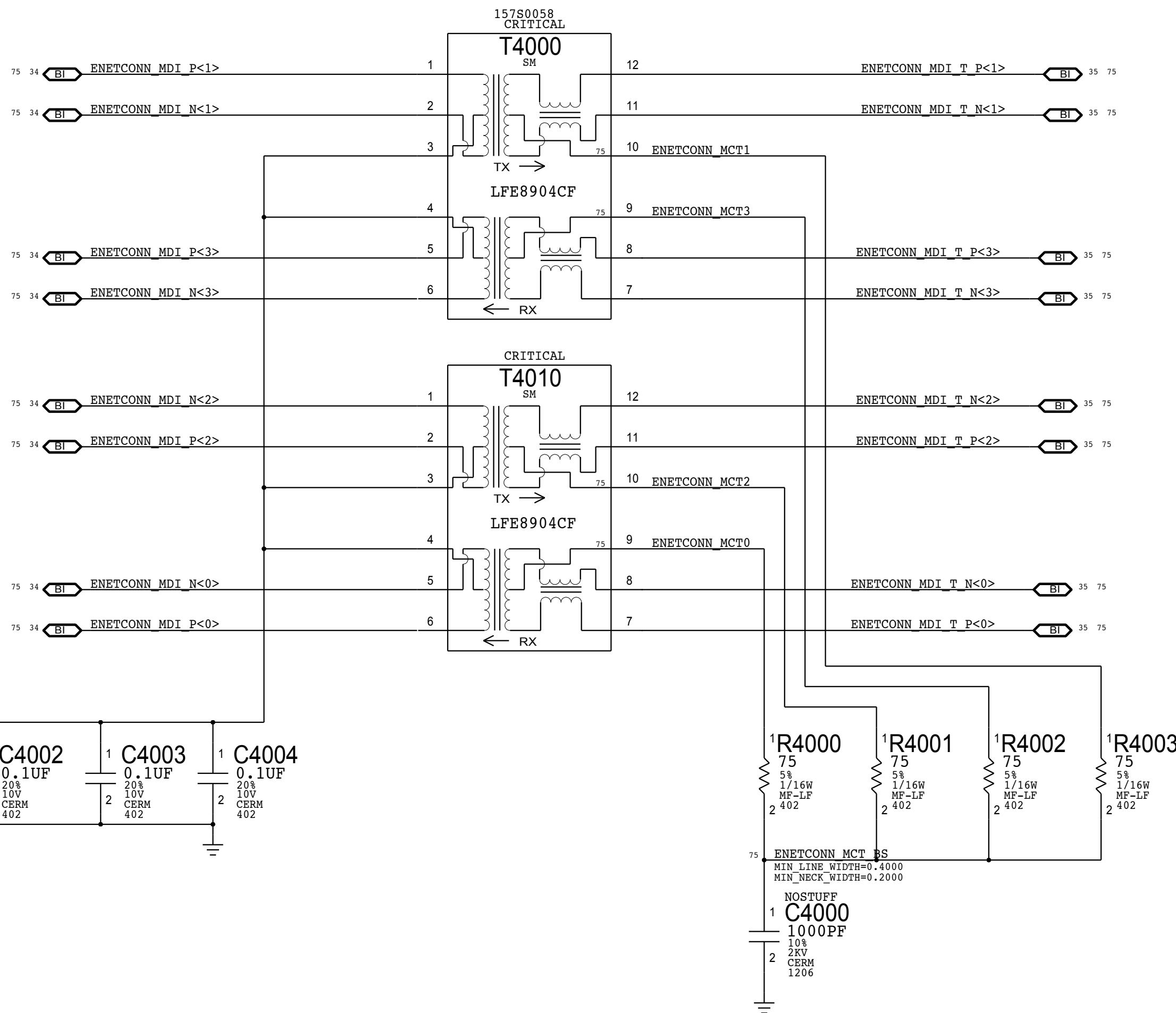
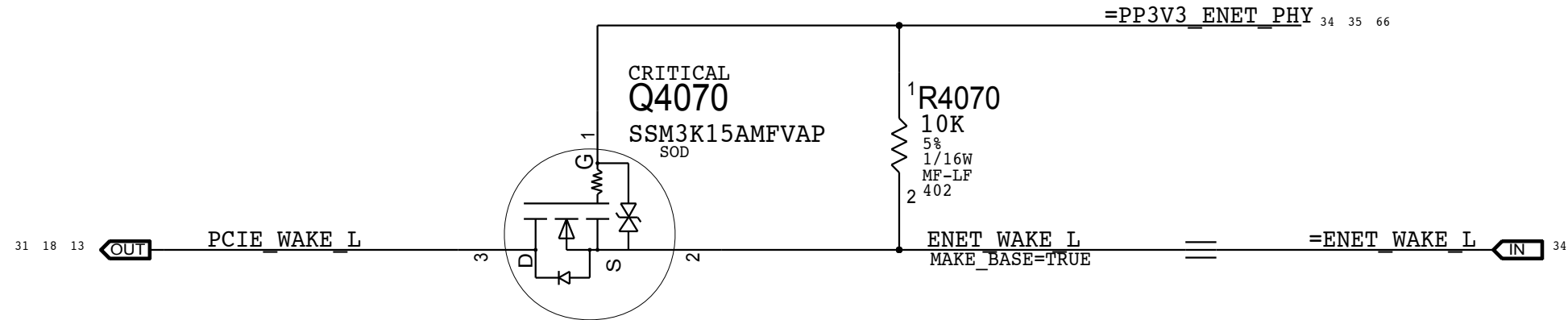
CRITICAL
Q4020
NTR4101P
SOT-23-HF



CAESAR IV ACTIVITY LED



CAESAR IV WAKE# ISOLATION



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Ethernet Support & Connector			
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		PAGE	40 OF 123
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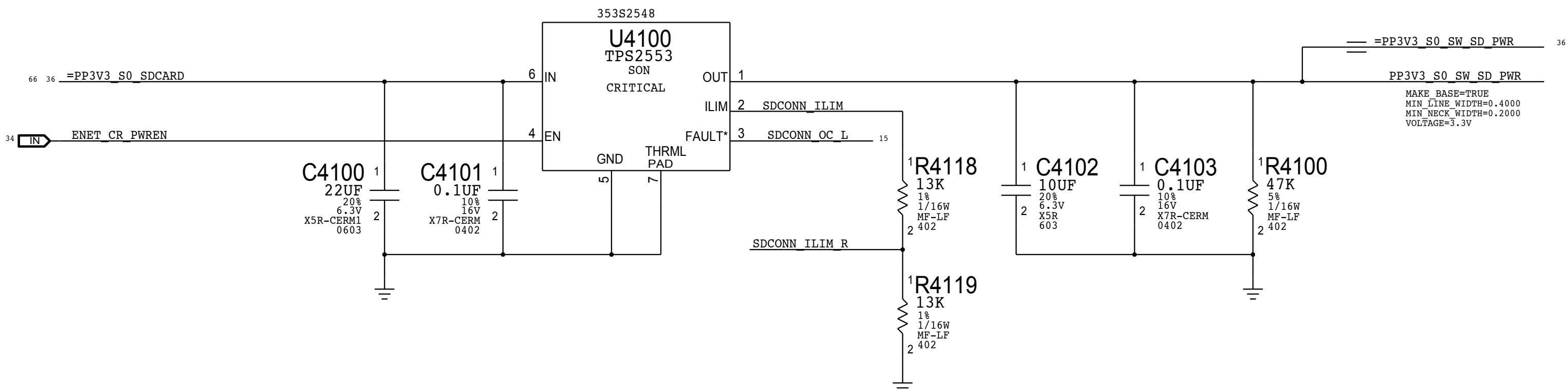
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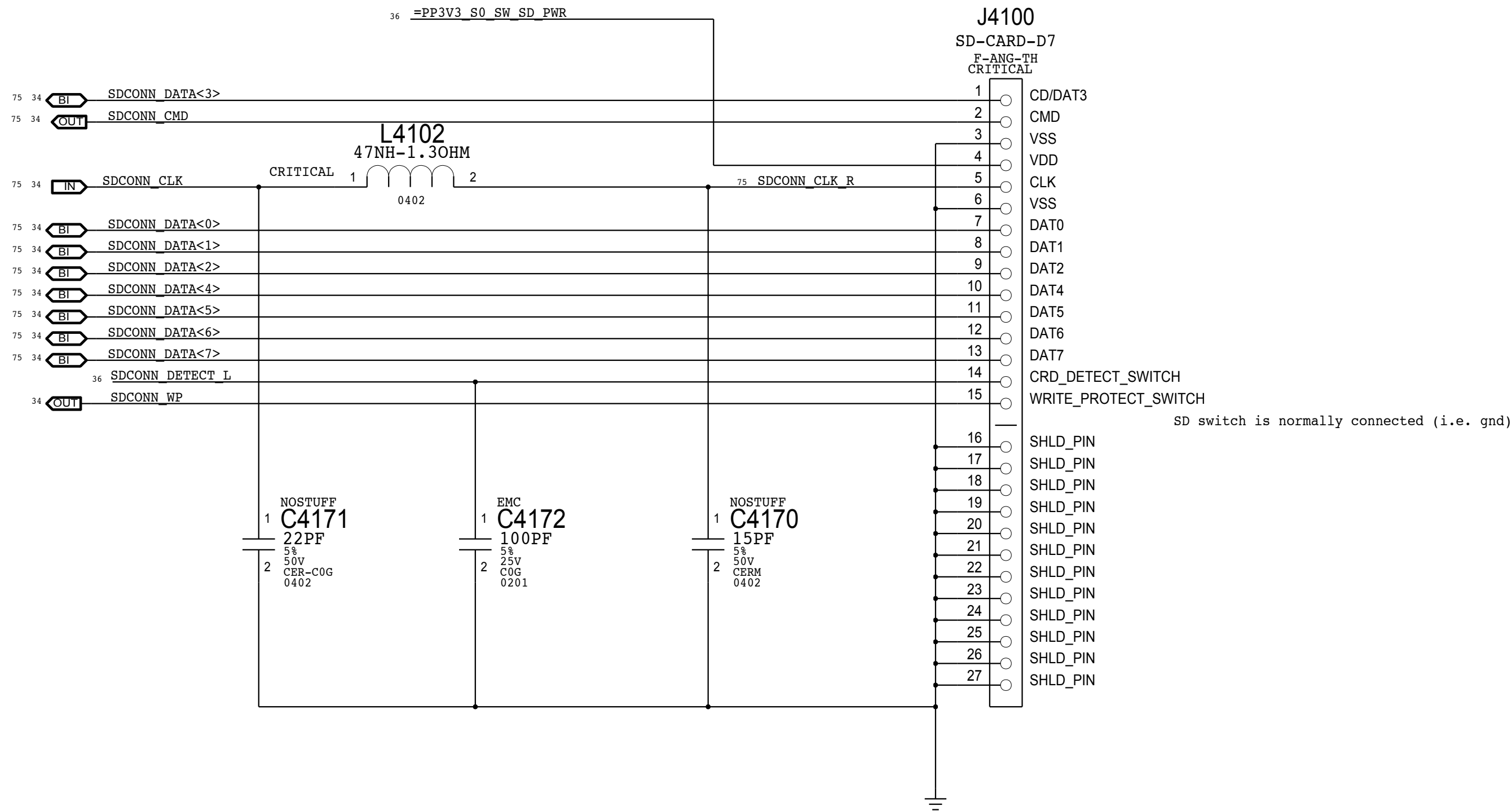
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SD CARD 3.3V OVERCURRENT PROTECTION CHIP

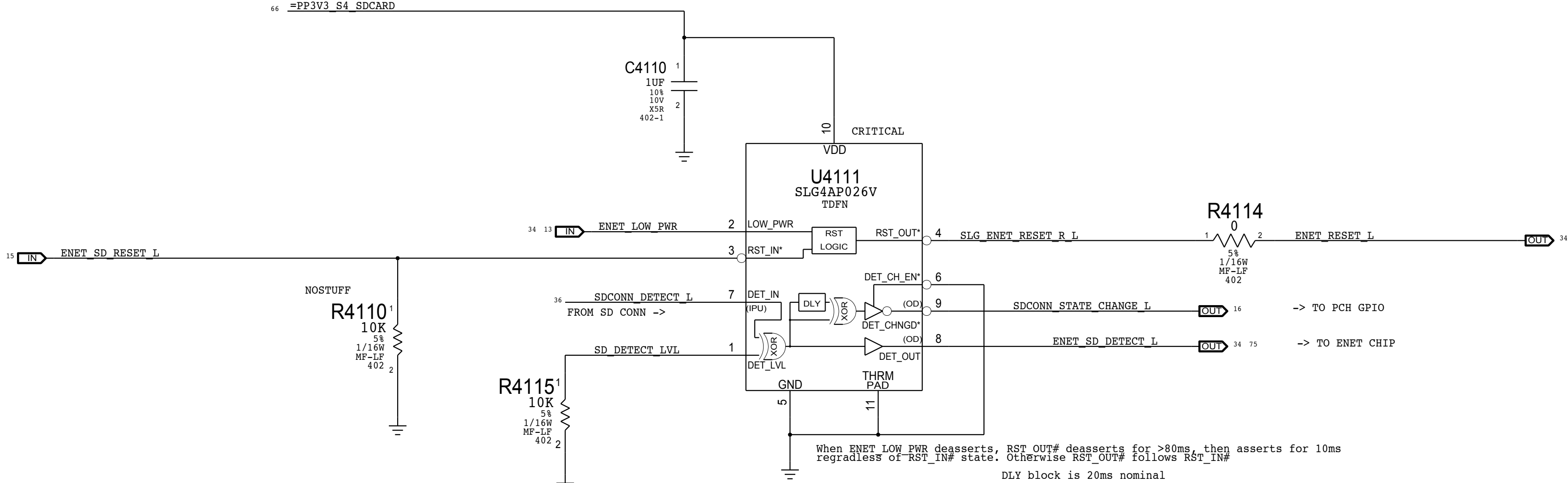


J16:516-0249 / J17:512-0038

SD CARD CONNECTOR



SDCONN DETECT DEBOUNCE. ENET_RESET AND DETECT-CHANGED PCH GPIO PULSE GENERATION.



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PAGE TITLE			
SD READER CONNECTOR			
	Apple Inc.	DRAWING NUMBER	051-00081
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Camera Controller			
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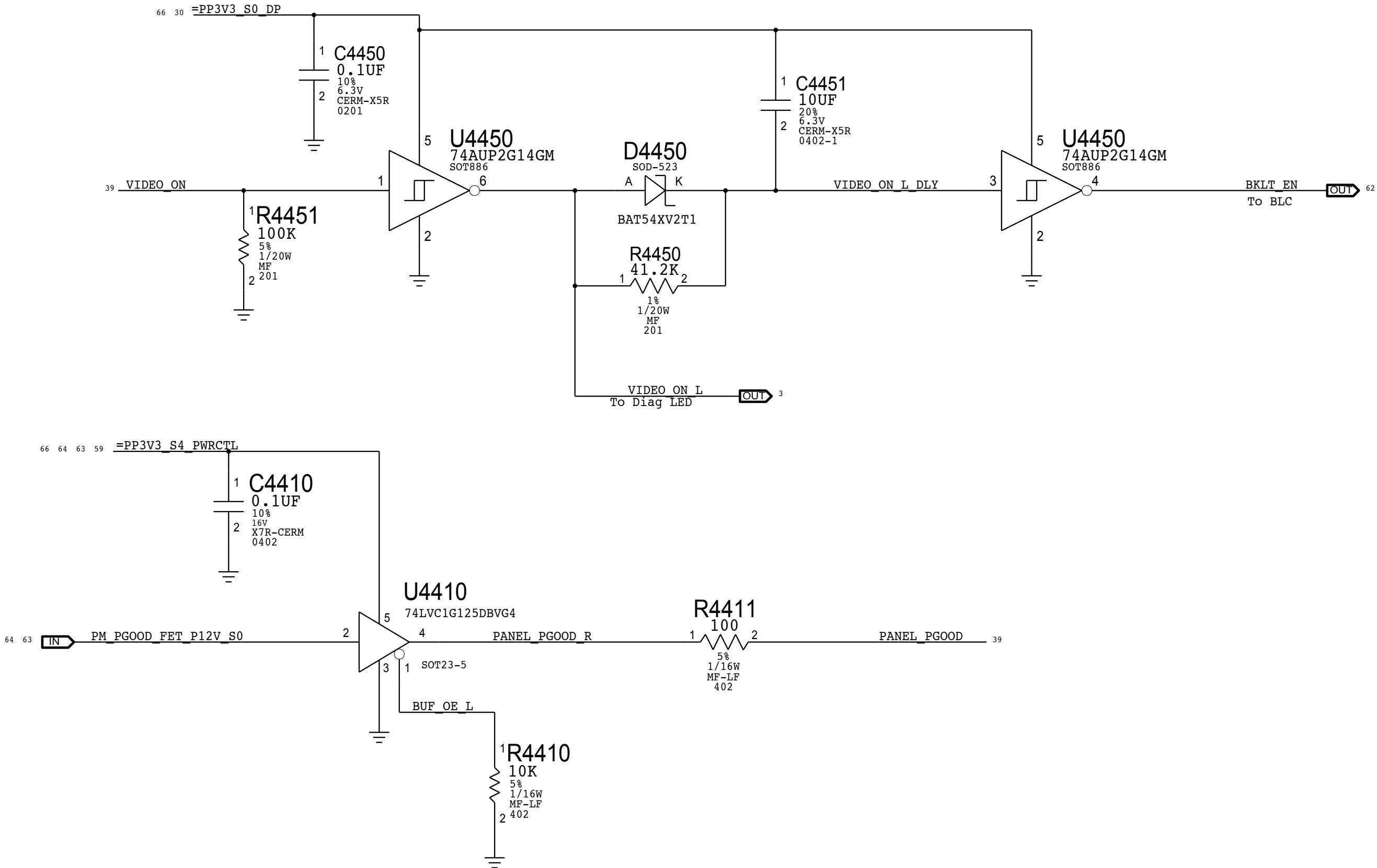
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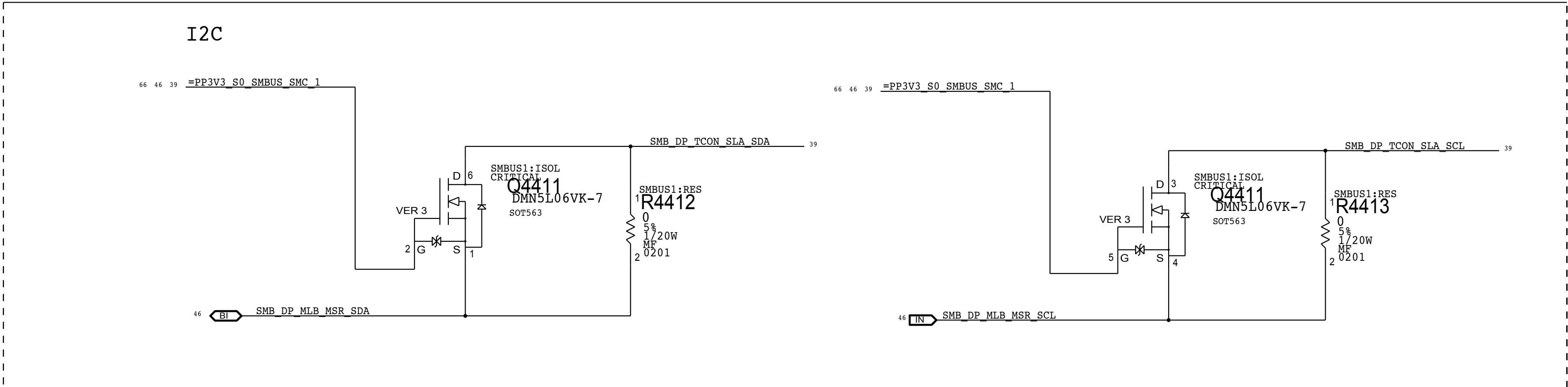
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Backlight Control

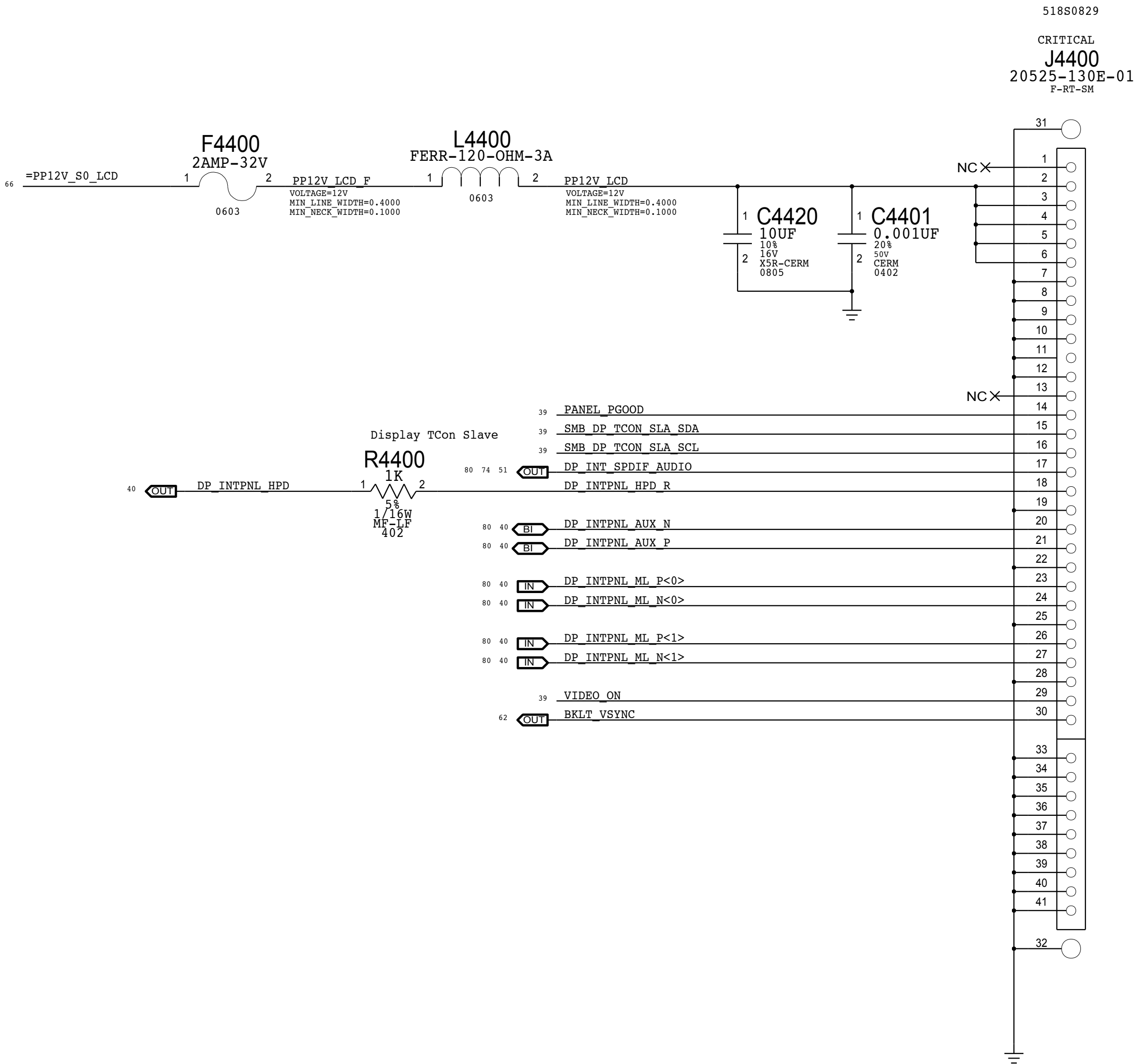
Delay applies only on a L->H transition on VIDEO_ON. This guarantees video is valid before the backlight is enabled.
On a H->L transition, output follows with standard logic propagation delay. This ensures the backlight is off immediately after loss of video

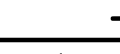


SMBus Isolation



Internal DP Connector



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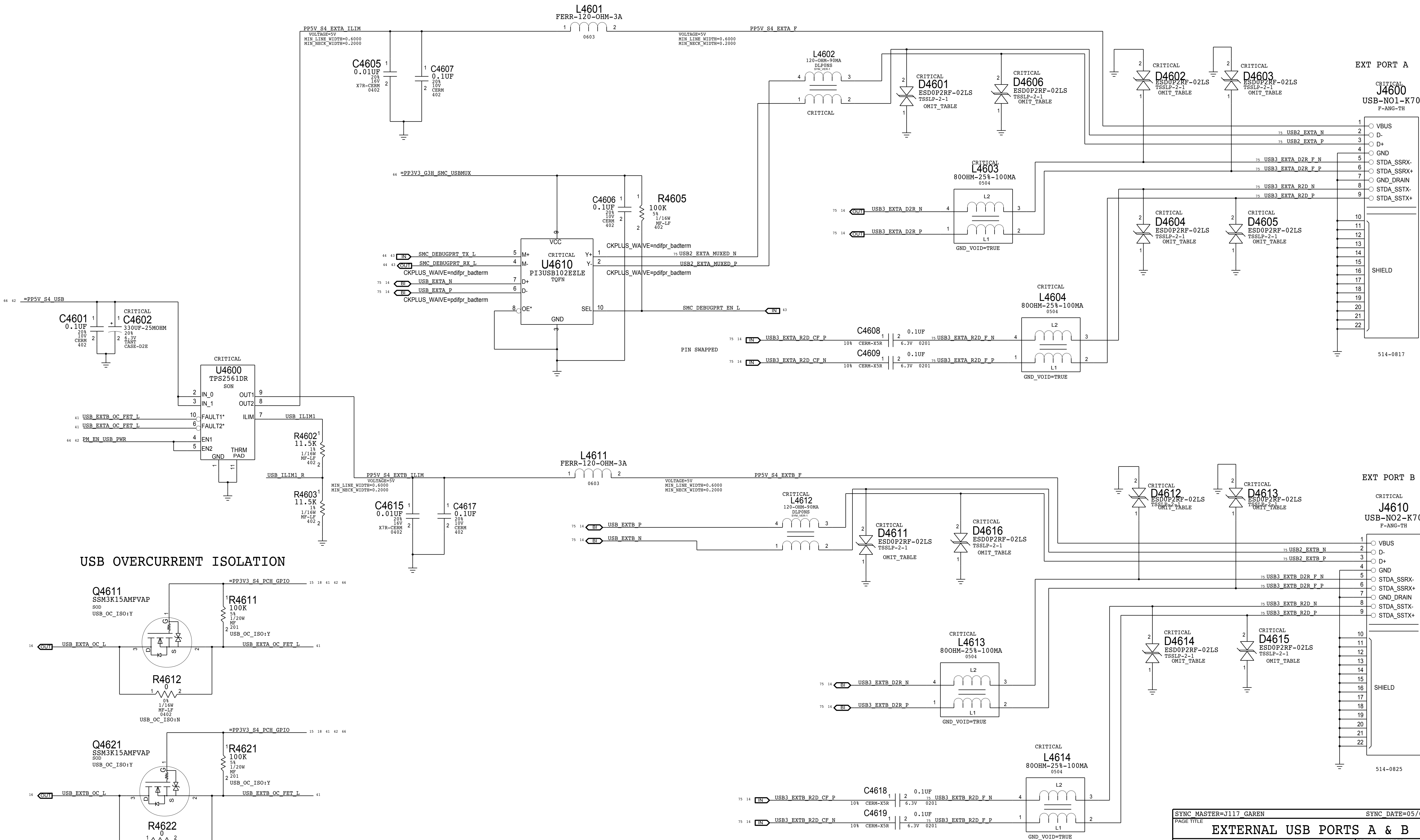
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37780155	6	USB3 diodes	D4601,D4602,D4603,D4604,D4605,D4606	
37780155	6	USB3 diodes	D4611,D4612,D4613,D4614,D4615,D4616	

P and N pins swapped for cleaner routing
Lane polarity inversion supported by USB3
See Section 6.4.2 of USB3 Spec

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EXTERNAL USB PORTS A & B			
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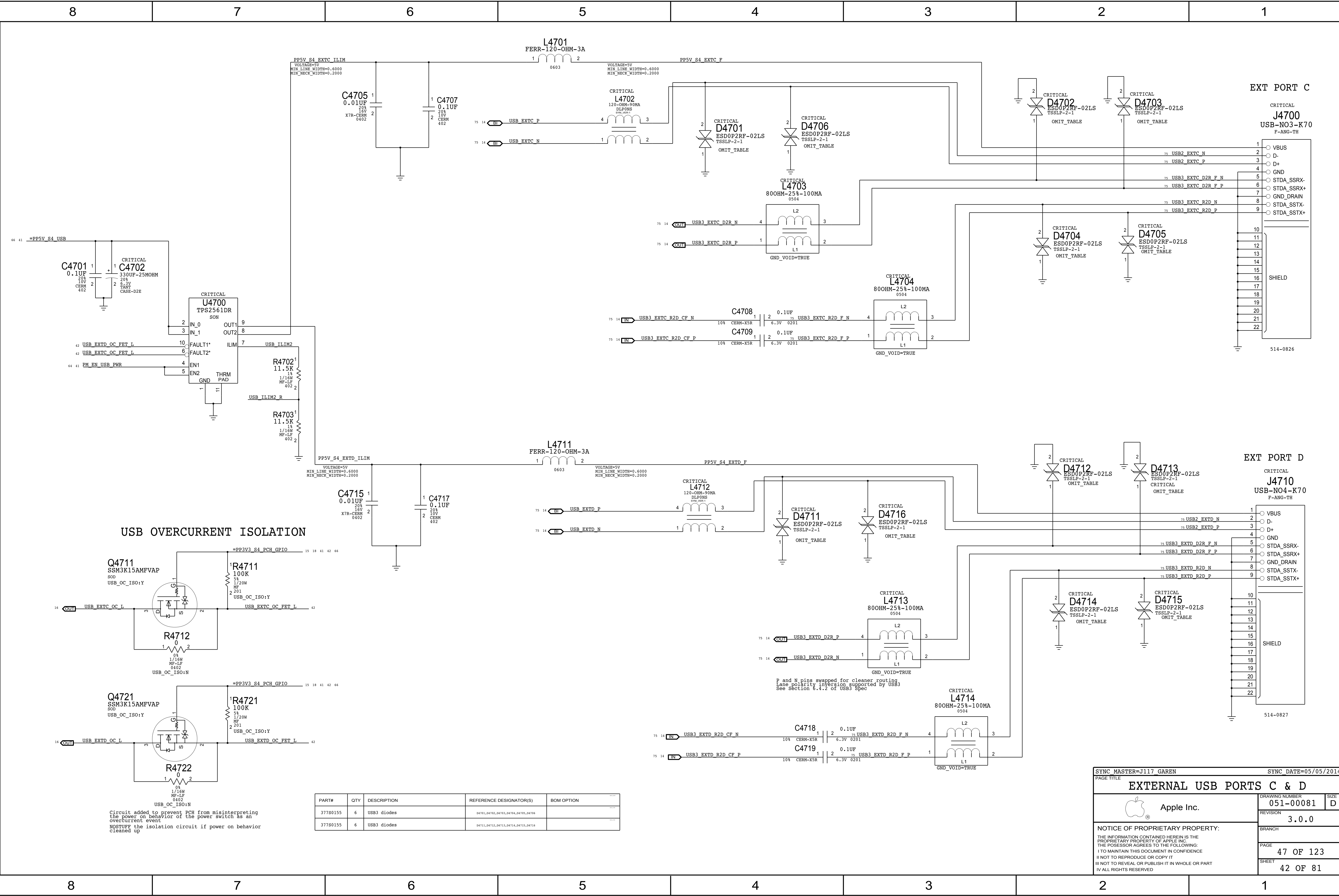
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PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	BOM OPTION
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37780155	6	USB3 diodes	D4711,D4712,D4713,D4714,D4715,D4716	

SYNC_MASTER=J117_GAREN

SYNC_DATE=05/05/2014

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EXTERNAL USB PORTS C & D

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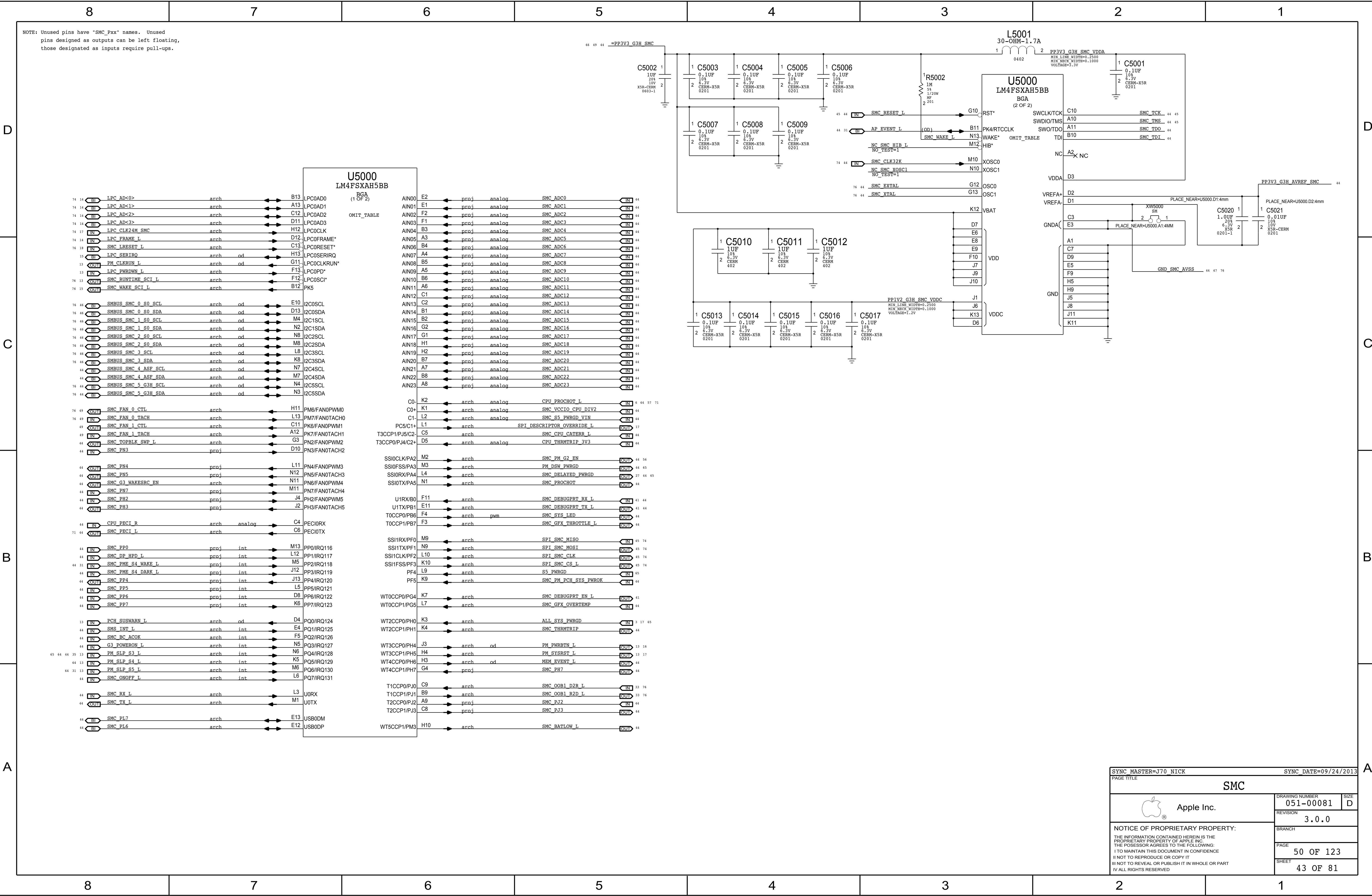
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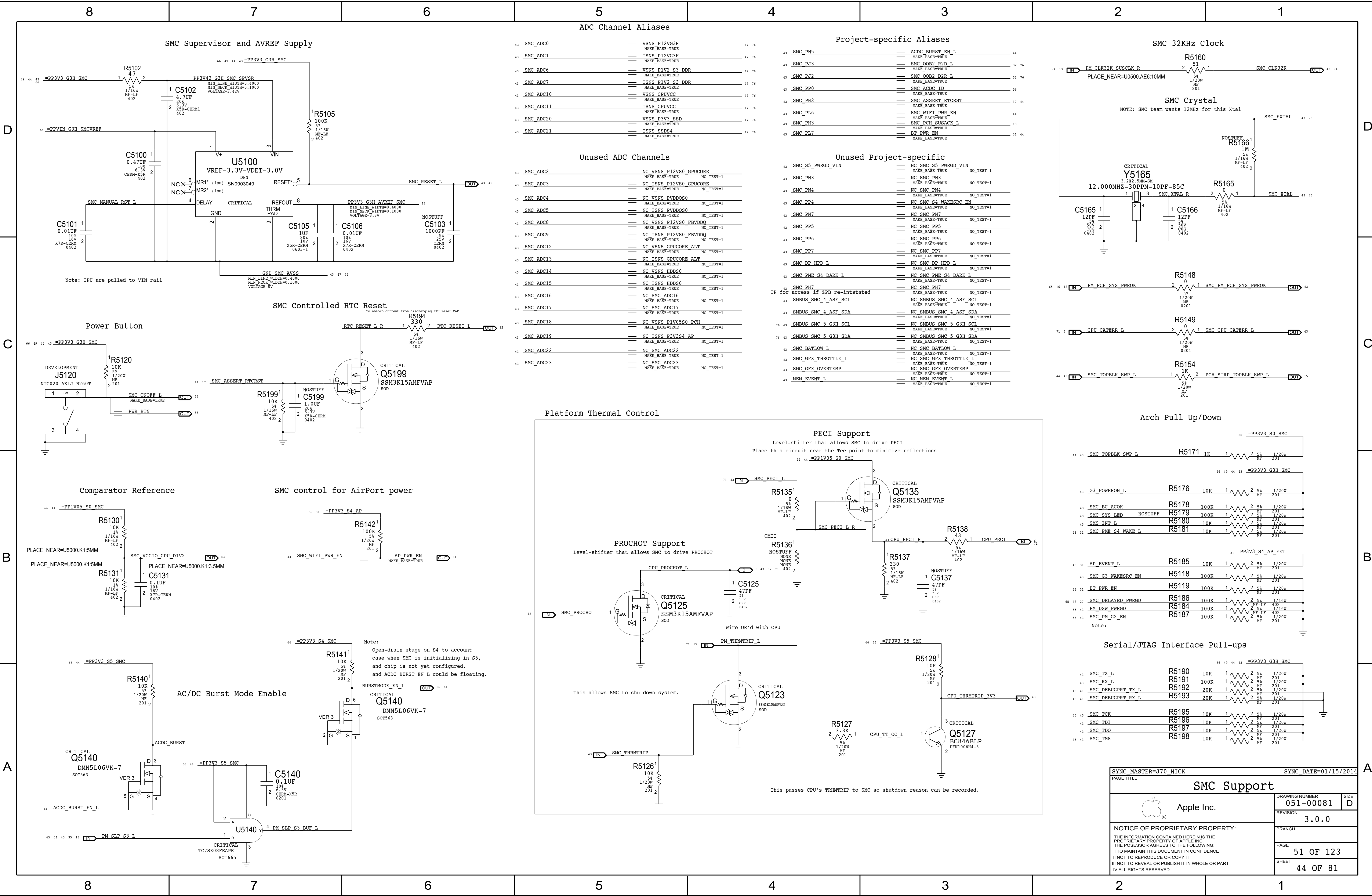
SHEET

42 OF 81

USB OVERCURRENT ISOLATION

Circuit added to prevent PCB from misinterpreting the power on behavior of the power switch as an overcurrent event
NOSTUFF the isolation circuit if power on behavior cleaned up





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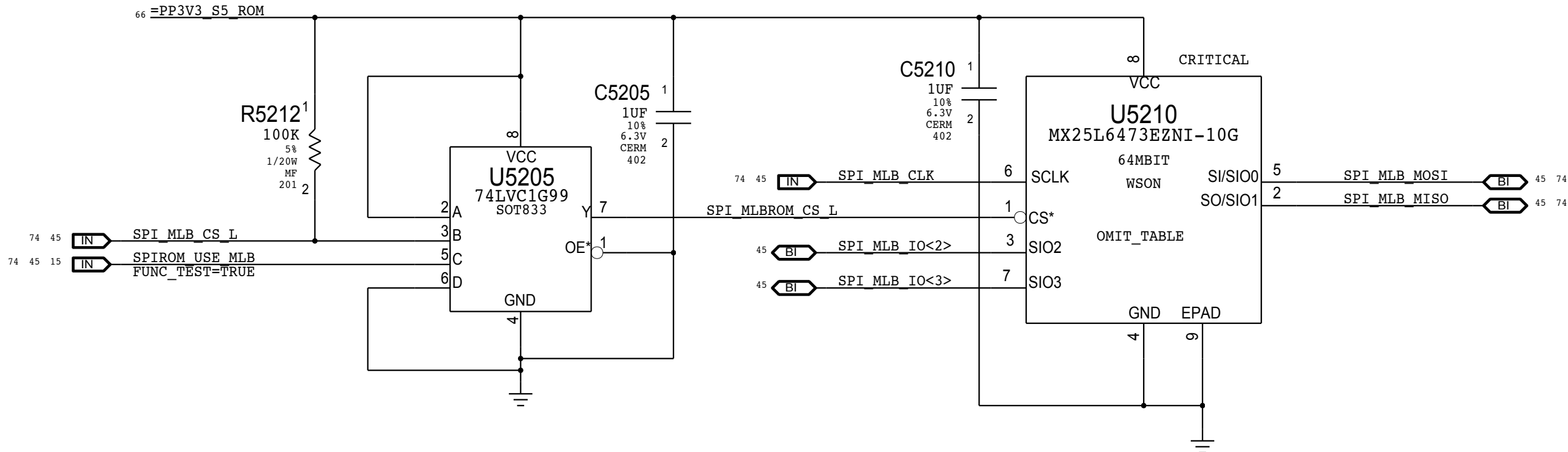
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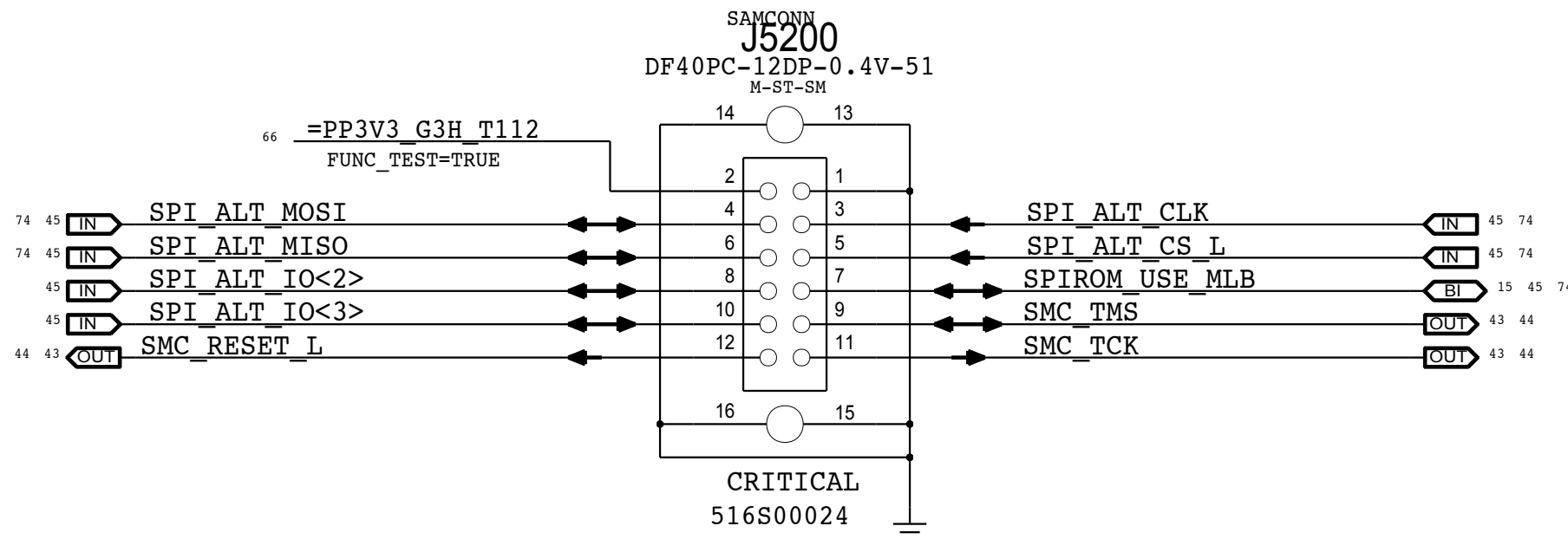
A

SPI BootROM

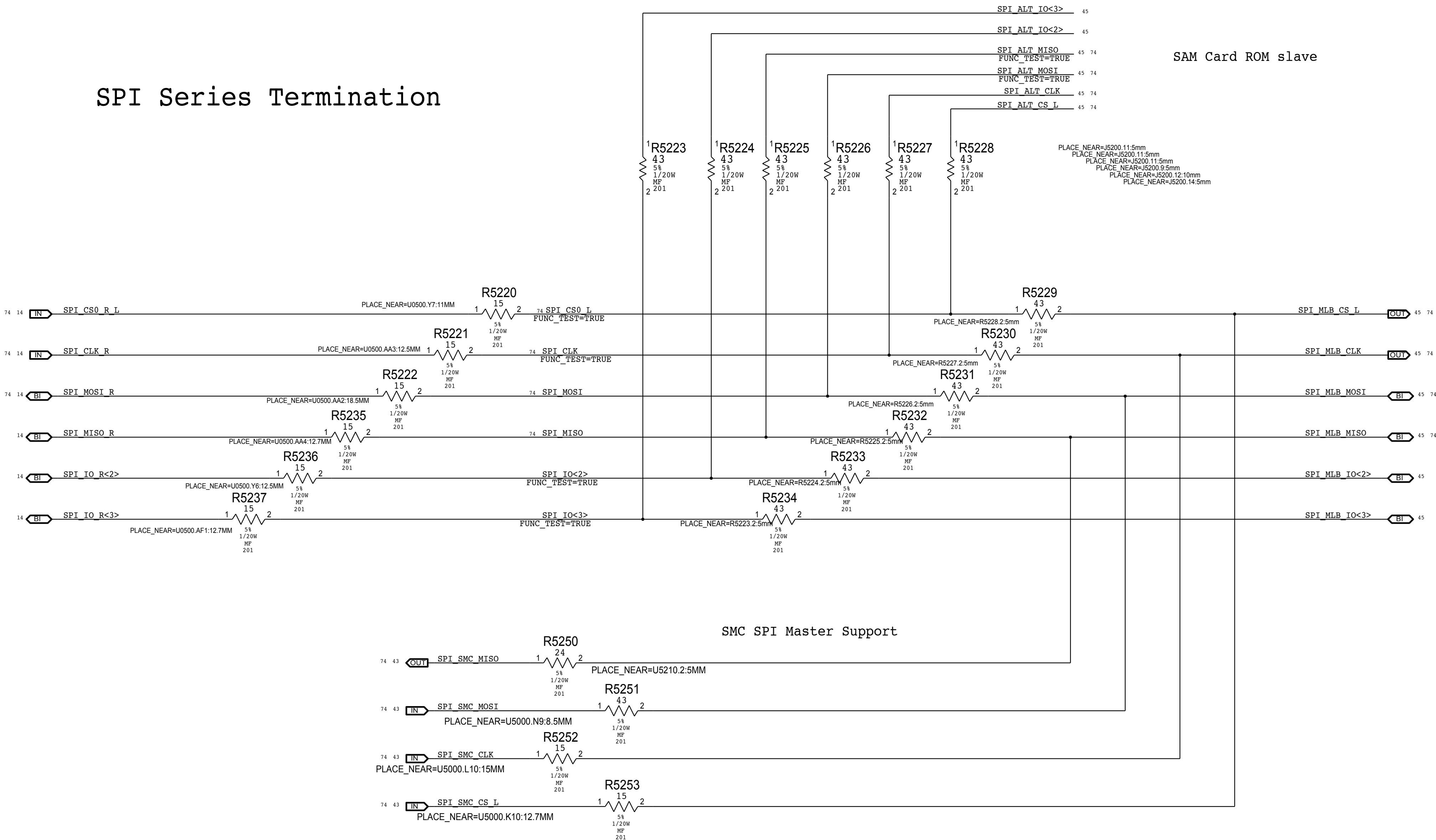
Quad_IO Mode (Mode 0 & 3) supported
SPI Freq: 50MHz for PCH, 20 MHz for SMC

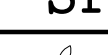


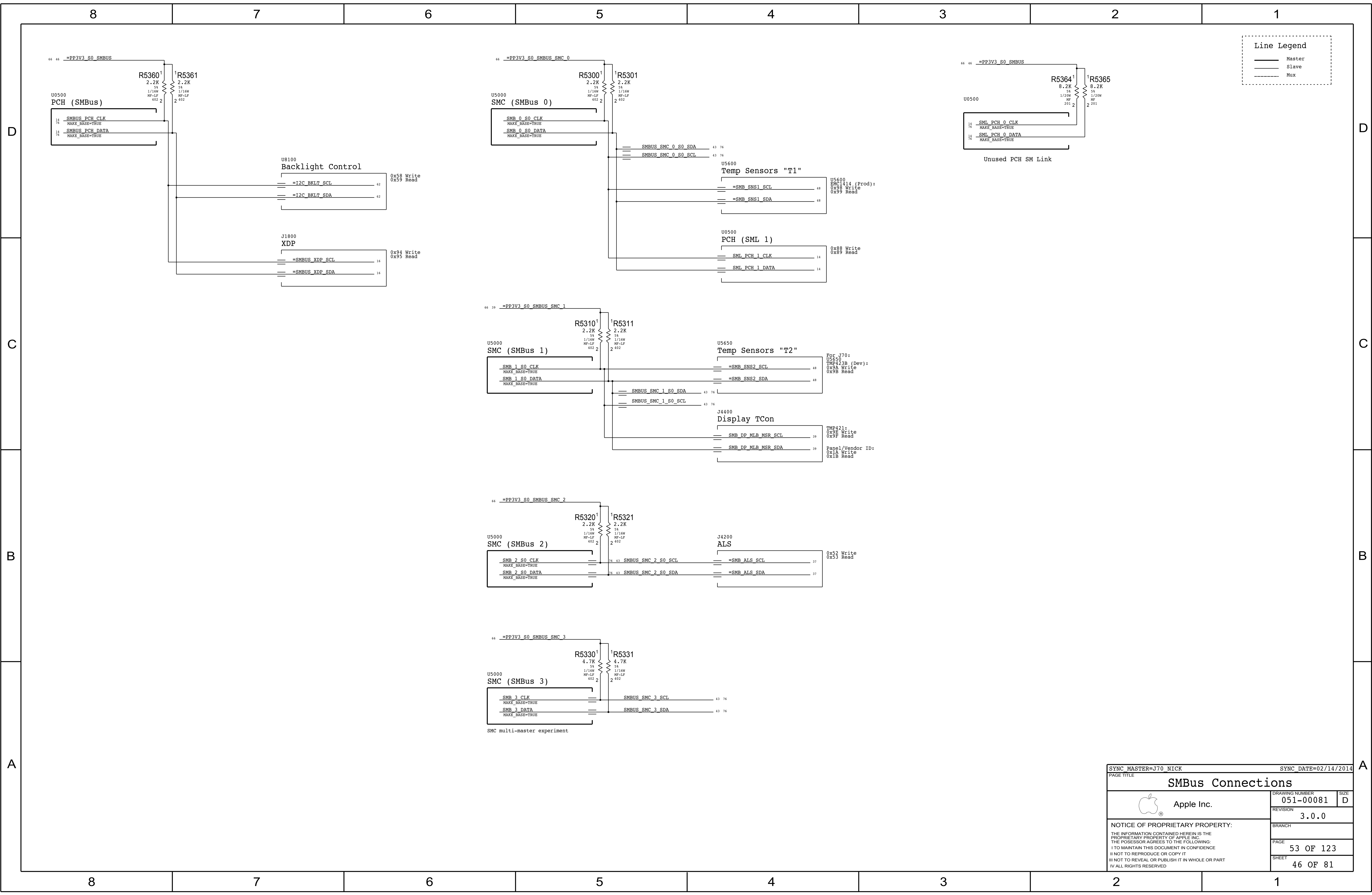
SPI+SWD SAM Connector



SPI Series Termination

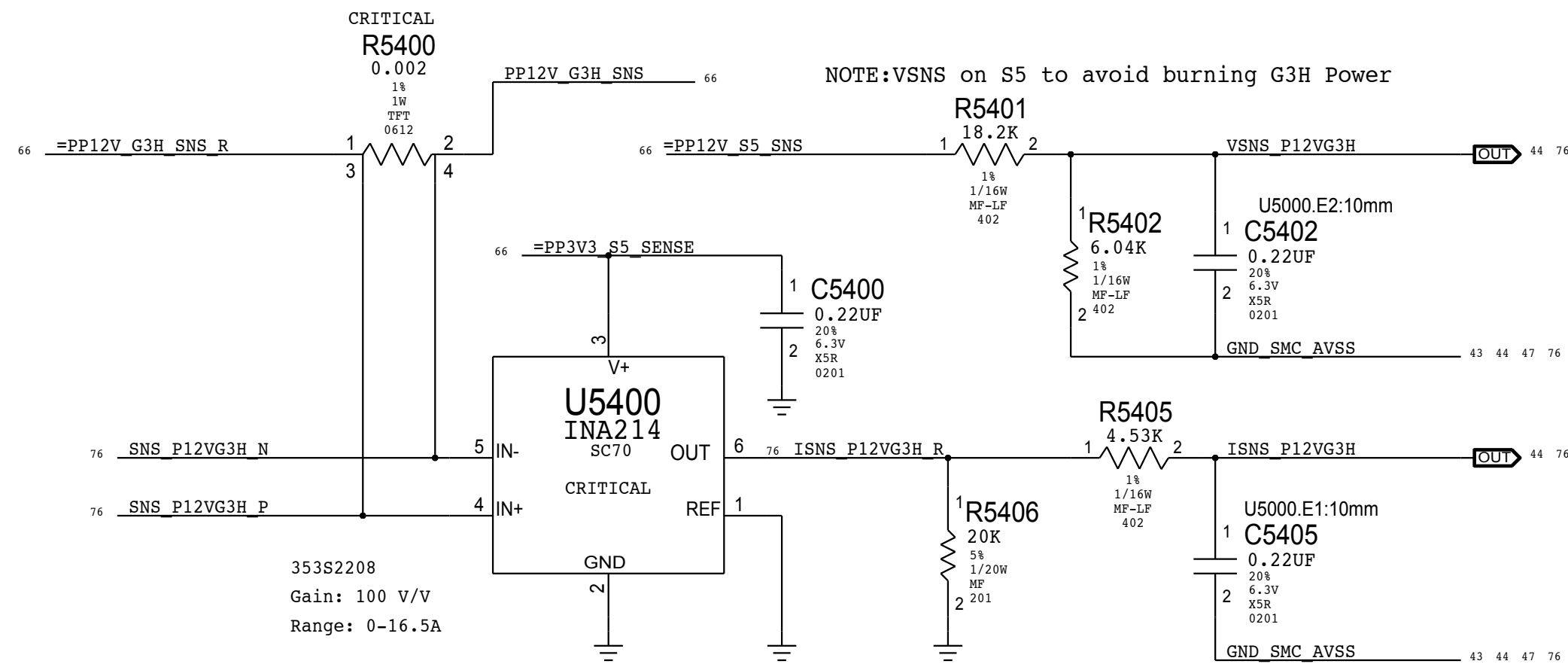


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SPI and Debug Connector			
	Apple Inc.	DRAWING NUMBER	051-00081
		SIZE	D
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		PAGE	52 OF 123
		SHEET	45 OF 81



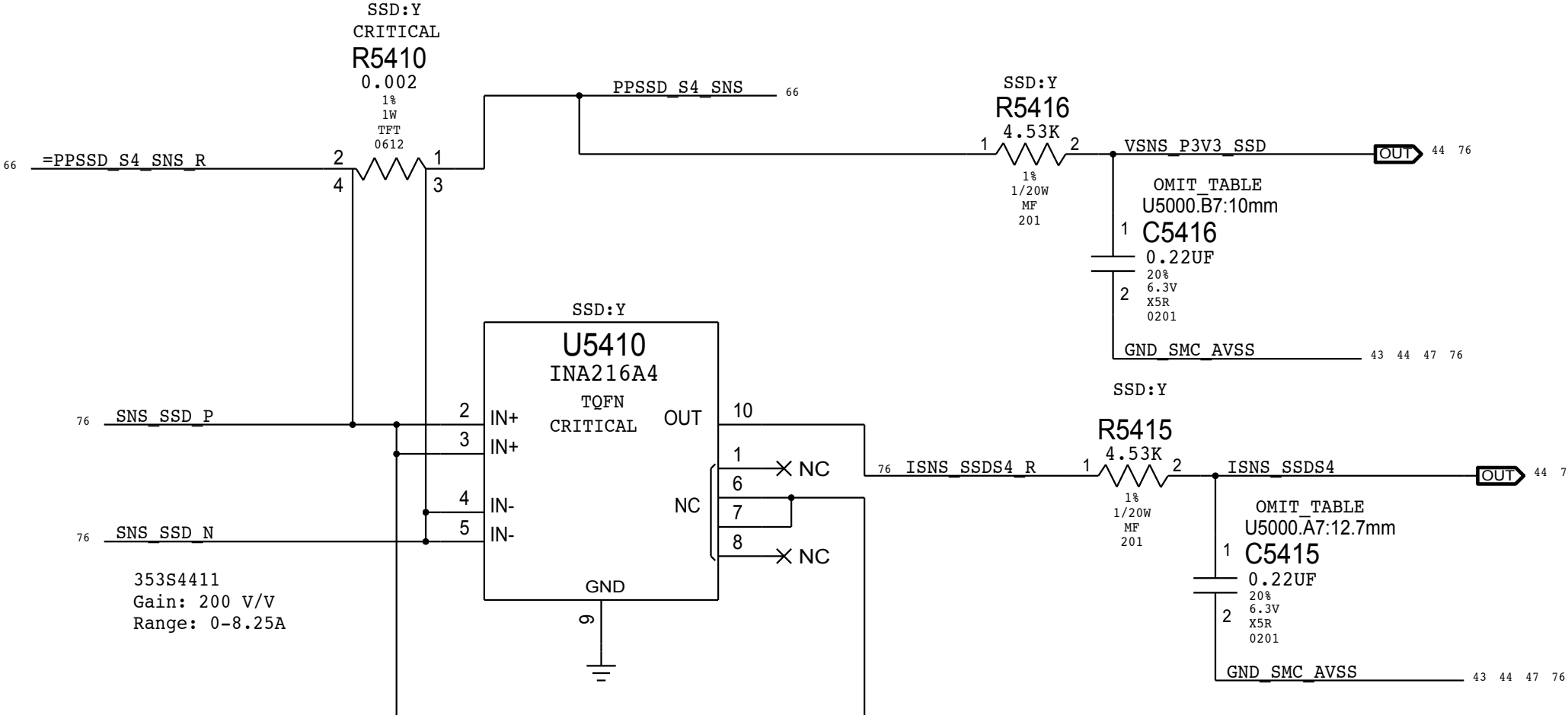
12V G3H (VD2R:ADC0/ID2R:ADC1)

AC/DC lowside sense (System total)



SSD S4 (IH1R:ADC21/VR3R:ADC20)

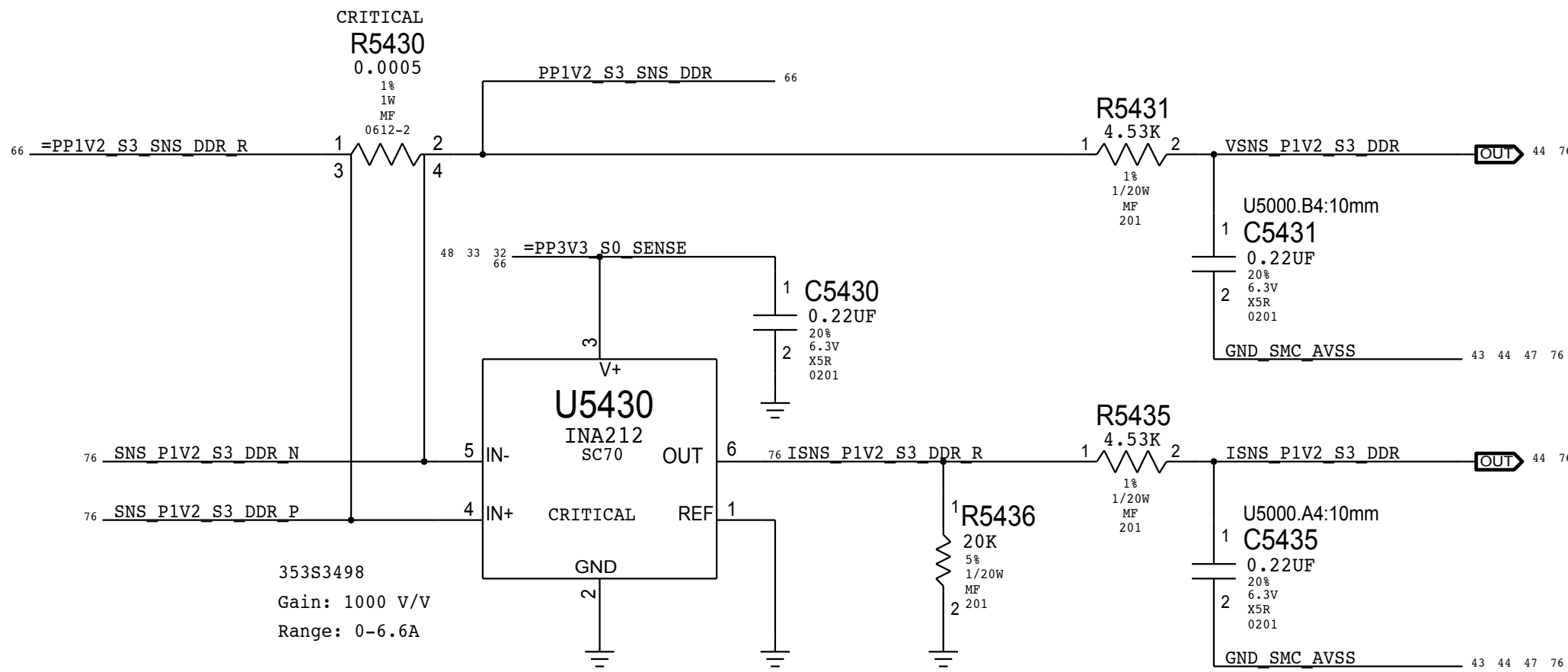
I-sense / V-sense for SSD



PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	BOM OPTION
132S0304	2	CAP,0.22UF,201	C5415,C5416	SSD:Y
11780201	2	RES,0 OHM,201	C5415,C5416	SSD:N

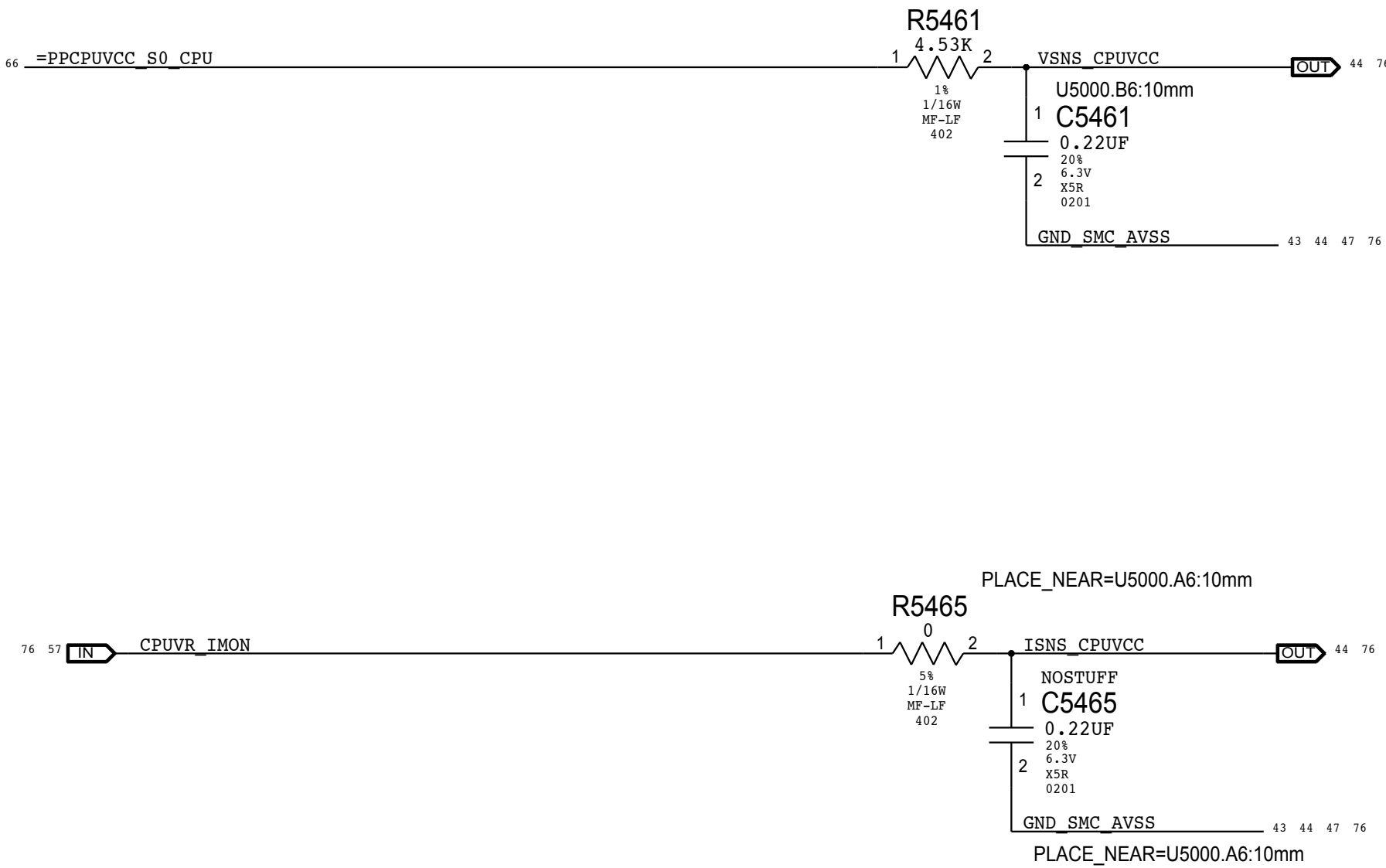
VDDQ S3 (VM0R:ADC6/IM0R:ADC7)

VDDQ lowside sense for DDR



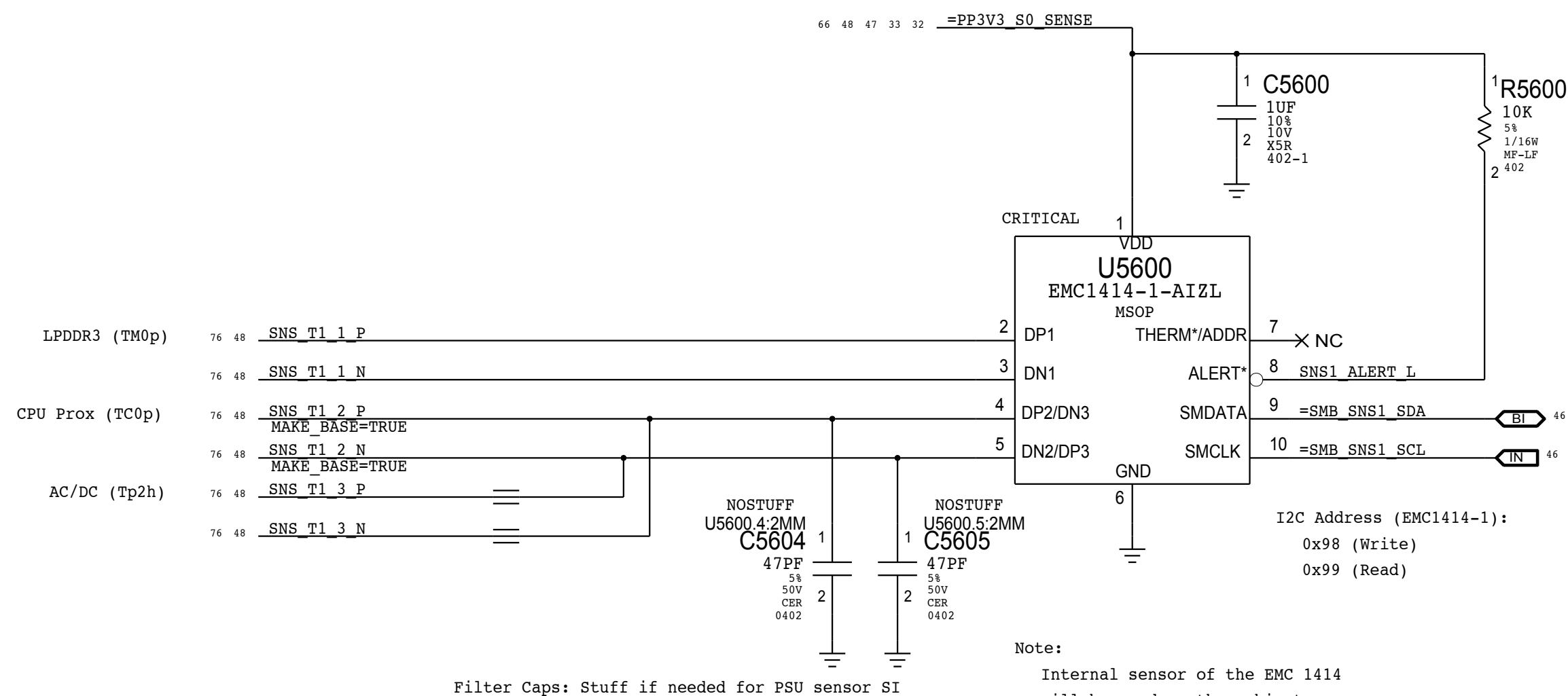
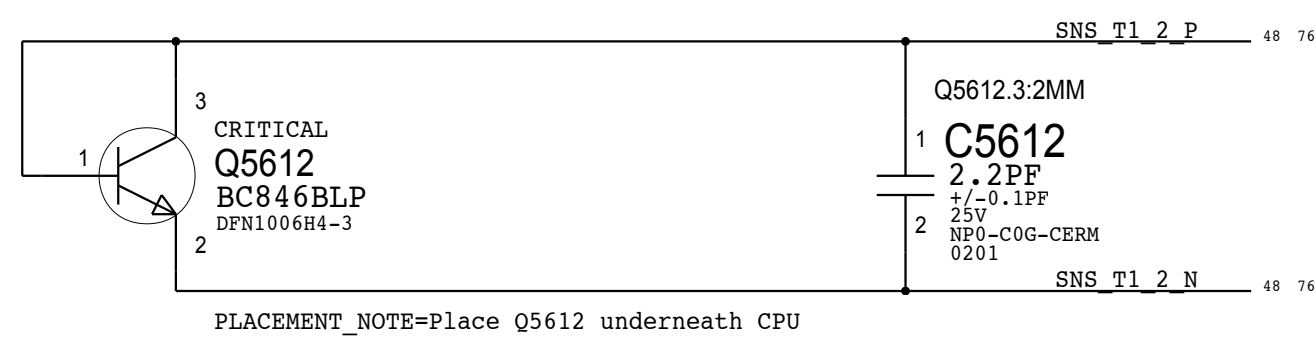
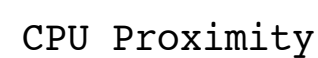
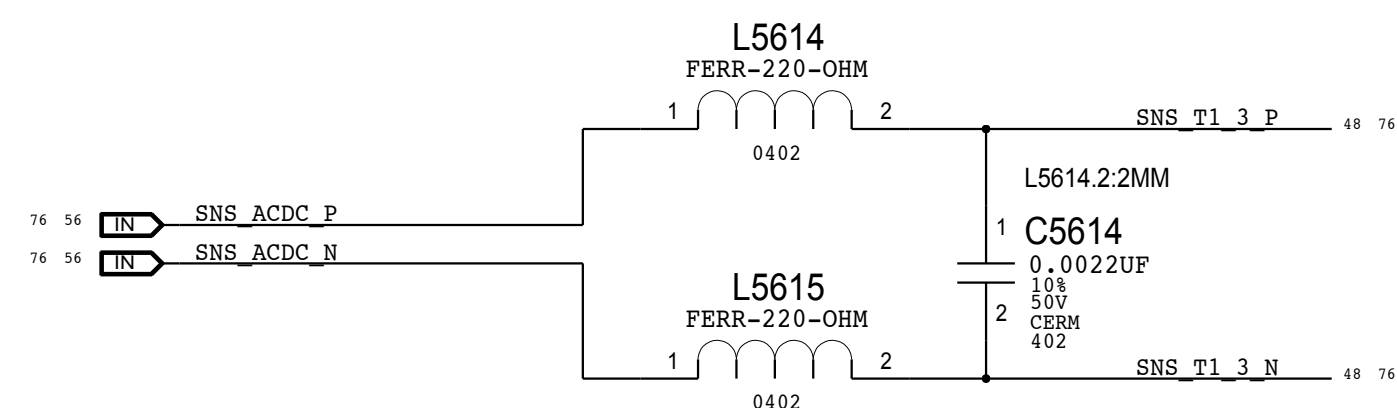
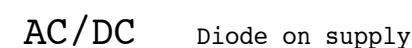
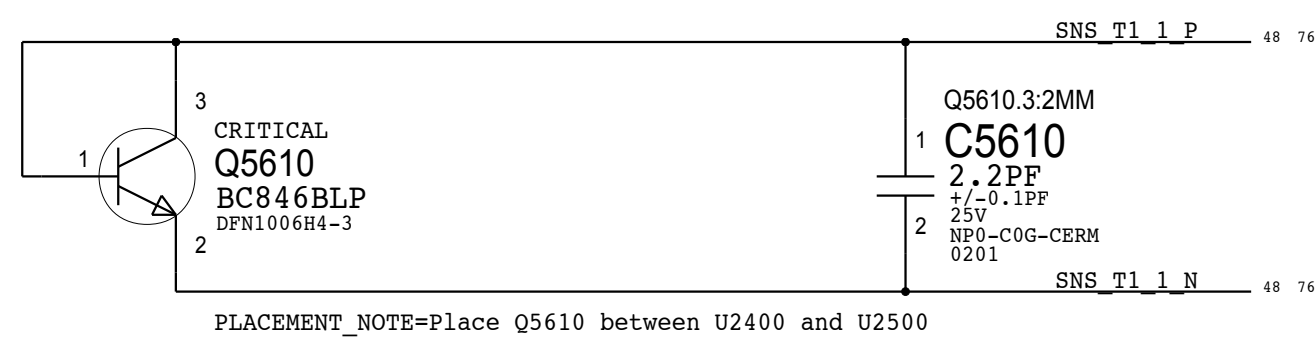
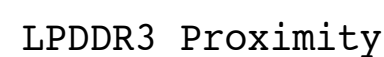
CPU Core (VC0C:ADC10/IC0C:ADC11)

Voltage sense and IMON amp (VC0C, IC0C)



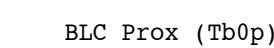
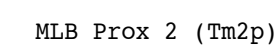
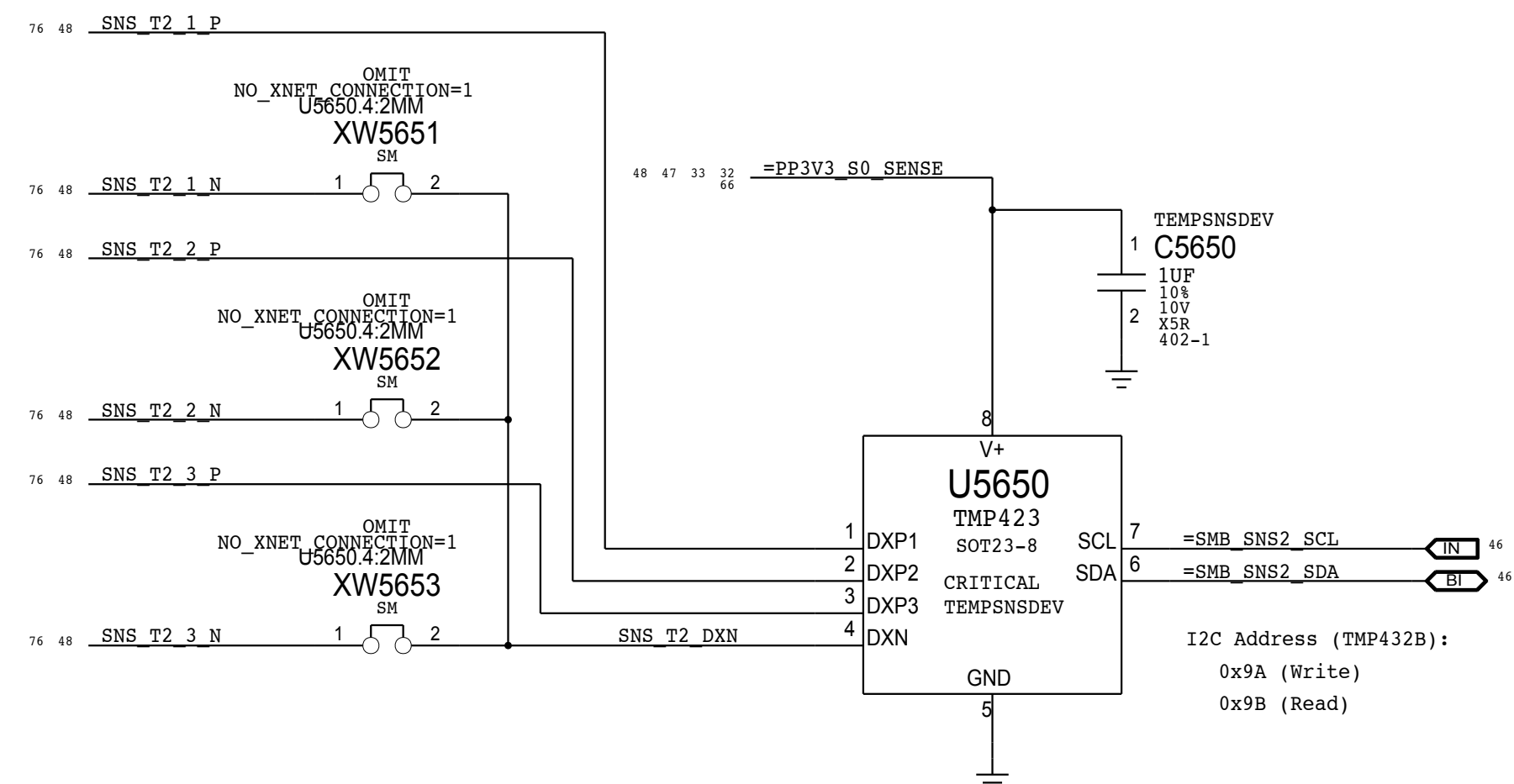
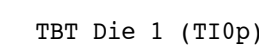
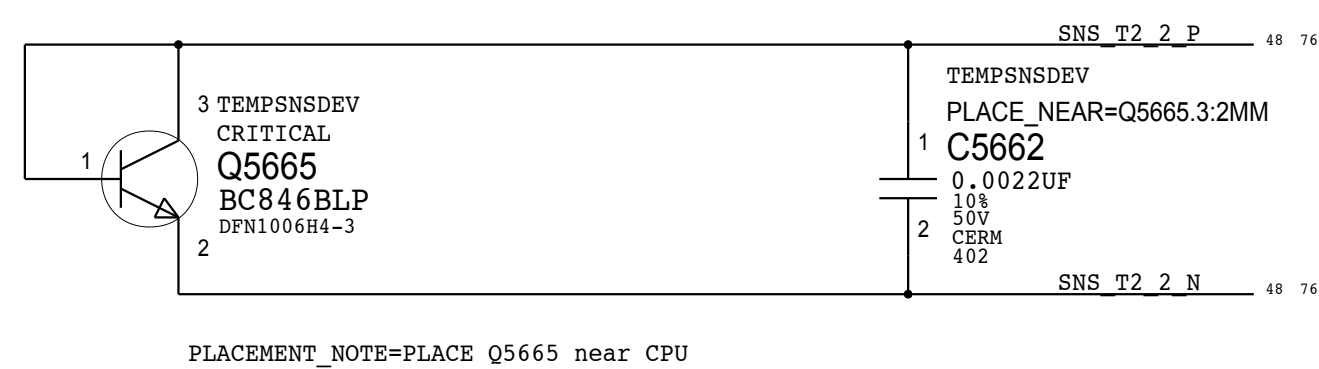
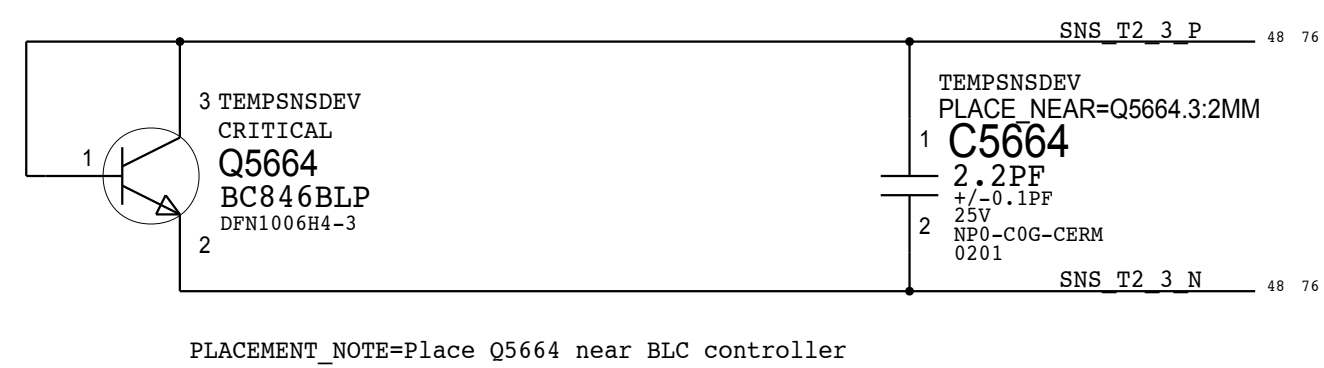
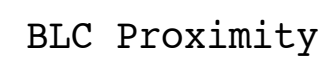
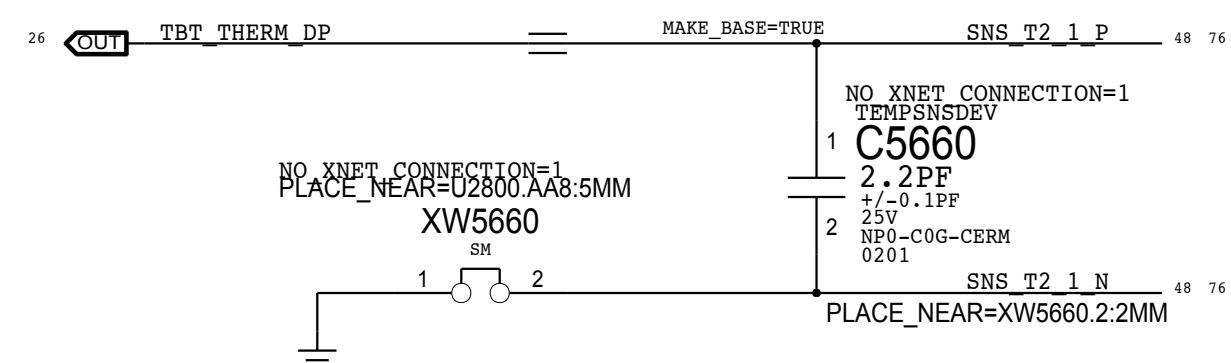
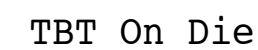
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I and V Sense			
 Apple Inc.	DRAWING NUMBER		SIZE
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Temperature Sensor T1



PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
372S0186	372S0185		ALL	Alternate Temp Diode

Temperature Sensor T2



SYNC MASTER=J70 NICK		SYNC DATE=11/05/2013	
PAGE TITLE			
 Temperature Sensors Apple Inc.			
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		REVISION 3.0.0	
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PAGE		56 OF 123	
SHEET		48 OF 81	

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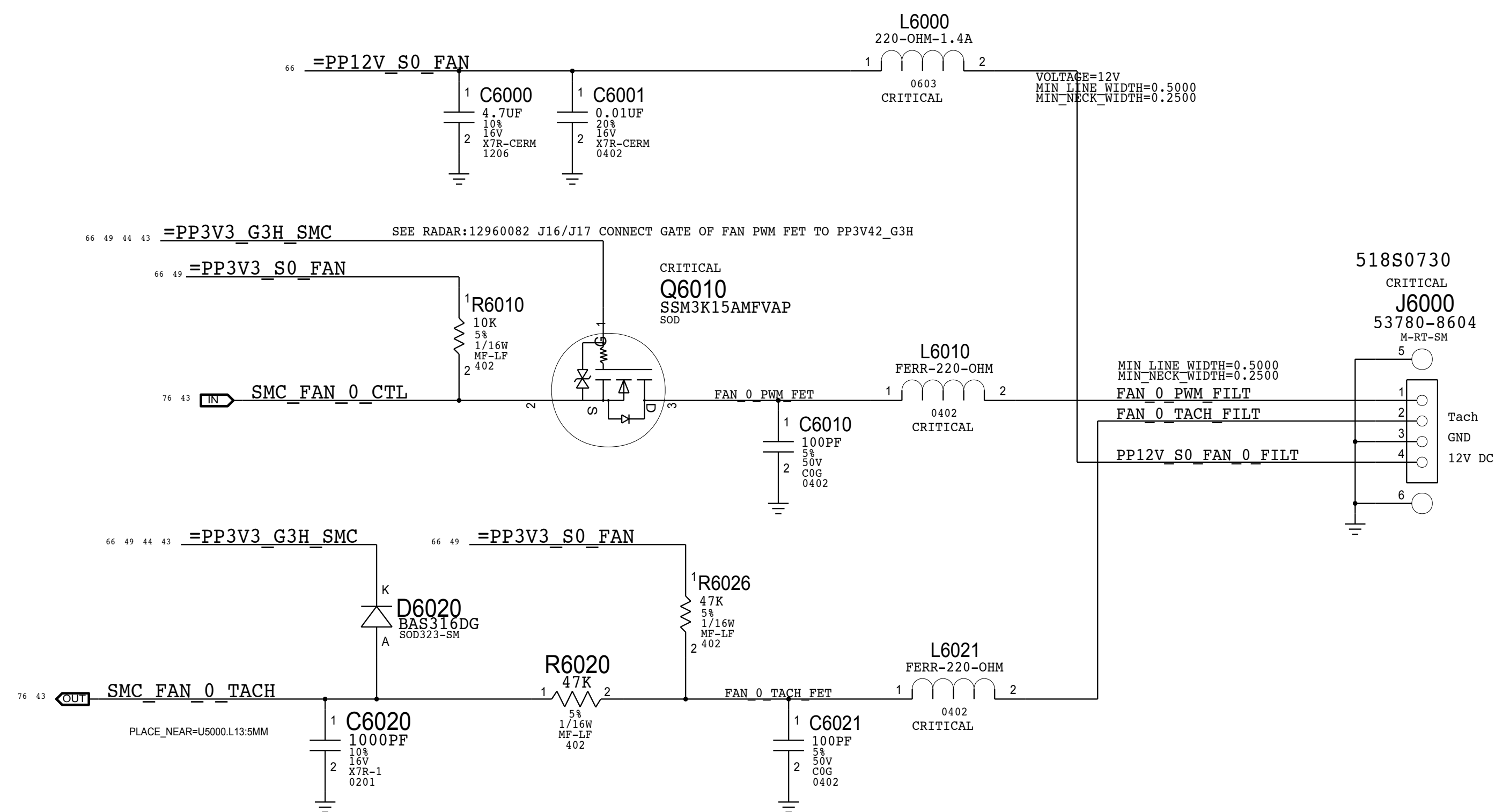
SMC Fan 0 (System)

Note:

The circuit for the PWM input to the fan acts as a non-inverting level-shifter to protect the SMC. It is assumed there is a pull-up to 5V/12V inside the fan, otherwise when the SMC PWM goes low and Q6010 turns on, there would be 5V/12V present on the SMC pin! Then by definition, the drain of Q6010 is at common and the SMC sinks current when Q6010 is on.

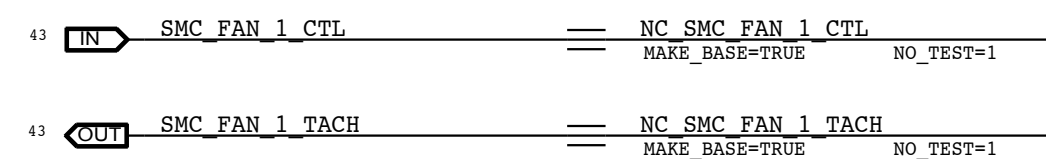
This resembles an open-drain if there is a pull-up, going to a Hi-Z FET input.

Otherwise, this is simply a pass-FET.
See RADAR: 10565825- D7: Need schematic and PCB file of fan(All Vendors).



Add C6020 1000pF Cap, Change R6020 to 47K -- Radar 11661918 D8 Proto1 Fan Tach instability.

SMC Fan 1 (Unused)



SYNC MASTER=J16 MLB IG		SYNC DATE=08/27/2013	
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		BRANCH	
		PAGE	60 OF 123
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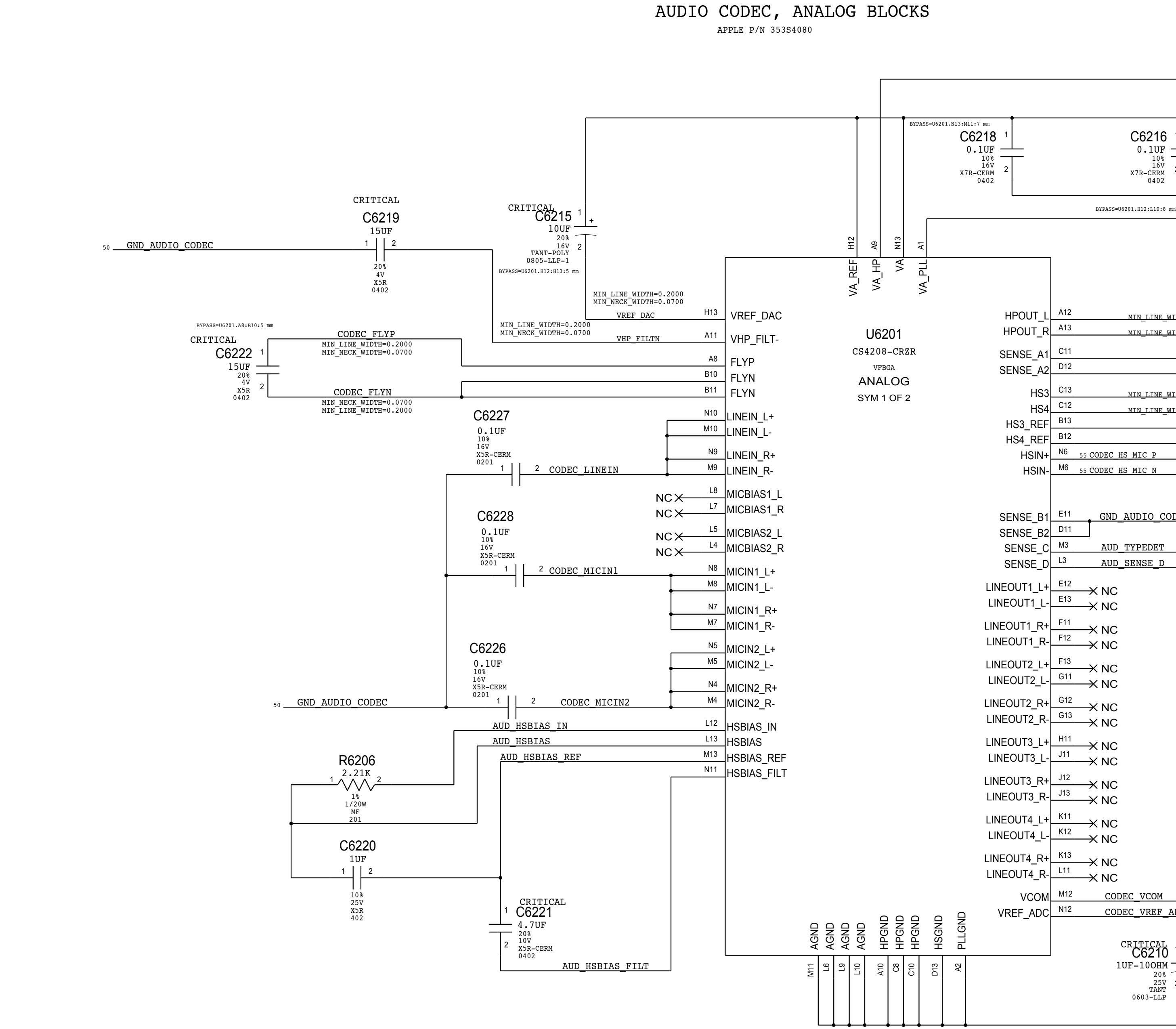
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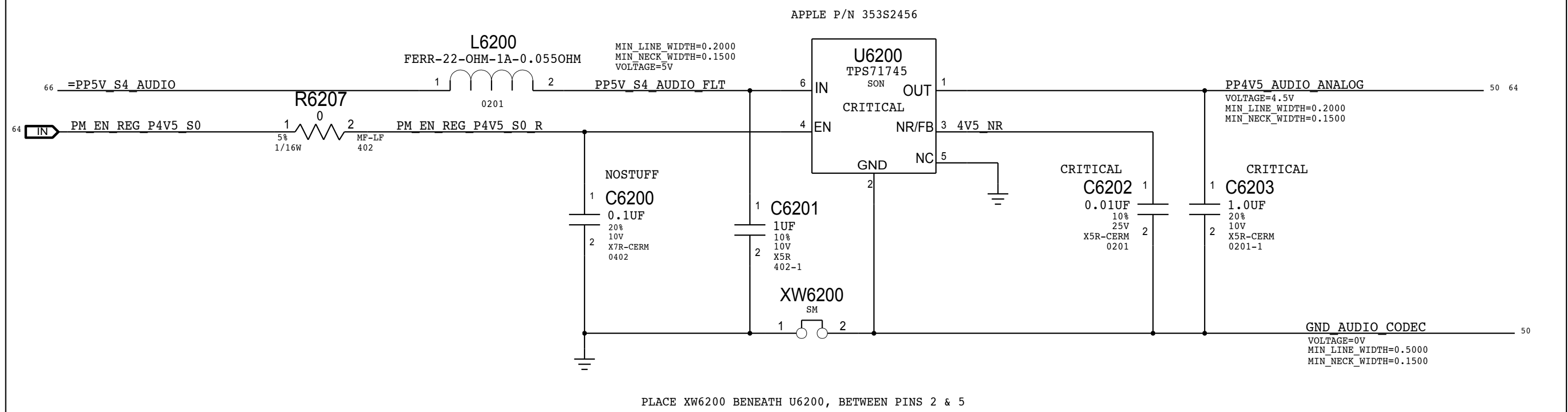
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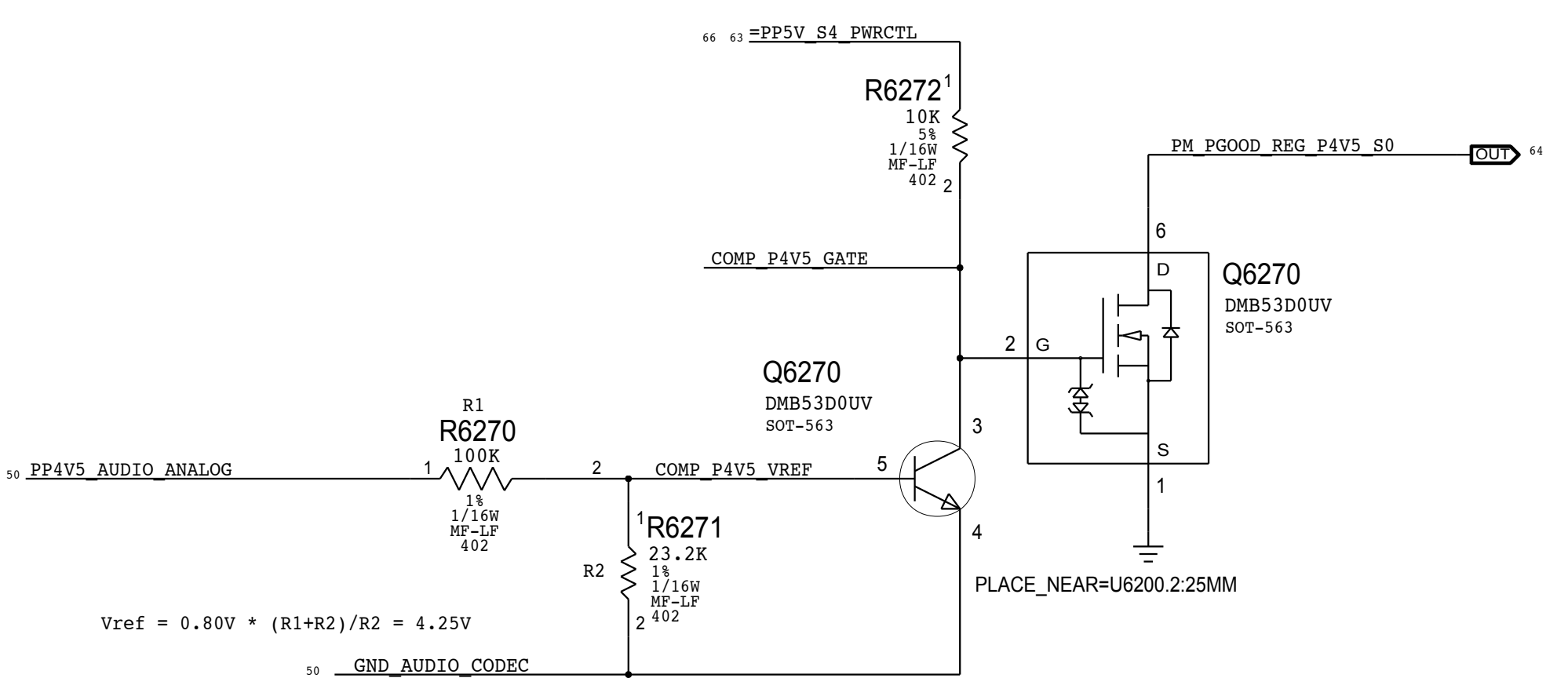
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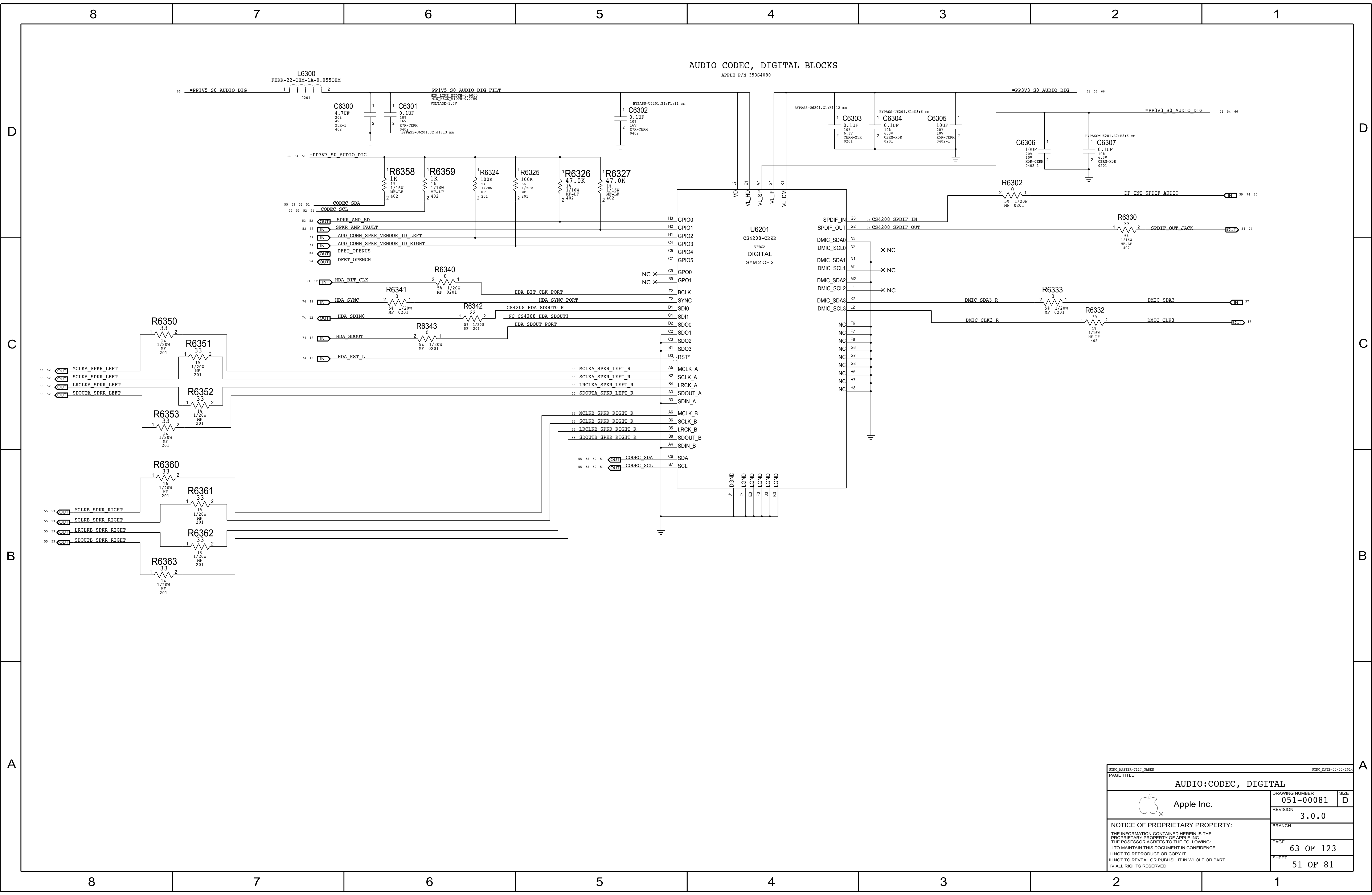
4.5V POWER SUPPLY FOR CODEC

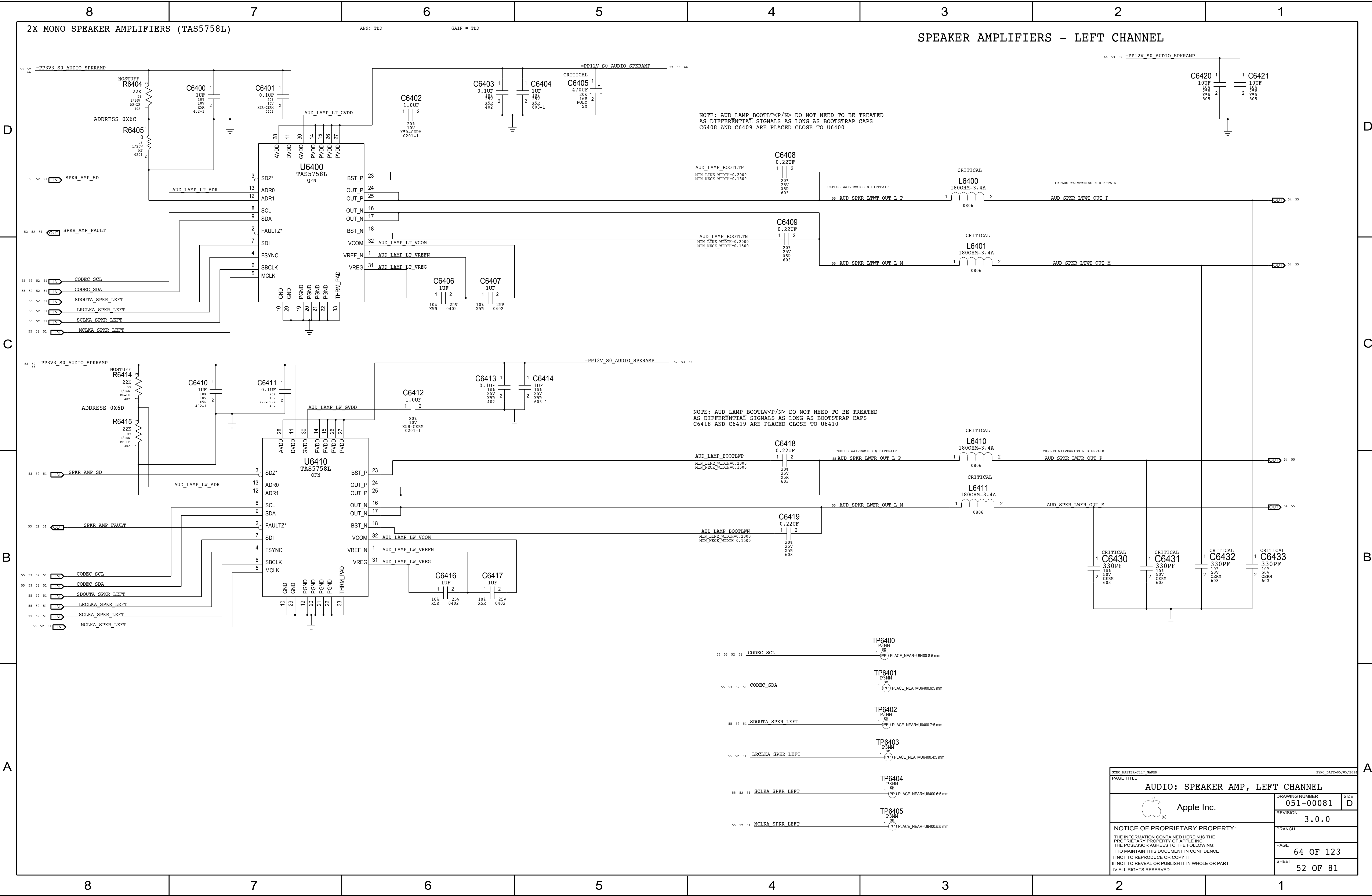


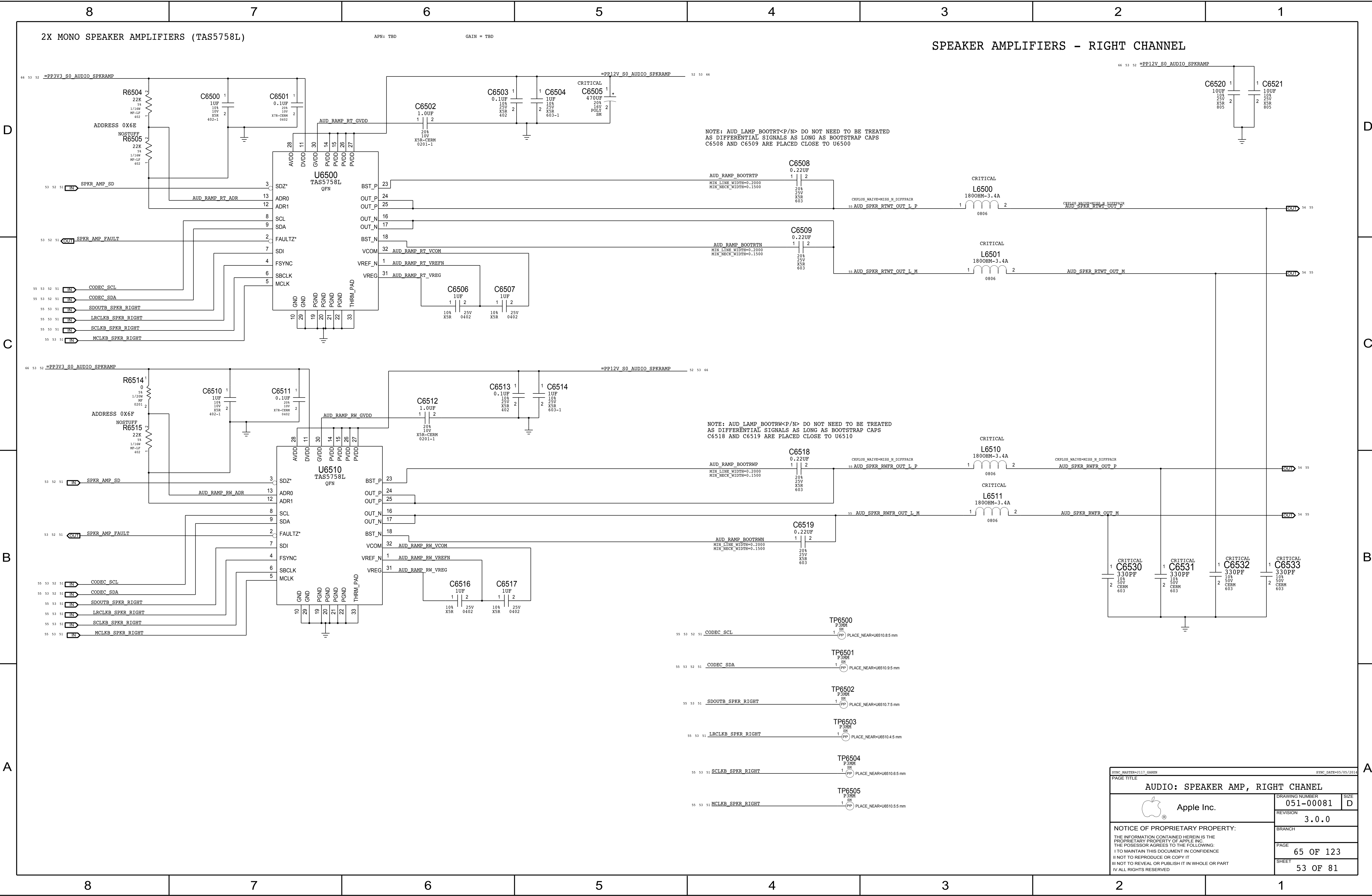
4.5V COMPARATOR



SYNC_MASTER=J117_GAREN		SYNC_DATE=05/05/2014	
PAGE TITLE			
AUDIOIO: Codec (Analog)			
 Apple Inc.	DRAWING NUMBER 051-00081		SIZE D
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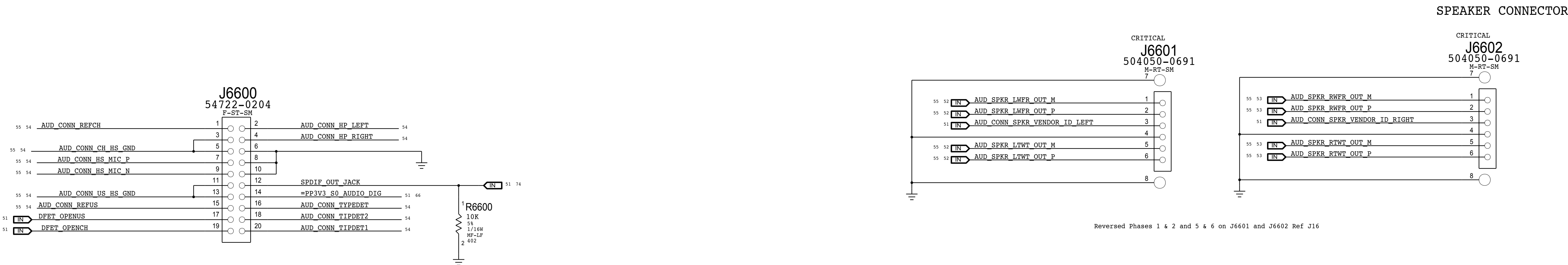


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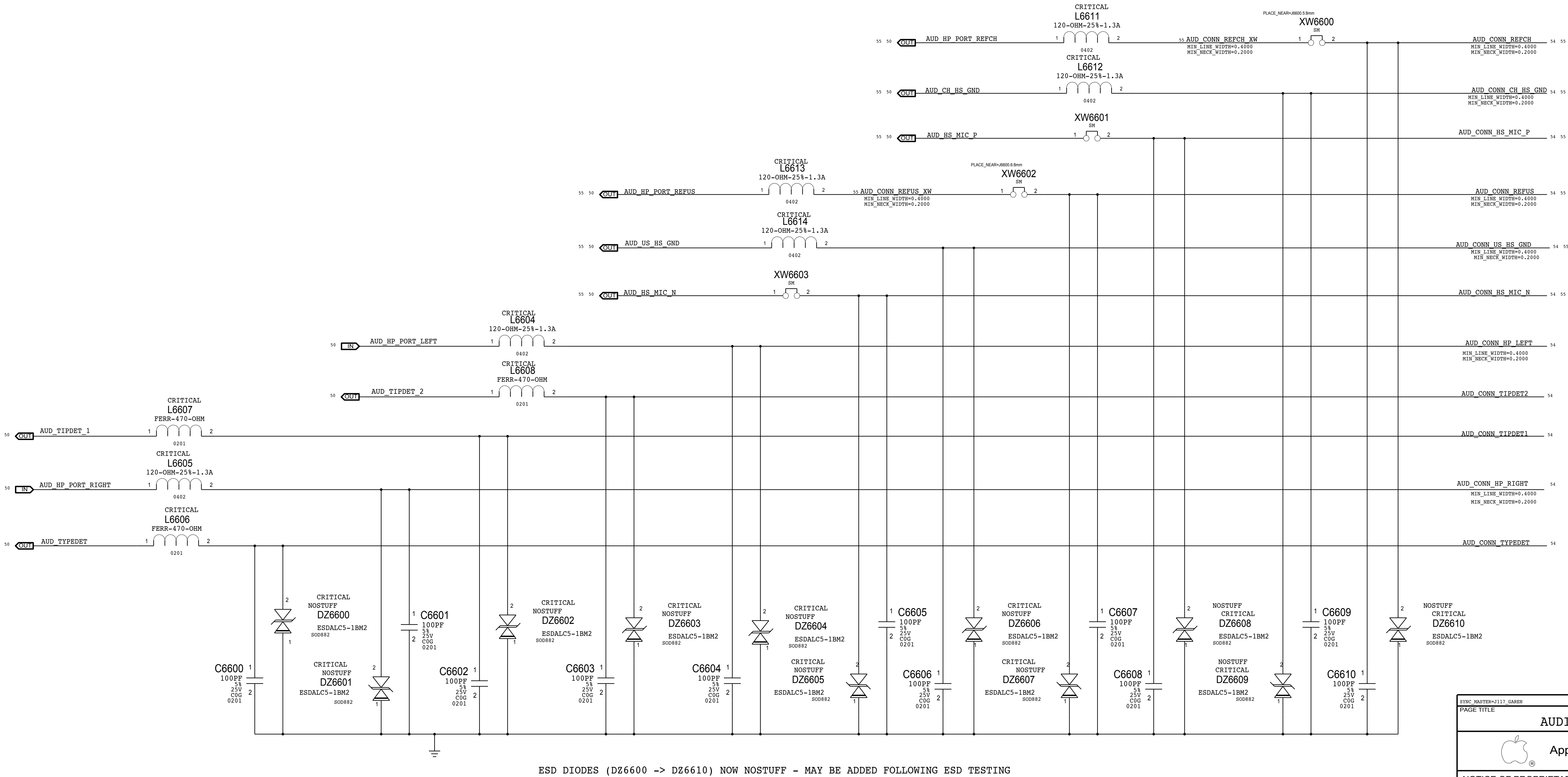


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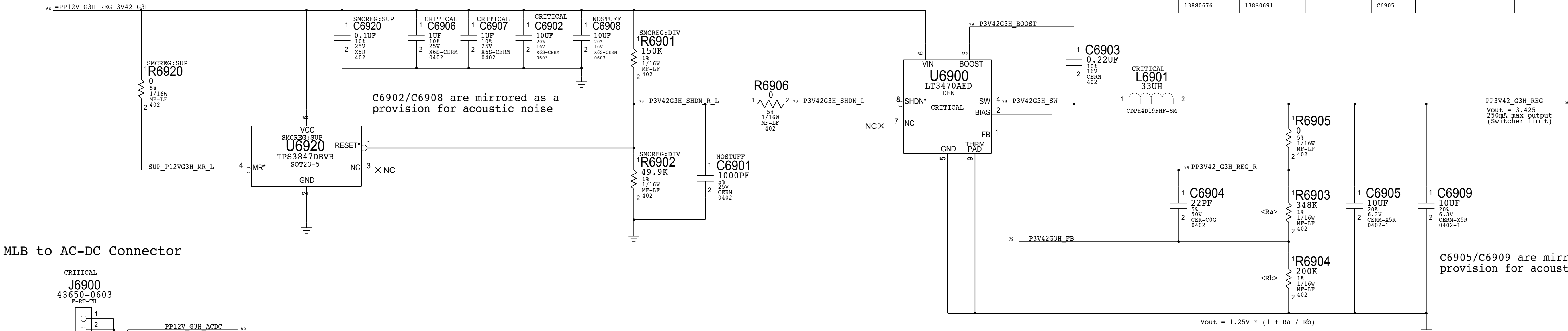


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AUDIO: JACK TRANSLATORS			
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		PAGE	66 OF 123
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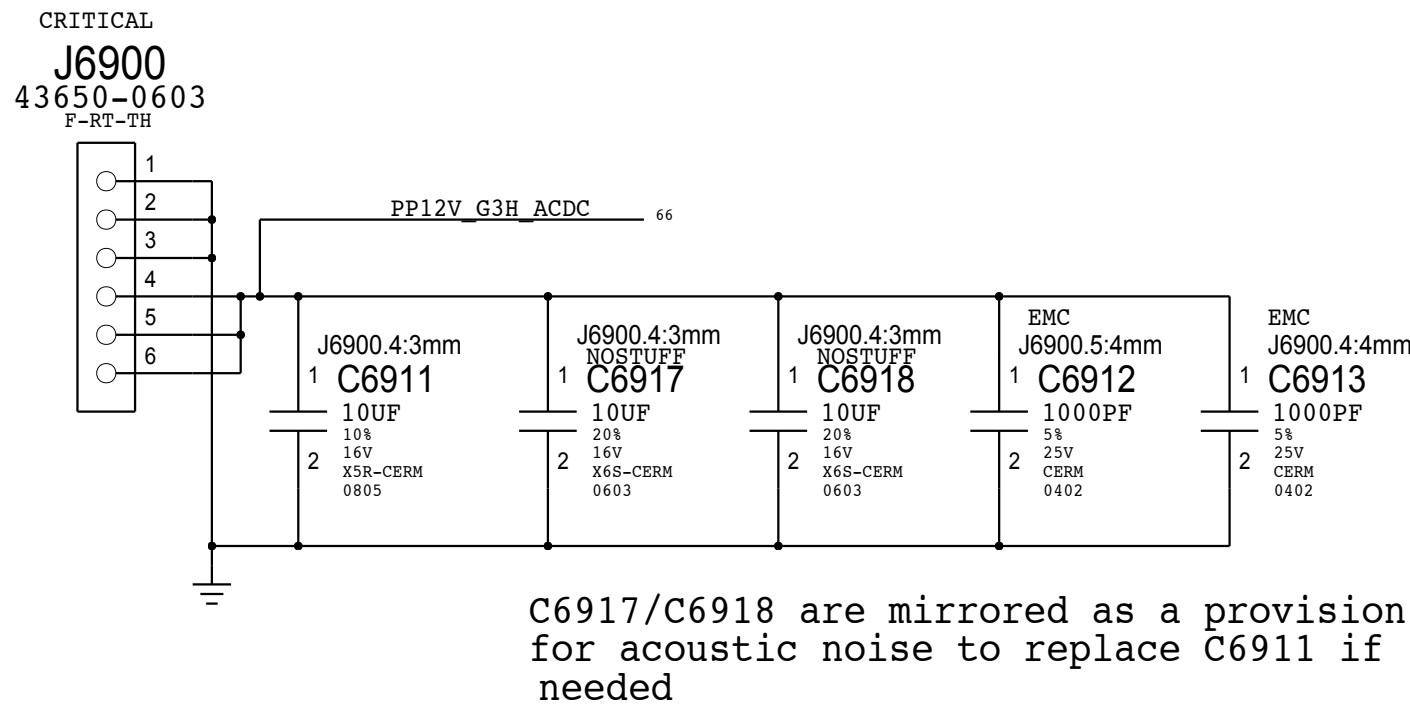
3.425V "G3Hot" Regulator

Switching freq: 409 kHz = $\frac{13.5}{L6901}$

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
138S0676	138S0691		C6905	

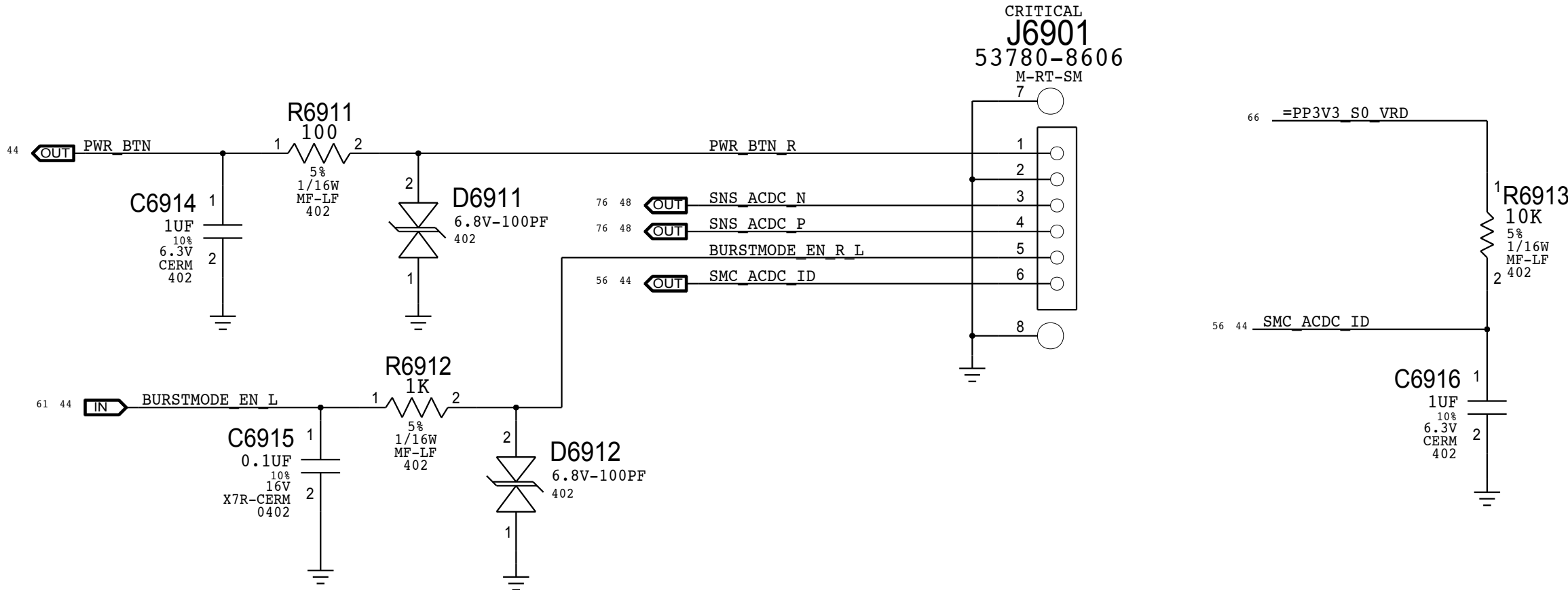


MLB to AC-DC Connector

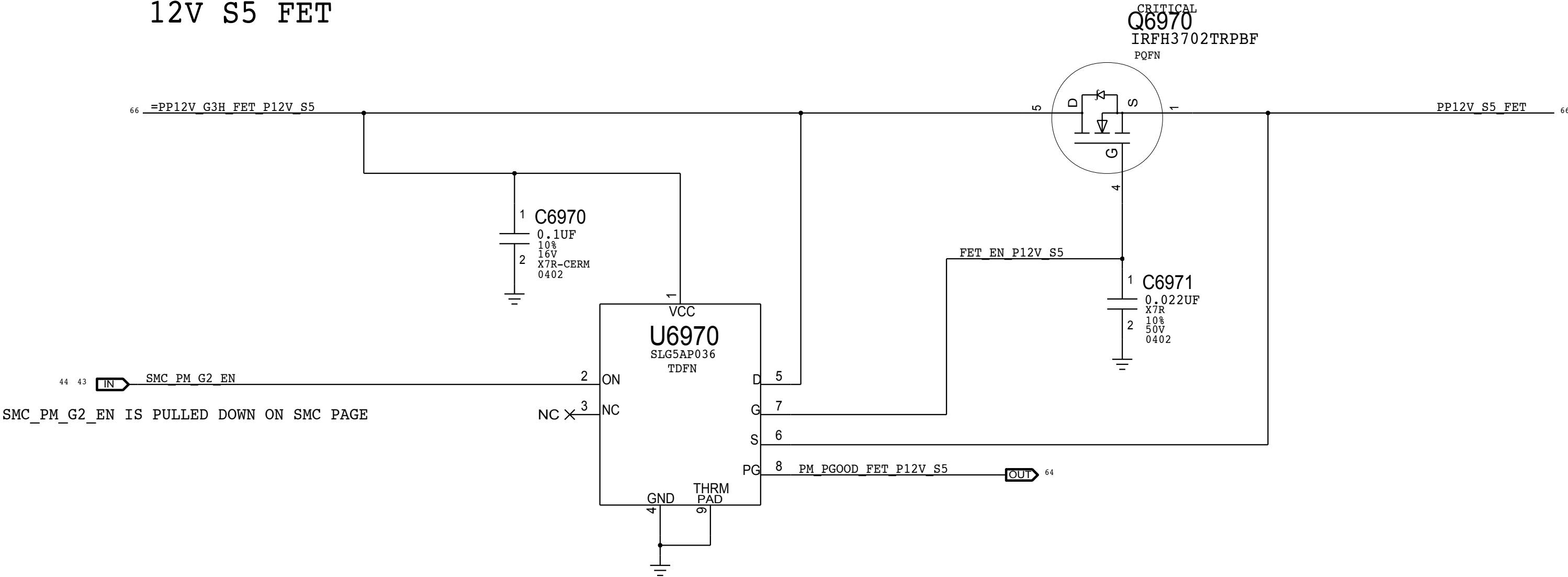


C6917/C6918 are mirrored as a provision for acoustic noise to replace C6911 if needed

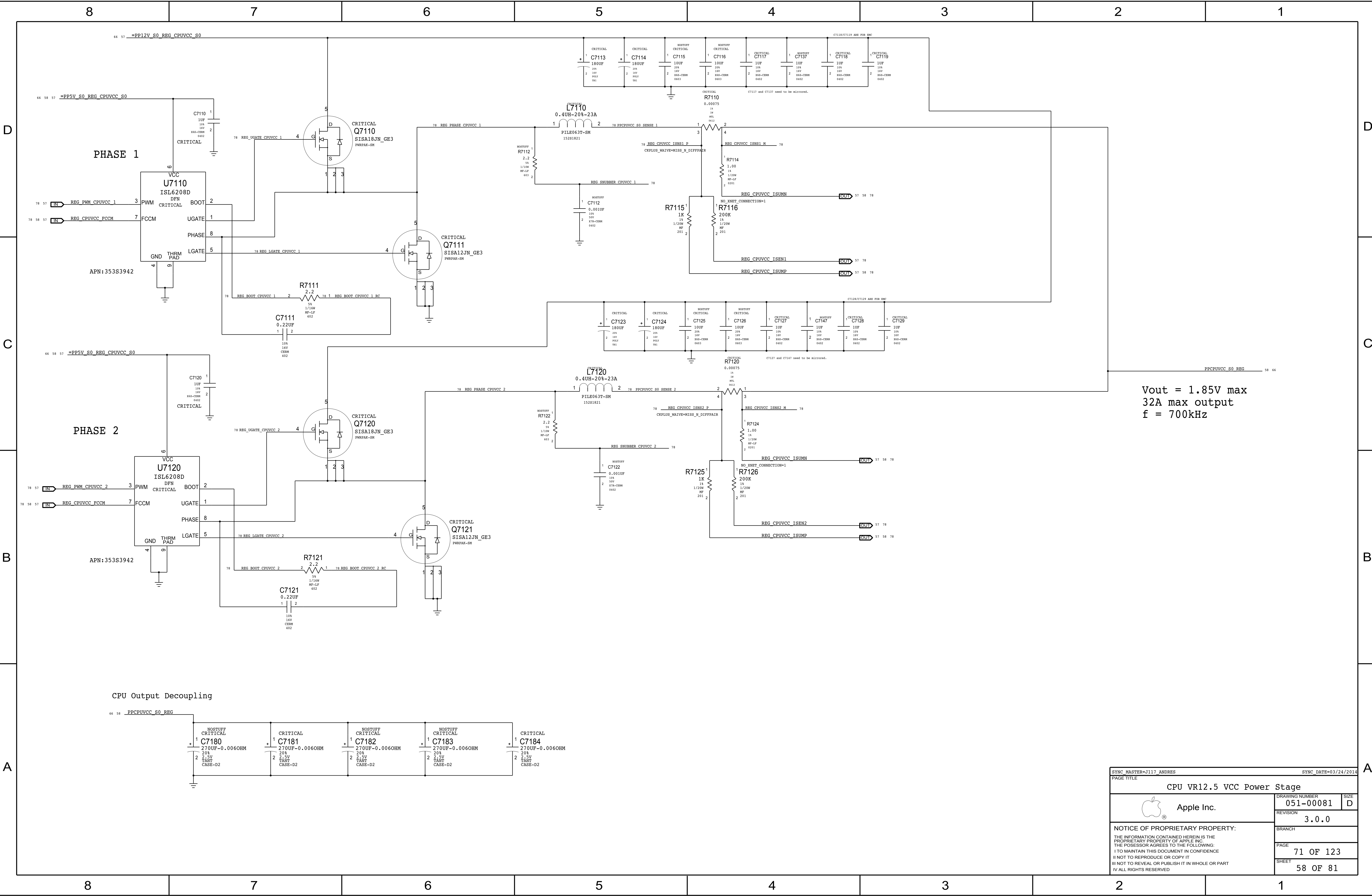
MLB to AC-DC Supplemental Signal Connector



12V S5 FET



SYNC_MASTER=J117_ANDRES		SYNC_DATE=03/24/2014	
PAGE TITLE			
Power Connectors / VReg G3Hot			
	Apple Inc.	DRAWING NUMBER	051-00081
		SIZE	D
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Vout = 1.85V max
32A max output
f = 700kHz

SYNC_MASTER=J117_ANDRES		SYNC_DATE=03/24/2014	
PAGE TITLE			
CPU VR12.5 VCC Power Stage			
 Apple Inc.	DRAWING NUMBER		SIZE
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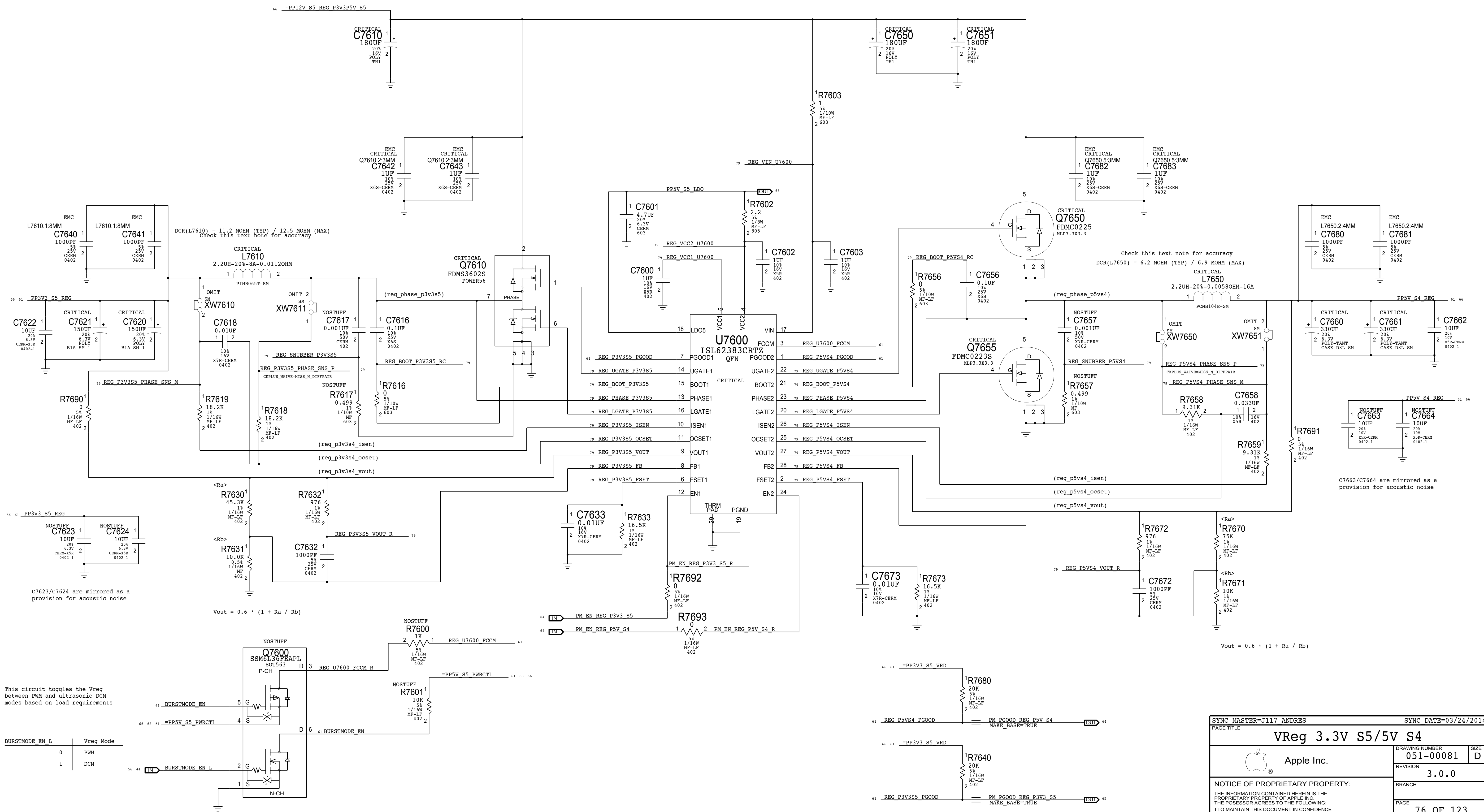
A

3.3V S5 Regulator

OC trip point:	12.5 A =	$\frac{R7618 * 10 \text{ E-6}}{DCR(L7610)}$
Switching freq:	356 kHz =	$\frac{1}{170 \text{ E-12} * R7633}$

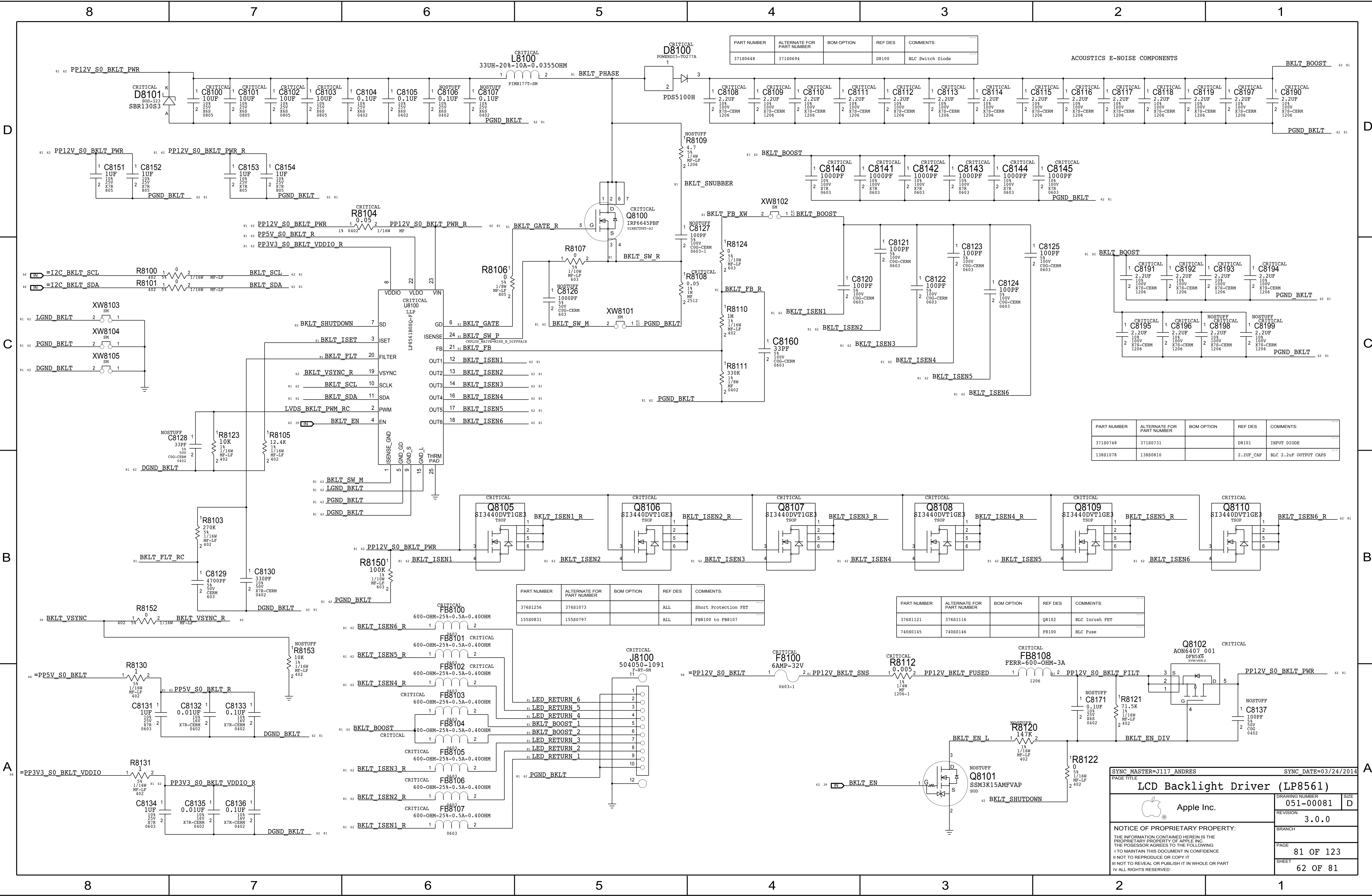
5V S4 Regulator

OC trip point:	14.1 A =	$\frac{R7658 * 10 \text{ E-6}}{\text{DCR(L7650)}}$
Switching freq:	356 kHz =	$\frac{1}{170 \text{ E-12} * R7673}$



BURSTMODE_EN_L	Vreg Mode
0	PWM
1	DCM

SYNC MASTER=J117 ANDRES		SYNC DATE=03/24/2014	
PAGE TITLE			
VReg 3.3V S5/5V S4			
 Apple Inc.	DRAWING NUMBER		SIZE
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PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
371S0648	371S0694		D8100	BLC Switch Diode

ACOUSTICS E-NOISE COMPONENTS

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
371S0748	371S0731		D8101	INPUT DIODE
138S1078	138S0810		2.2UF_CAP	BLC 2.2uF OUTPUT CAPS

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
376S1256	376S1073		ALL	Short Protection FET
155S0831	155S0797		ALL	FB8100 to FB8107

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
376S1121	376S1116		Q8102	BLC Inrush FET
740S0145	740S0146		FB100	BLC Fuse

SYNC_MASTER=J117 ANDRES
PAGE TITLE

LCD Backlight Driver (LP8561)

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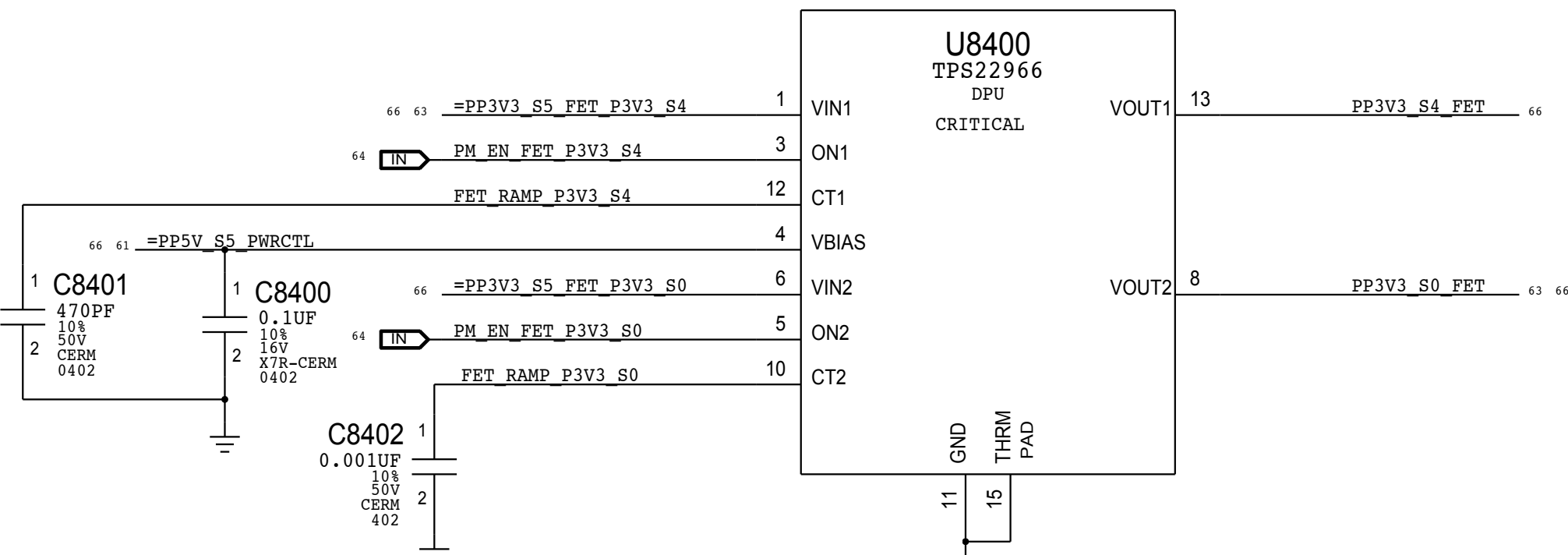
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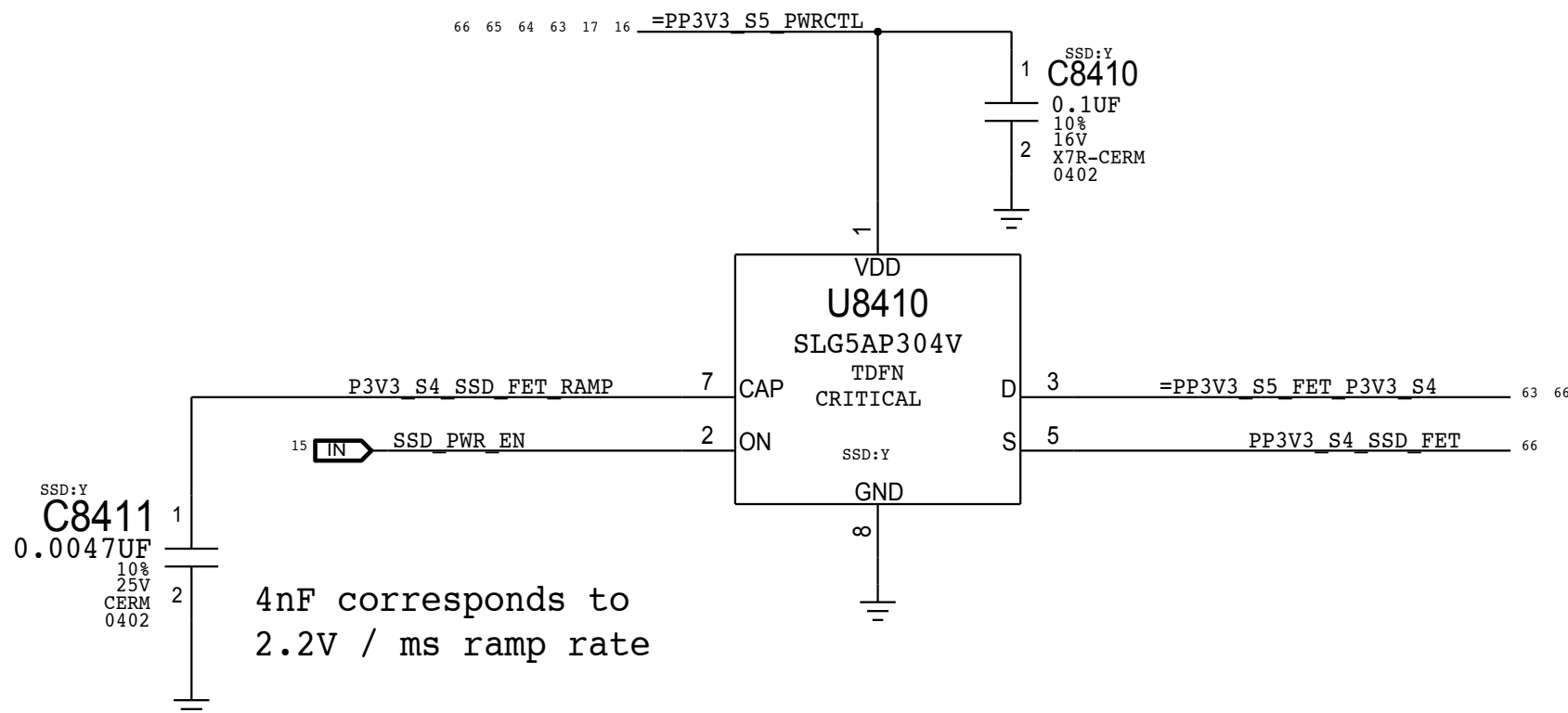
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3.3V S4/S0 FET

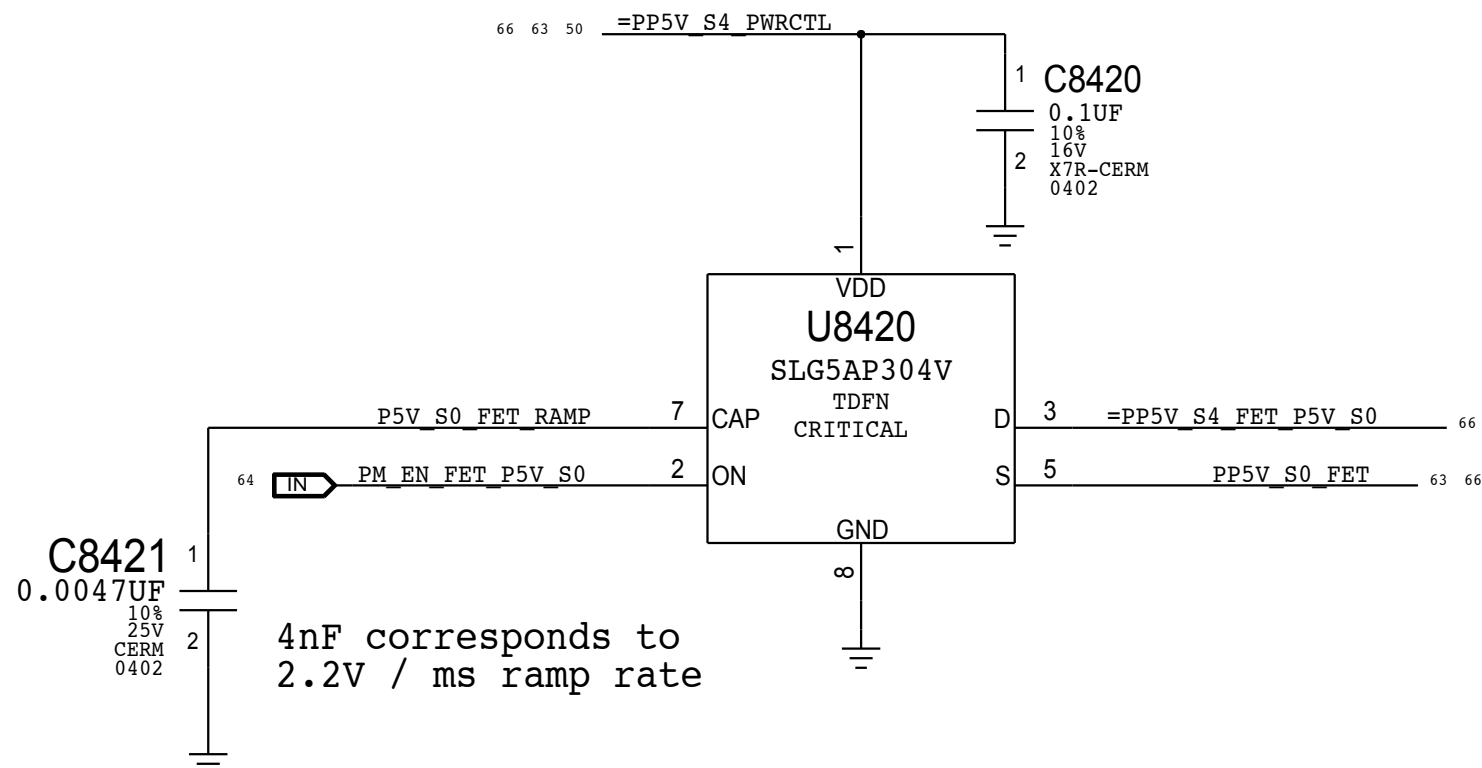


Rise Time For VD = 3.3V:
470 pF -- 603 us
1000 pF -- 1185 us

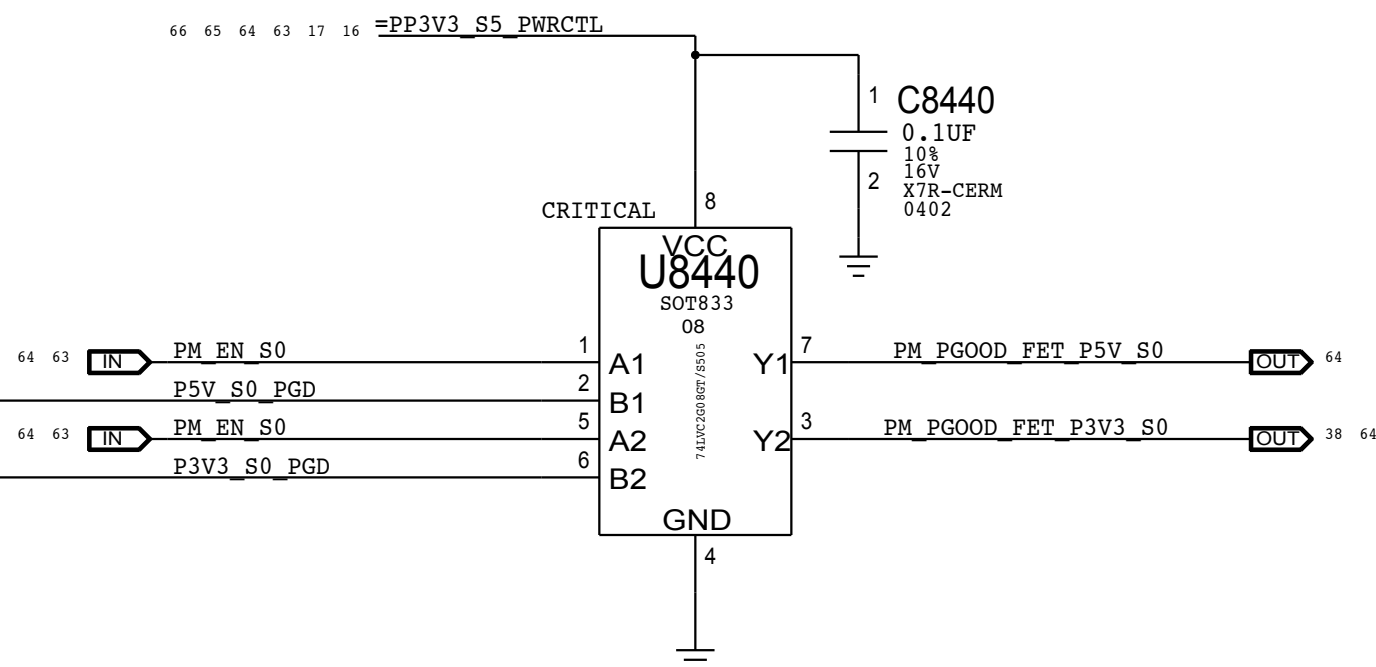
3V3 S4 SSD



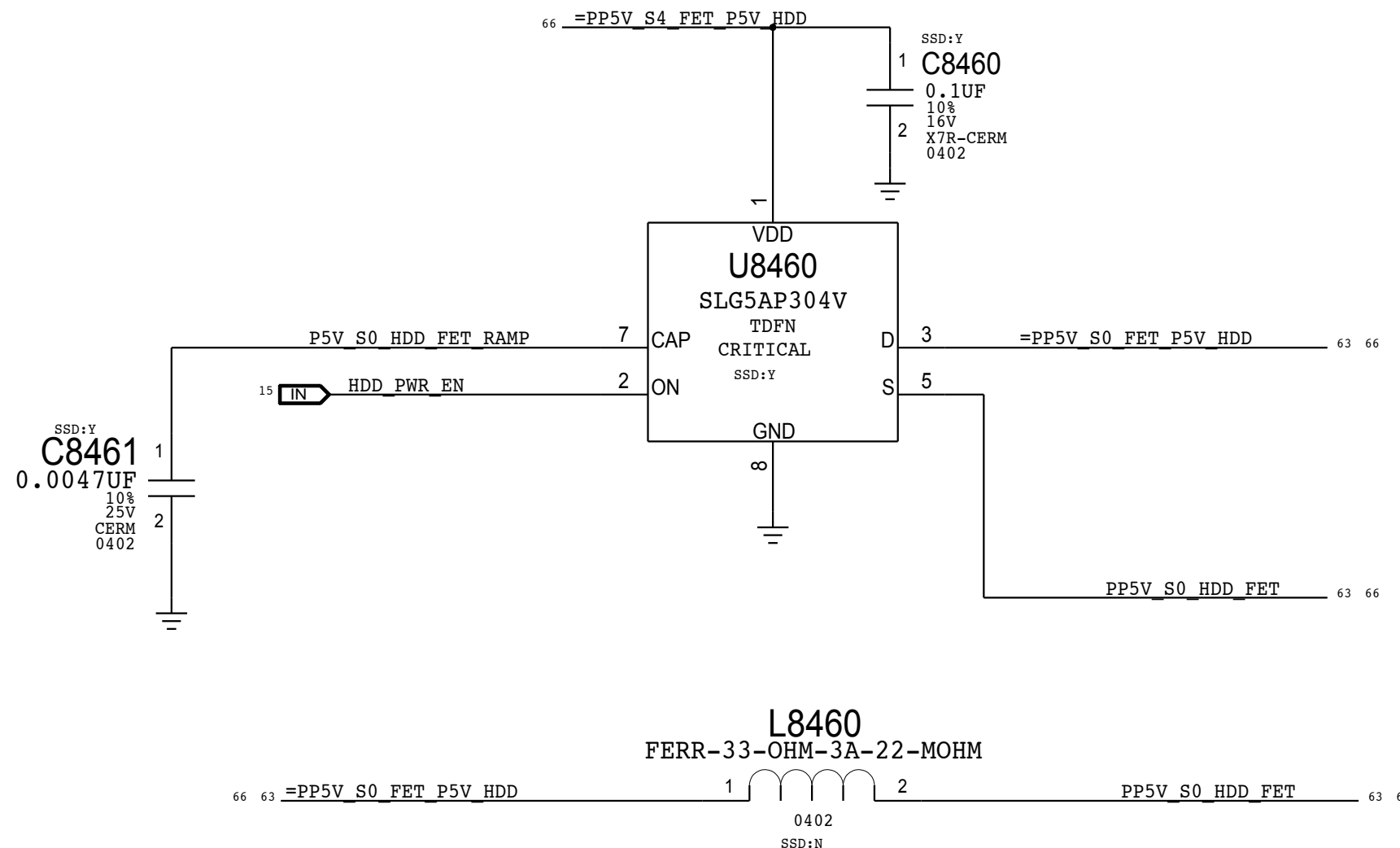
5V S0 FET



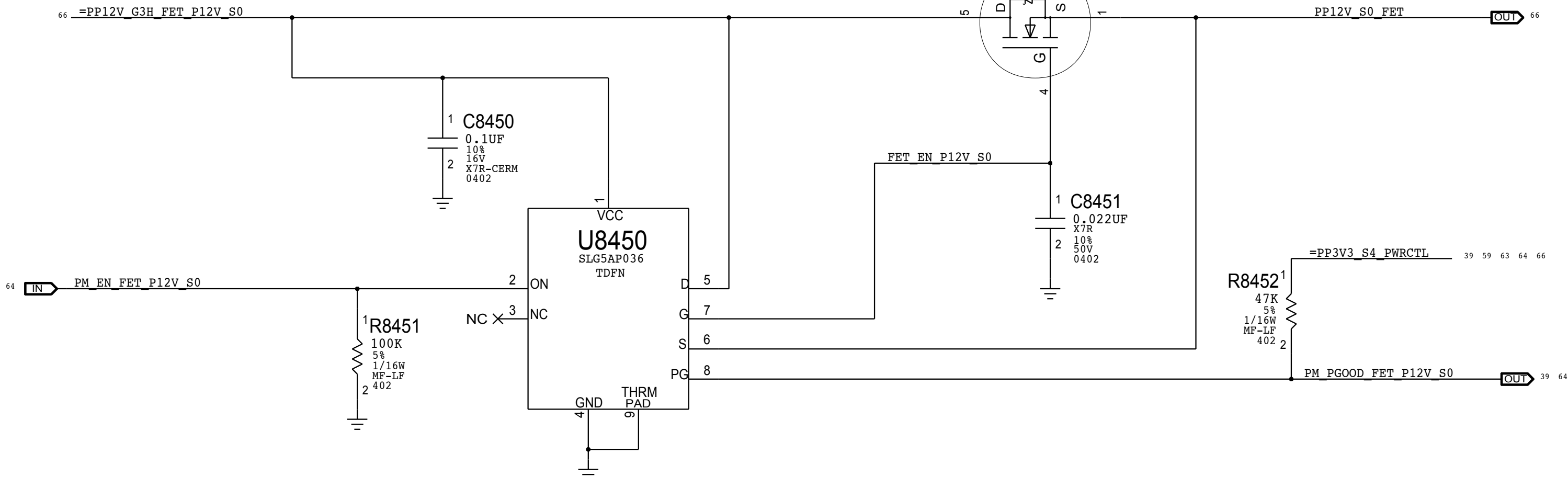
5V / 3V3 S0 PGOODs



5V HDD FET

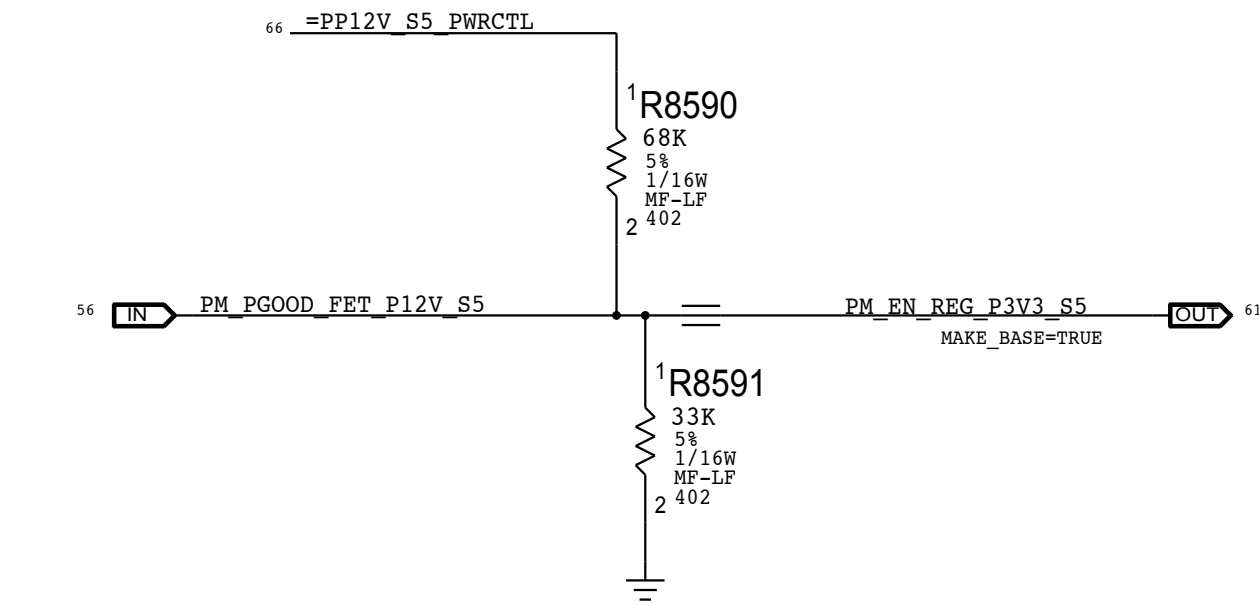


12V S0 FET

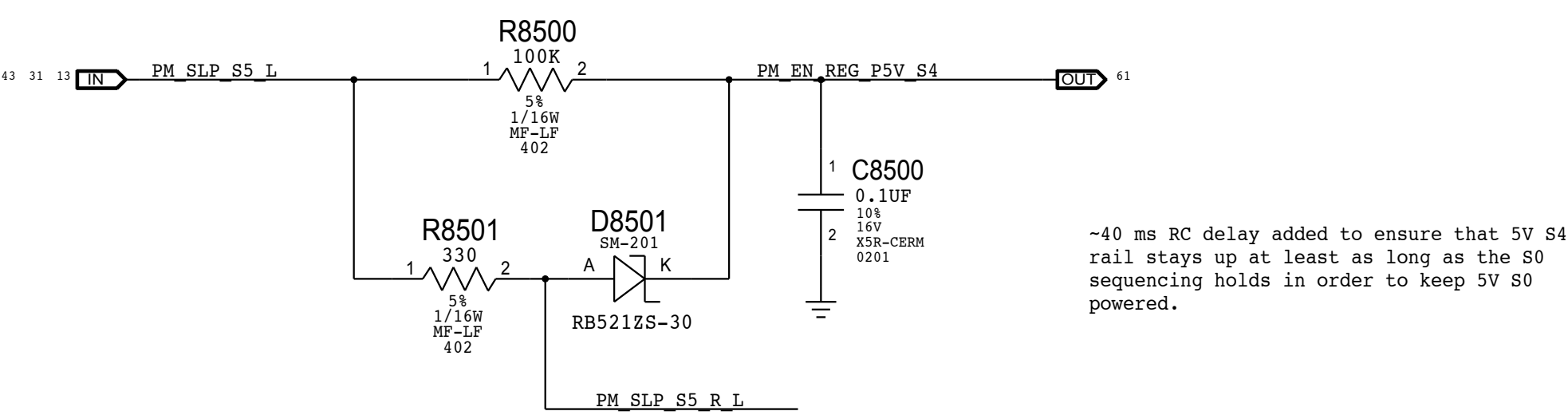


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PAGE TITLE			
PM FETs/LDOs			
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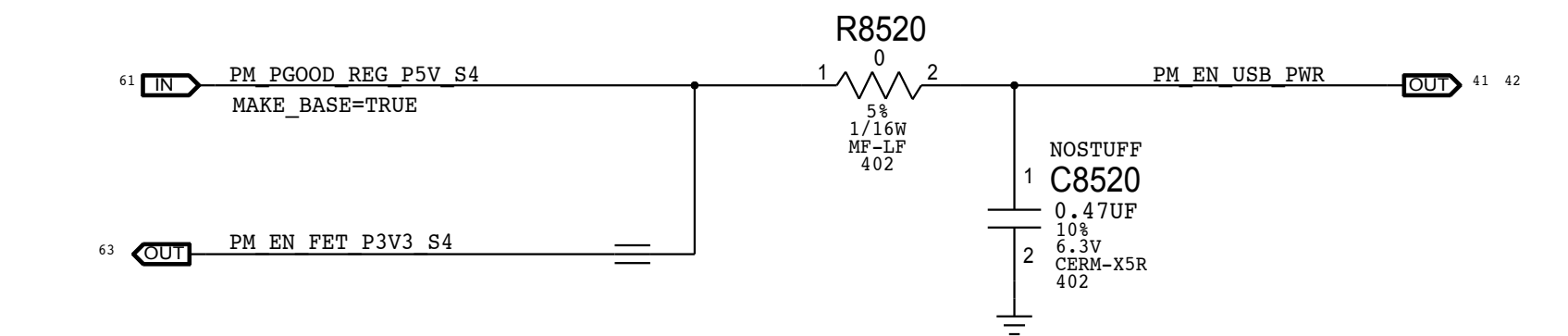
S5 Enable



S4 Enables



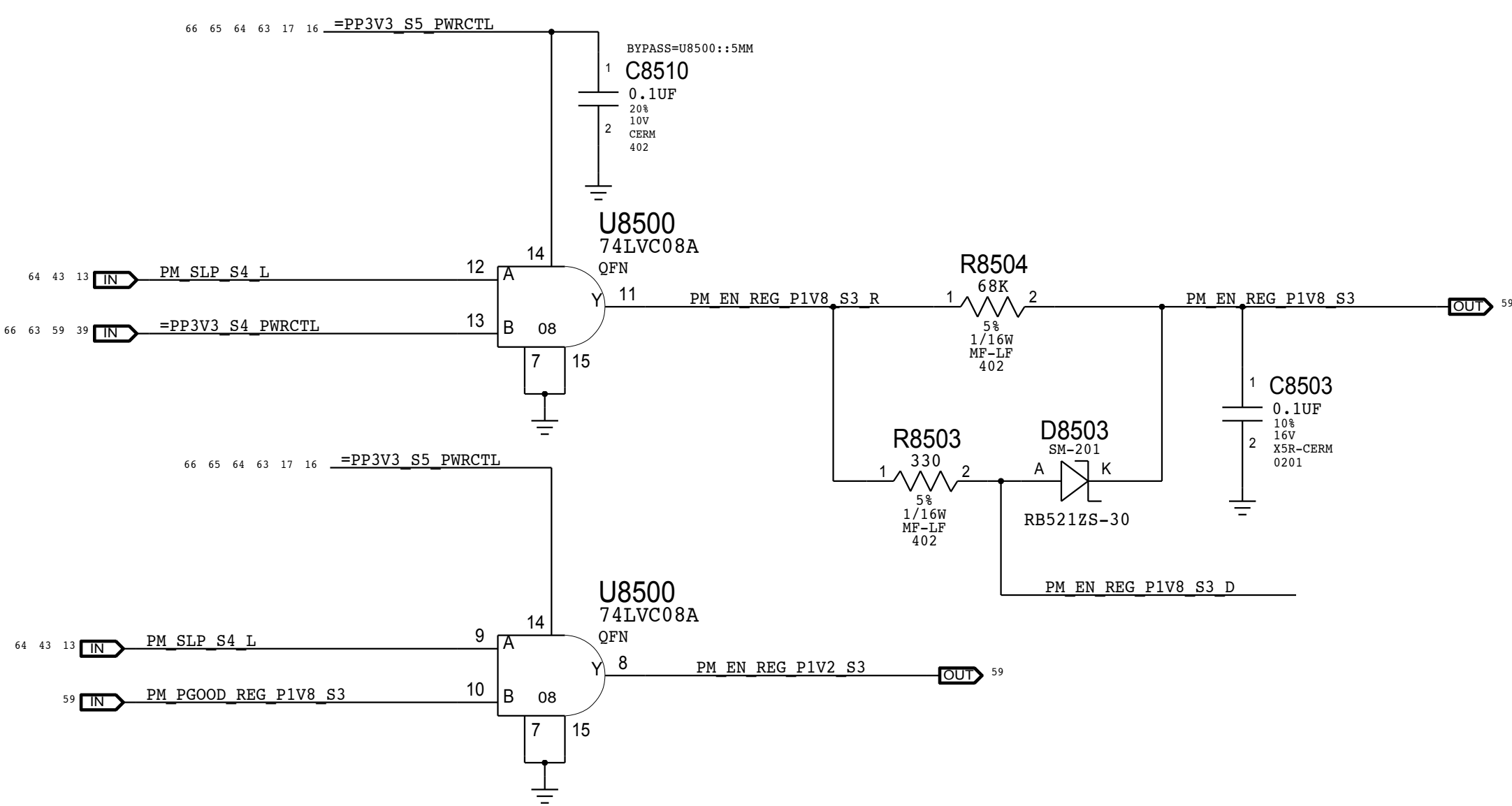
S4 USB Enable



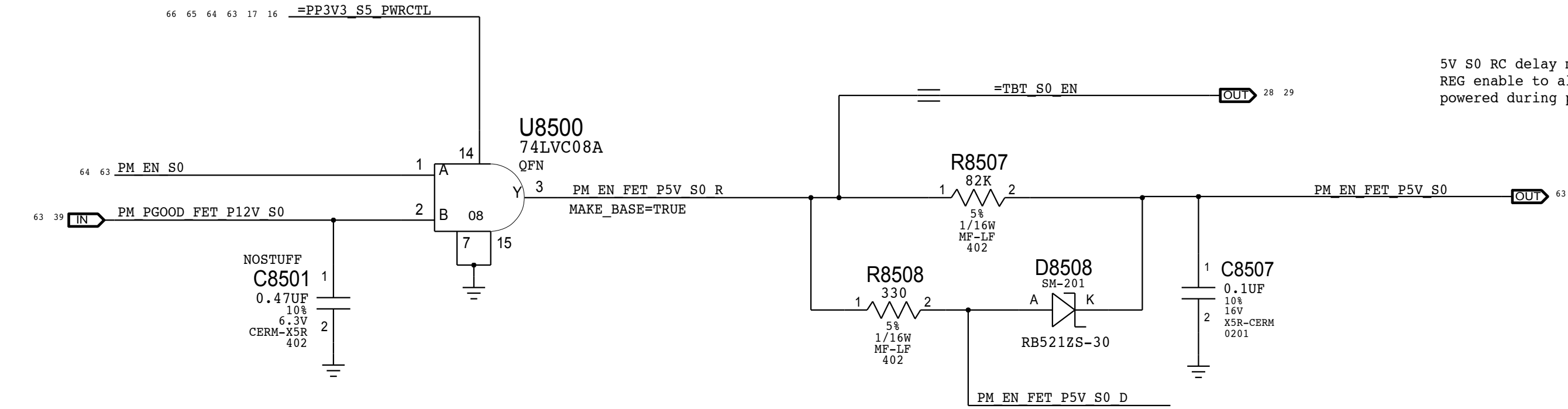
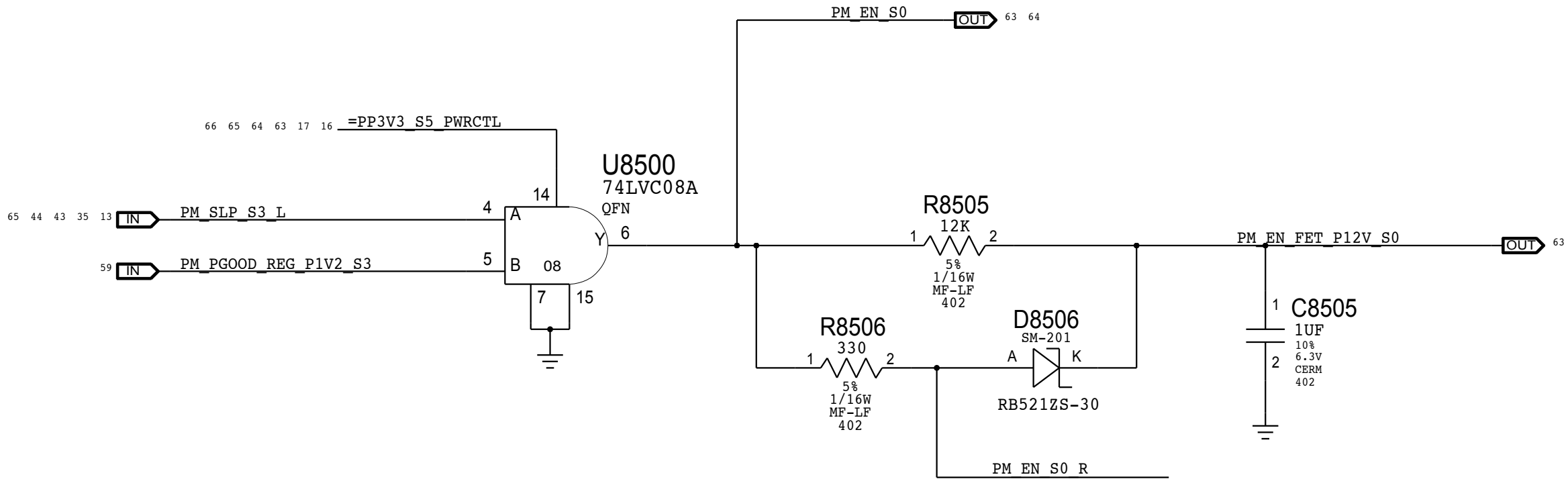
S4 TBT Port Enable



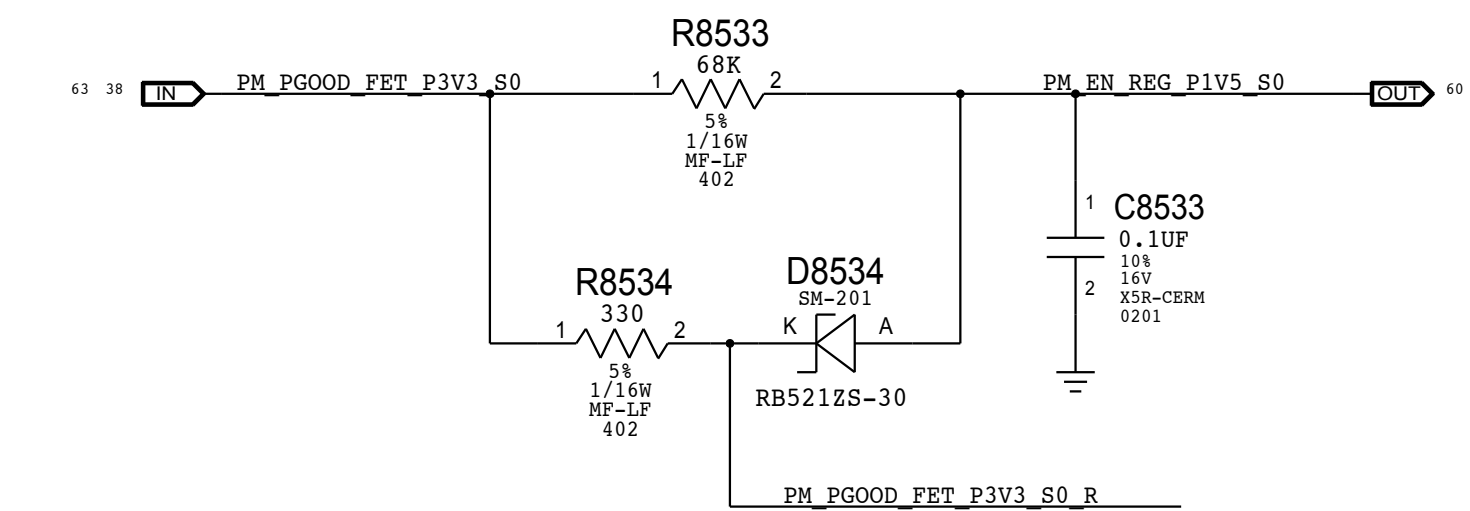
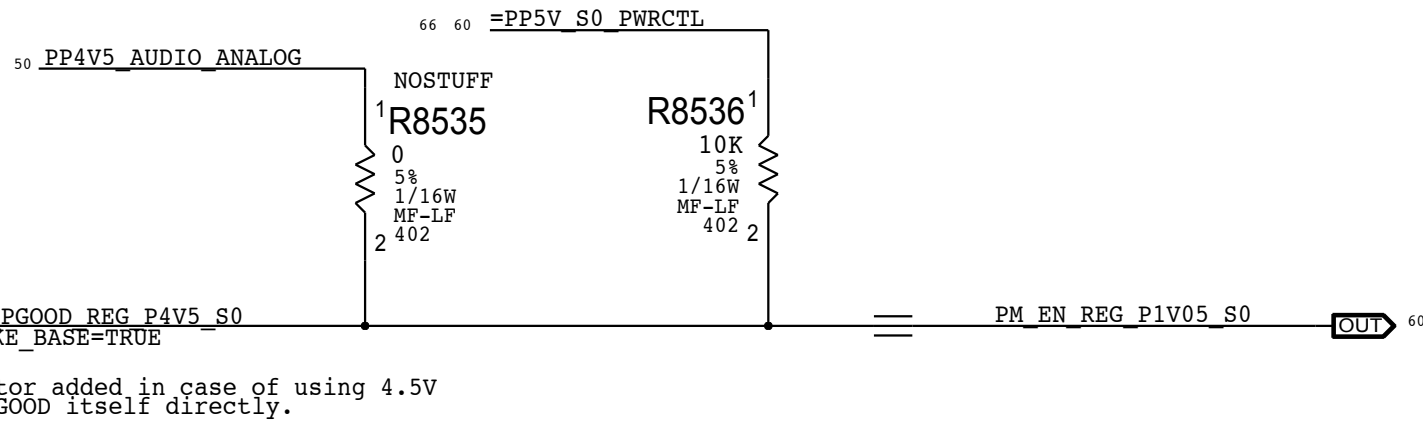
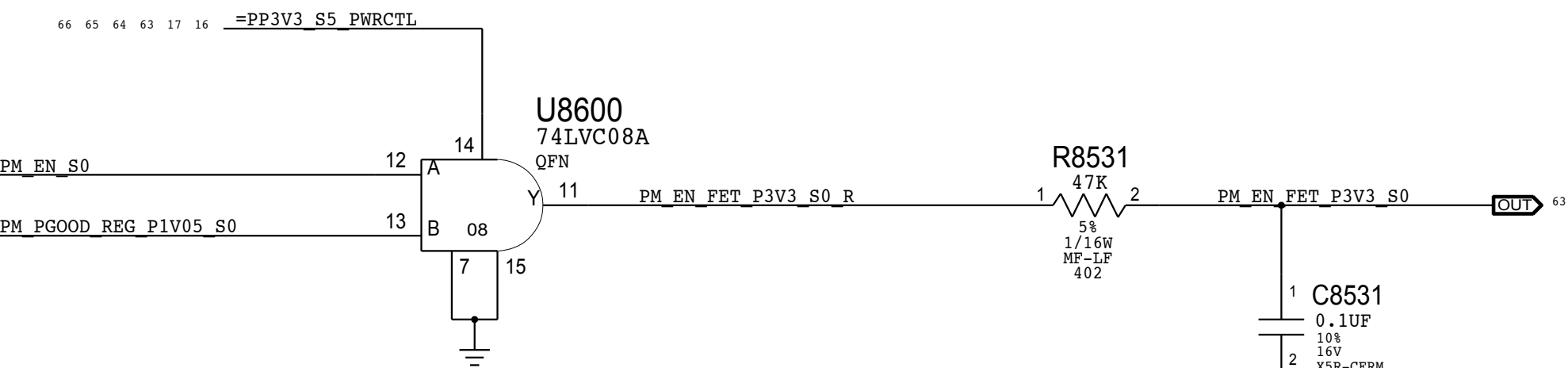
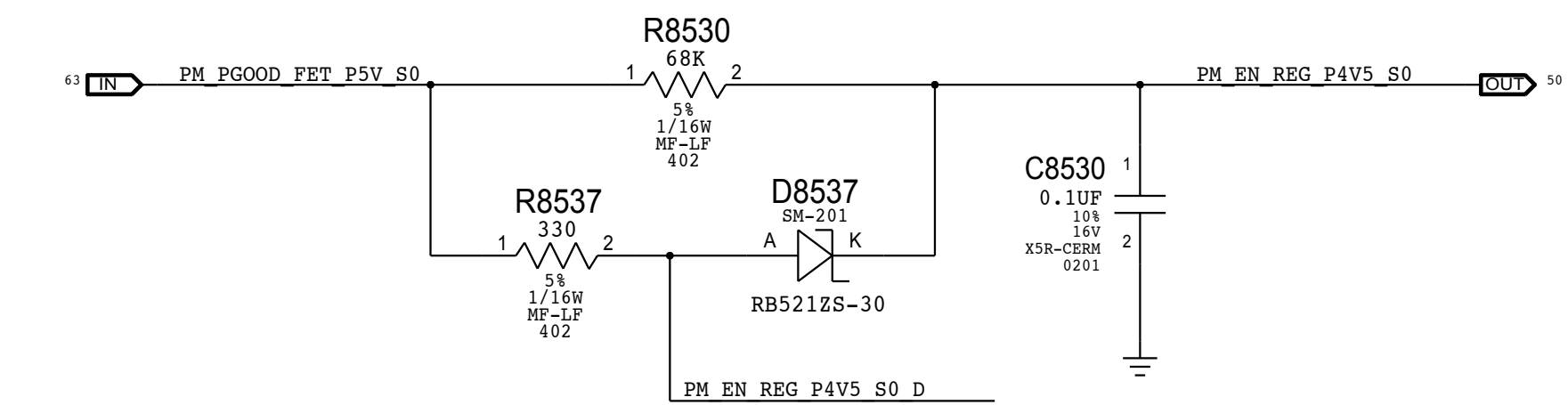
S3 Enables

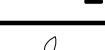


S0 Enables

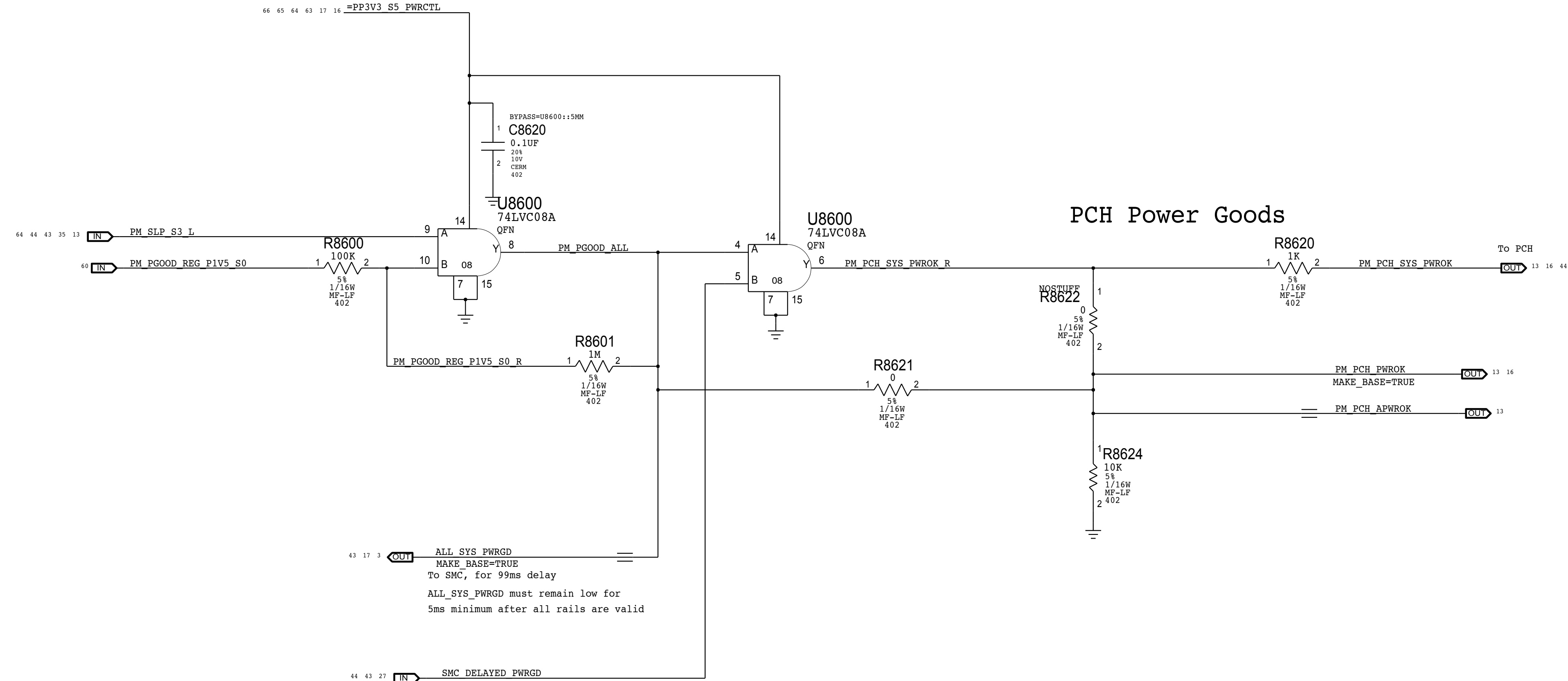


Audio + PCH Sequencing Requirements:
4.5V -> 1.05V -> 3.3V -> 1.5V -> ALL SYS GOOD



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PM Regulator Enables			
 Apple Inc.		DRAWING NUMBER	051-00081
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ALL_SYS_PWRGD,PCH_PWROK & SYS_PWROK Generation



Rail definitions

Platform: All processor non-Core and non-Graphics (5V, 3.3V, 1.5V, 1.05V for PCH/TBT/GPU)
Uncore: 1.8V and 1.2V for DDR3

Notes on sequencing requirements

- Intel:
1. No hard specification on platform rails
 2. SMC guarantees timing on PCH DPWROK and PWROK
 3. VCC3_3 may power up before VCC, VCC must ramp to 0.6V within 25ms of VCC3V3 ramping to 2.6V
 4. VCC1_5 may power up before VCC, VCC must ramp to 0.6V within 25ms of VCC1V5 ramping to 1.35V
 5. VCC may power down before VCC3_3, VCC3_3 must ramp down to 2.6V within 35ms
 6. VCC may power down before VCC1_5, VCC1_5 must ramp down to 1.35V within 35ms

Resume Reset

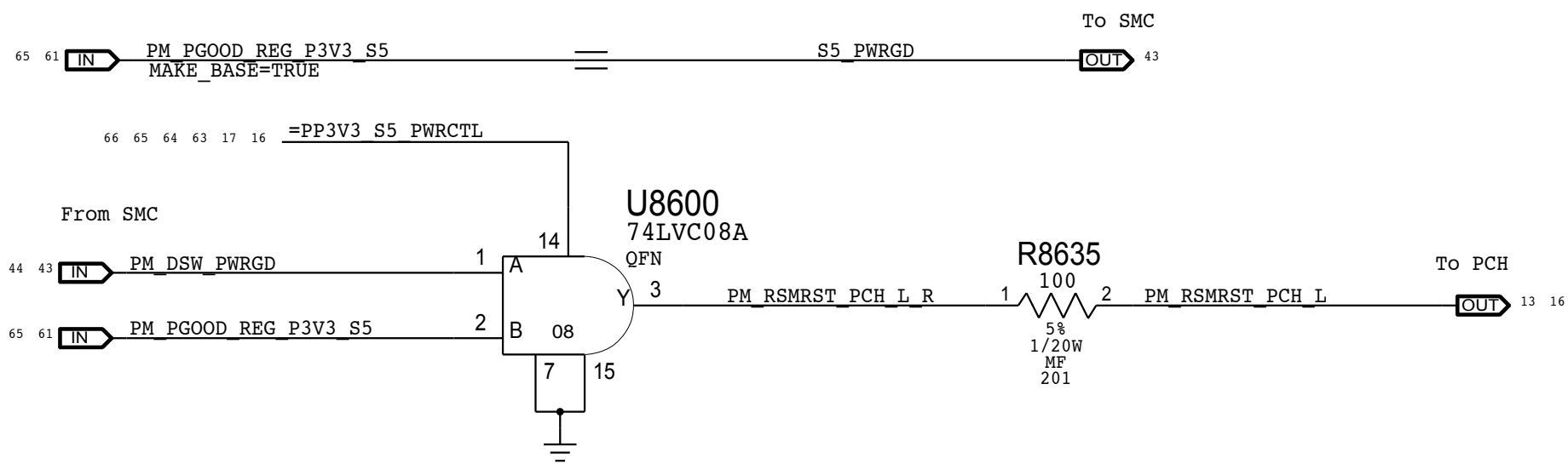
Intel Doc# 29517 Maho Bay PDG, Section 22.13
Intel Doc# 29562 Panther Point EDS, Section 8.7 and 8.8

- Note:
- The iMac J70 design does not support Deep Sx modes so both DPWROK and RSMRST# signals are shorted together
- Requirements:
- Power on:
- Asserted at least 10 ms after all suspend well power is valid
- Power off or loss of AC:
- Transition to 0.0V or less before VccSUS3_3 drops to 2.90 V to allow PCH to switch suspend well to battery without excessive loading

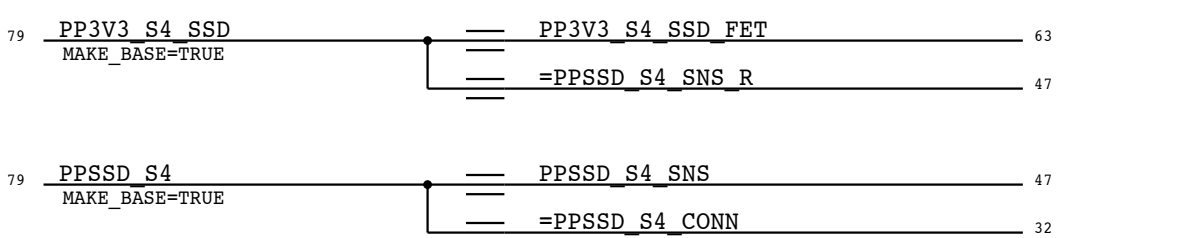
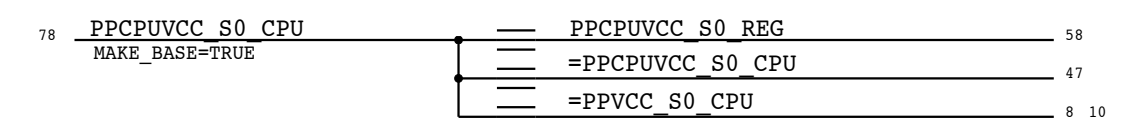
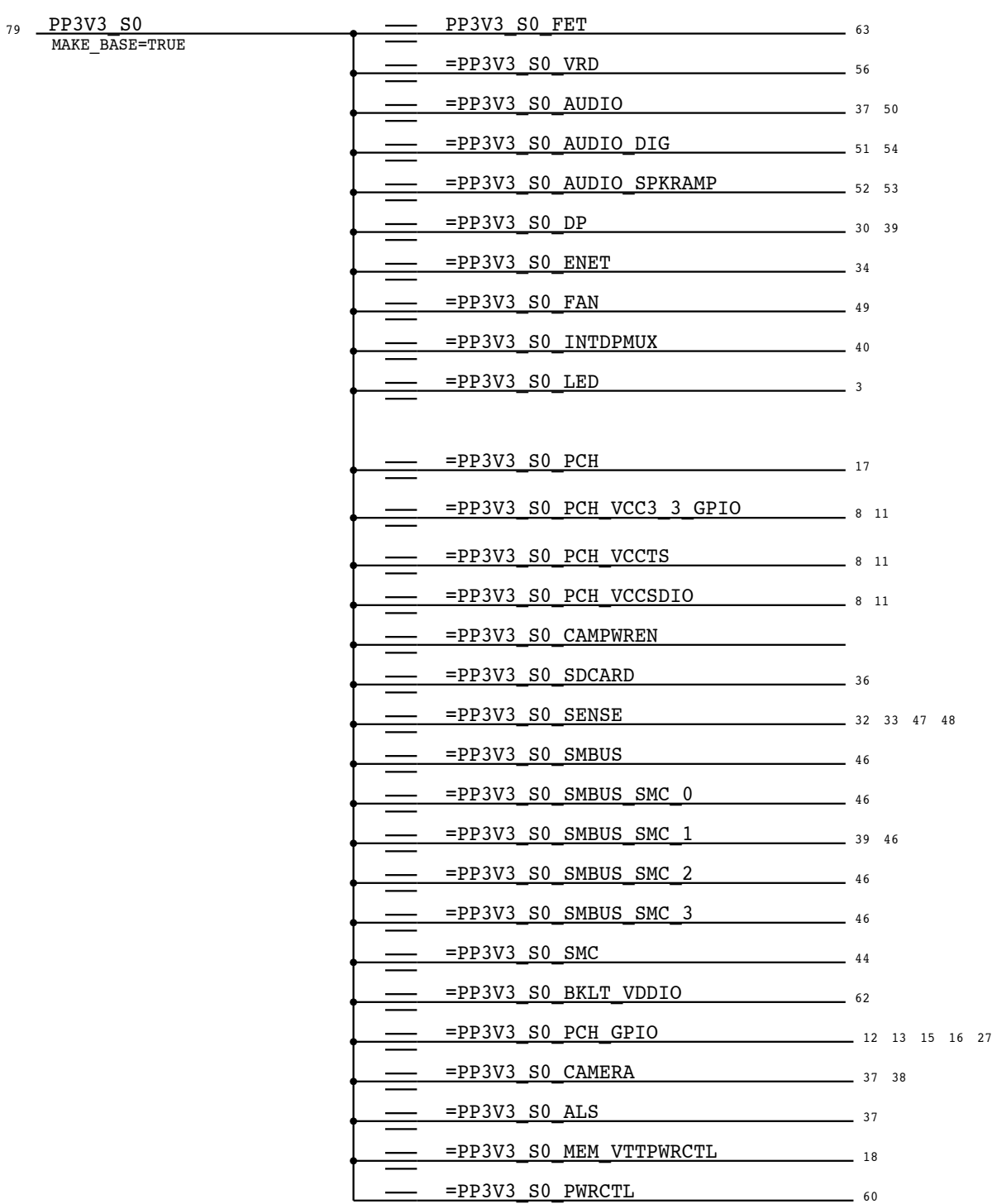
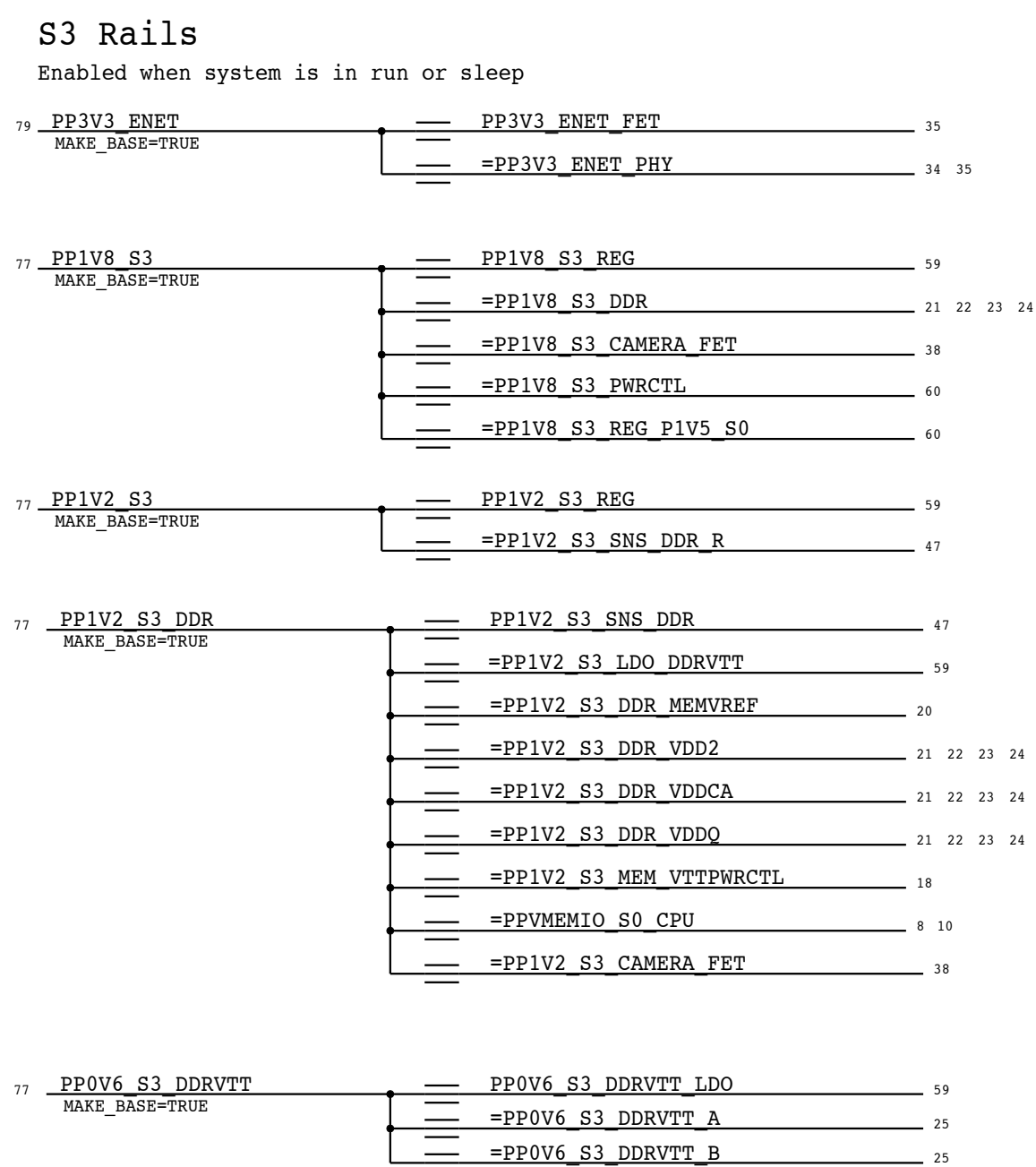
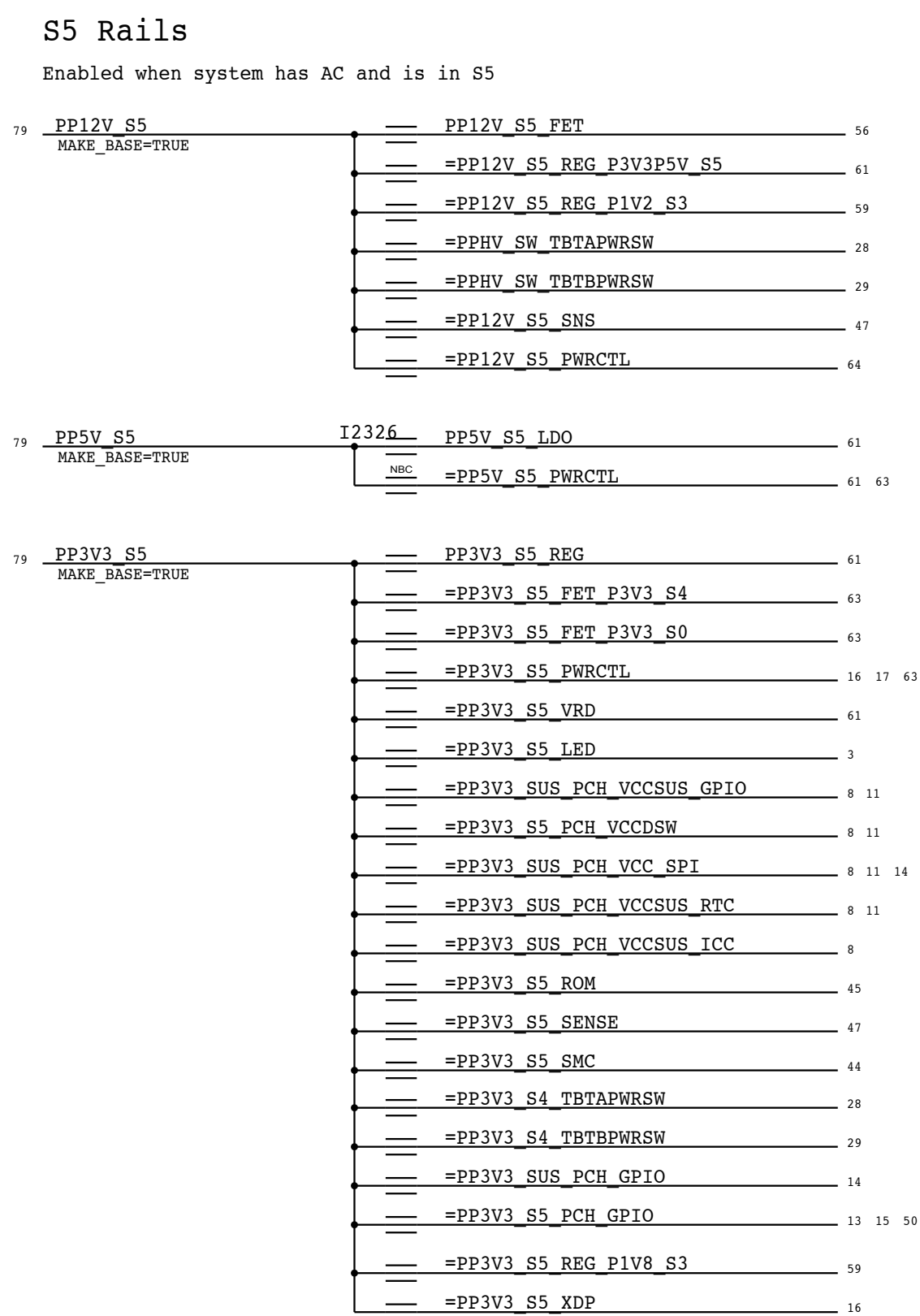
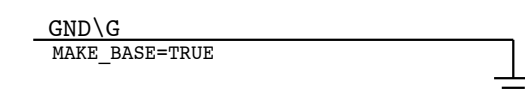
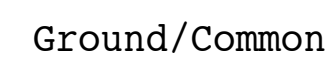
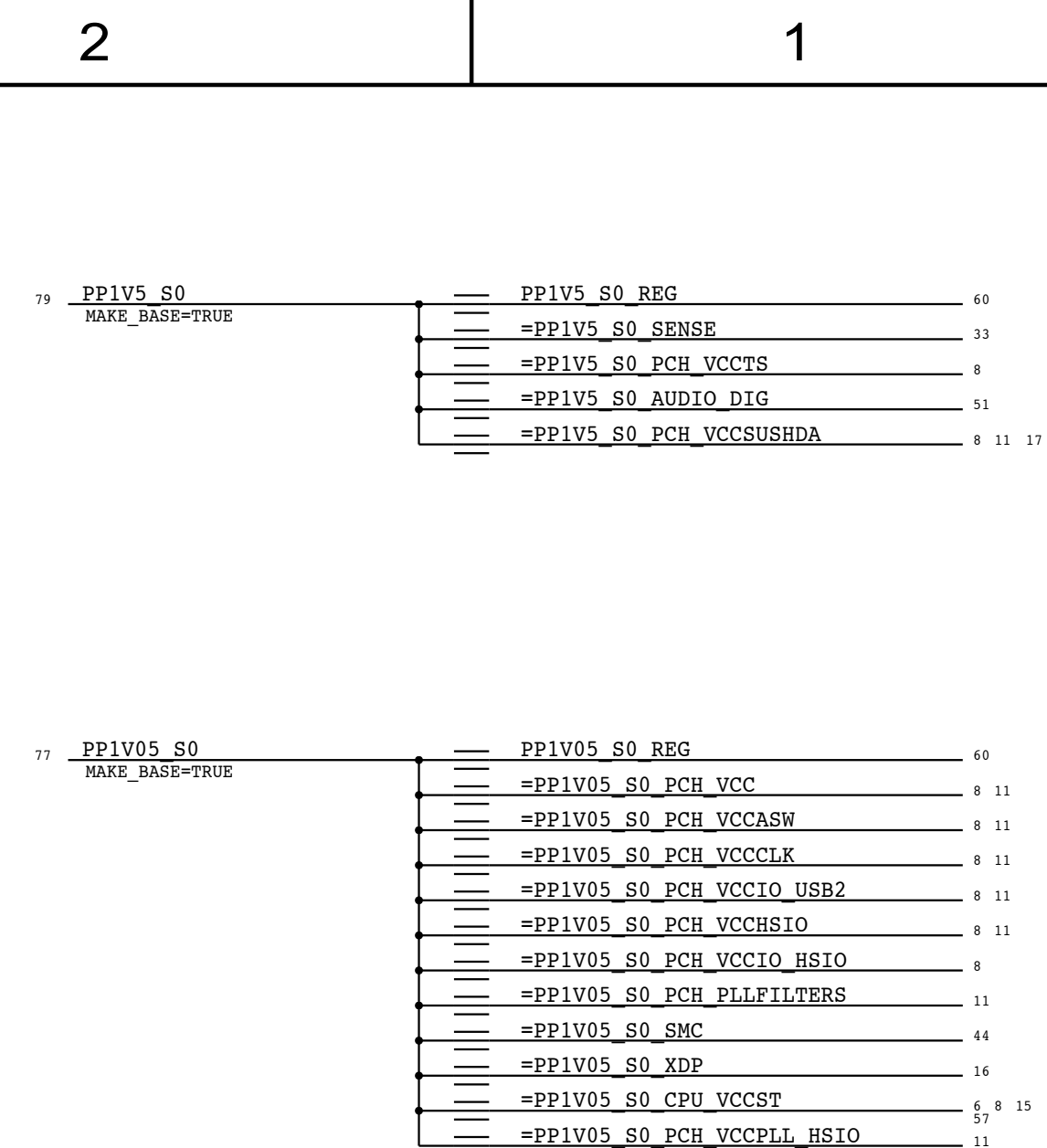
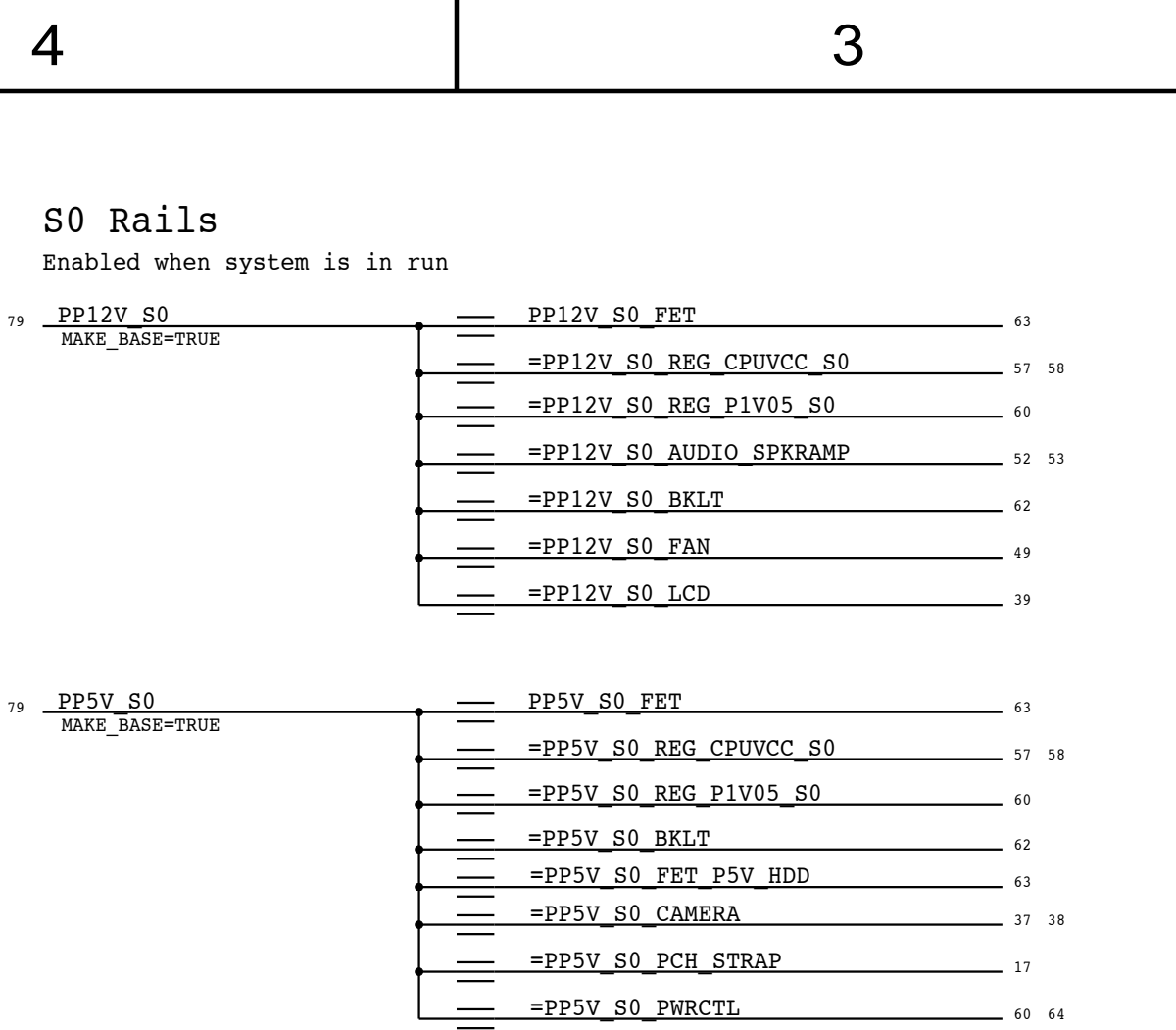
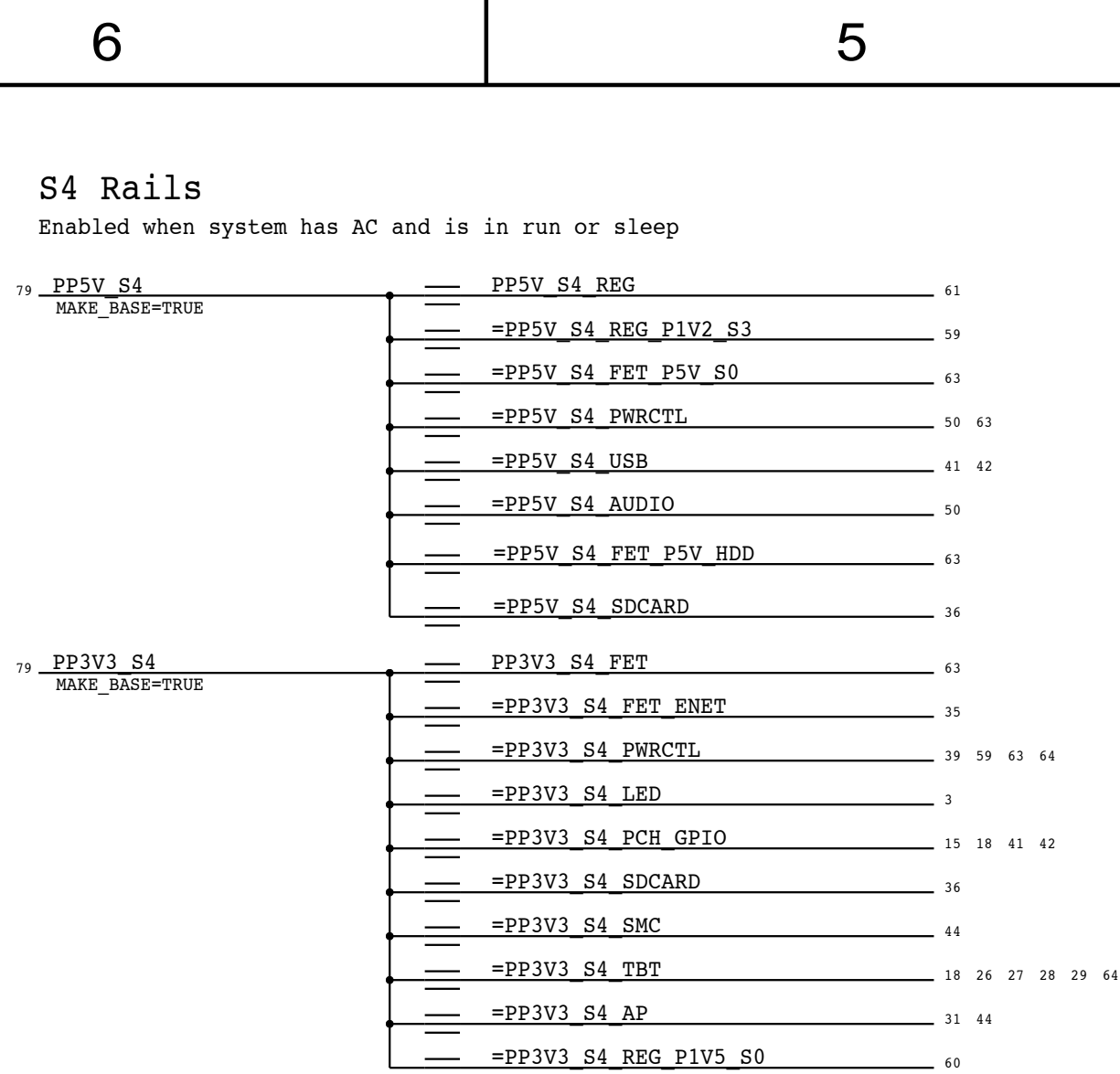
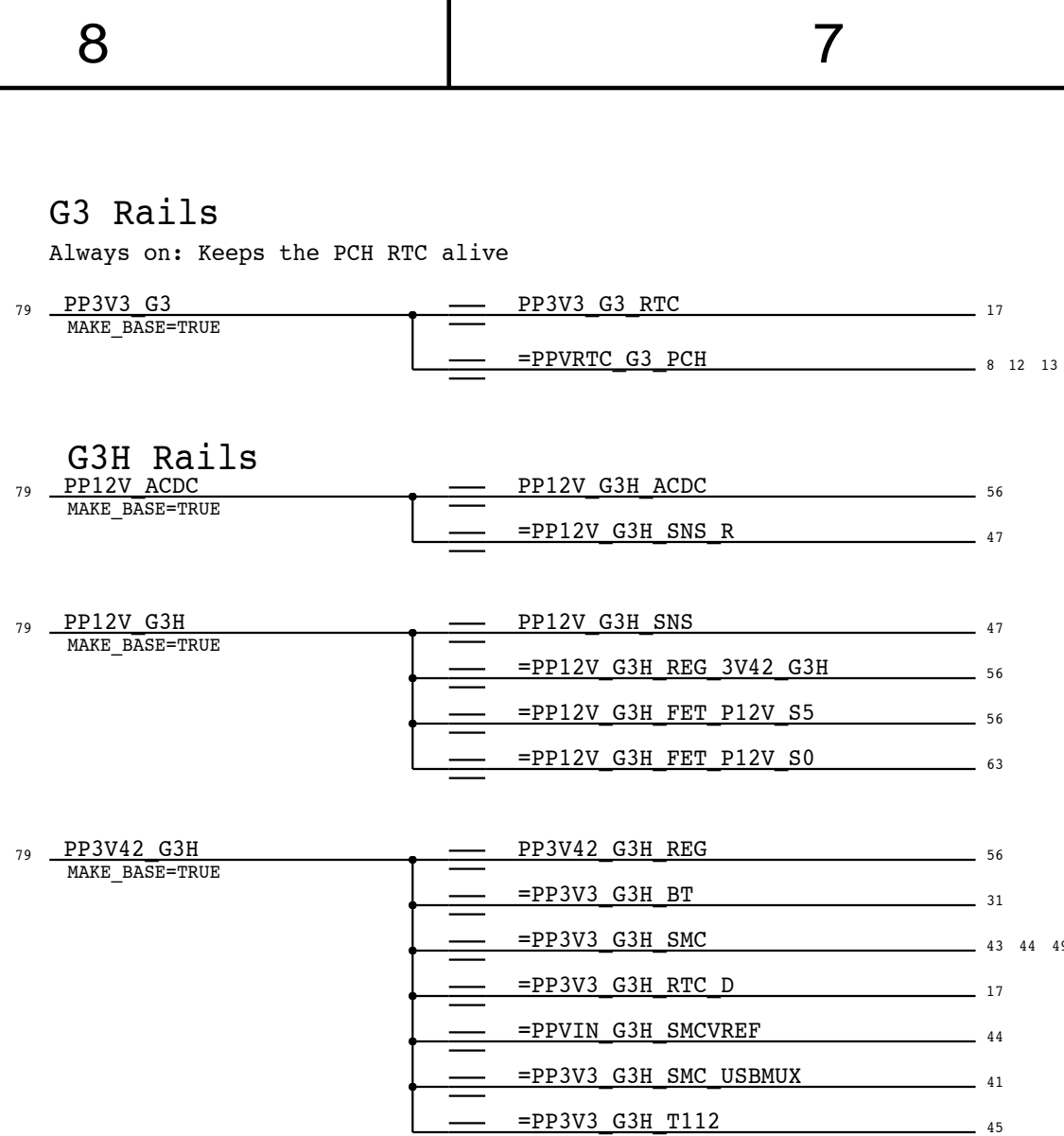
Method:

The SMC guarantees proper assertion and de-assertion of RSMRST# for normal operation via PM_DS_W_PWRGD.

RSMRST# is asserted when power good from regulator is de-asserted in the event AC is lost. Power good de-assertion should happen quickly enough to meet Intel spec.



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PM Power Good			
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D	PCH Miscellaneous 12 TP_HDA_SDIN1 == NC_HDA_SDIN1 MAKE_BASE=TRUE NO_TEST=1 12 TP_PCIE_CLK100M_CAMERAP == NC_PCIE_CLK100M_CAMERAP MAKE_BASE=TRUE NO_TEST=1 12 TP_PCIE_CLK100M_CAMERAN == NC_PCIE_CLK100M_CAMERAN MAKE_BASE=TRUE NO_TEST=1 12 TP_PCIE_CLK100M_FWP == NC_PCIE_CLK100M_FWP MAKE_BASE=TRUE NO_TEST=1 12 TP_PCIE_CLK100M_FWN == NC_PCIE_CLK100M_FWN MAKE_BASE=TRUE NO_TEST=1 Unused Thunderbolt Aliases 24 TP_TBT_PCIE_RESETO_L == NC_TBT_PCIE_RESETO_L MAKE_BASE=TRUE NO_TEST=1														
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Unused Signal Aliases



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DDR3

DDR3-specific Physical Rules

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
DDR_40S	*	=40_OHM_SE	=40_OHM_SE	=40_OHM_SE	=40_OHM_SE	=STANDARD	=STANDARD
DDR_50S	*	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=STANDARD	=STANDARD
DDR_70D	*	=70_OHM_DIFF	=70_OHM_DIFF	=70_OHM_DIFF	=70_OHM_DIFF	=70_OHM_DIFF	=70_OHM_DIFF
DDR_73D	*	=73_OHM_DIFF	=73_OHM_DIFF	=73_OHM_DIFF	=73_OHM_DIFF	=73_OHM_DIFF	=73_OHM_DIFF
DDR_COMP	*	Y	0.305 MM	0.105 MM	=STANDARD	=STANDARD	=STANDARD

Minimum diff spacing is 4 mil
Table 4-5, Intel Doc# 486712

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
POWER_DDR_P4MM	*	Y	0.400 MM	0.100 MM	3.0 MM	=STANDARD	=STANDARD

Physical Net Type to Rule Map

NET_PHYSICAL_TYPE	AREA_TYPE	PHYSICAL_RULE_SET
POWER_DDR	*	POWER_DDR_P4MM
DDR_CLK_PHY	*	DDR_70D
DDR_CTRL_PHY	*	DDR_40S
DDR_CMD_PHY	*	DDR_40S
DDR_DQ_PHY	*	DDR_40S
DDR_DQS_PHY	*	DDR_70D
DDR_COMP_PHY	*	DDR_COMP

DDR3 Power-specific Spacing Definitions

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
POWER_DDR	*	=2:1_SPACING	?

DDR3-specific Spacing Definitions

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
DDR_CLK_ISO	*	=5:1_SPACING	?
DDR_CTRL_ISO	*	=3.5:1_SPACING	?
DDR_CTRL2CTRL	*	=2:1_SPACING	?
DDR_CMD_ISO	*	=3.5:1_SPACING	?
DDR_CMD2CMD	*	=2:1_SPACING	?
DDR_DATA_ISO	*	=4:1_SPACING	?
DDR_STROBE_ISO	*	=3:1_SPACING	?
DDR_DQ2DQ	*	=2:1_SPACING	900
DDR_DQ2DQS	*	=3:1_SPACING	?
DDR_BL2BL	*	=3:1_SPACING	?
DDR_CH2CH	*	=6.5:1_SPACING	?
DDR_COMP_ISO	*	0.381 MM	?

Main Segment Min Spacing Rules (mils) (HSW U/Y PDG, Intel Doc# 502636)

BDW does not have a separate power spec from HSW, so this document is still accurate

Table	Trace	Design	Iso	Design	Comments
6-14	4	(diff)	16	19.69	CLK trace spacing controlled by =70_OHM_DIFF.
6-14	7.5	7.87	12	13.78	
6-14	7.5	7.87	12	13.78	
6-14	7.5	7.87	16	11.81	DQ to other signals not in the same bytelane (but not ch)
			12	11.81	DQS to other signals of the same channel
			7.5	7.87	DQ to DQ in the same bytelane of the same channel
			12	11.81	DQ to DQS in the same bytelane of the same channel
			16	11.81	DQ or DQS in different bytelanes of the same channel
			?	25.59	DQ or DQS in different channels. ISO RULE NOT IN PDG
			-	25.59	DDR3 to any other signal not DDR3

Constraints

Clocks: CK[3:0], CK#[3:0]

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
DDR_CLK	*	*	DDR_CLK_ISO

Control: CS#[3:0], CKE[3:0], ODT[3:0]

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
DDR_CTRL	*	*	DDR_CTRL_ISO
DDR_CTRL	DDR_CTRL	*	DDR_CTRL2CTRL

Command: MA[15:0], RAS#, CAS#, WE# BS[2:0]

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
DDR_CMD	*	*	DDR_CMD_ISO
DDR_CMD	DDR_CMD	*	DDR_CMD2CMD

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
DDR_COMP	*	*	DDR_COMP_ISO

Data: DQS[7:0], DQS#[7:0], DQ[63:0]

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
DDR_A_DQ_BYTE*	*	*	DDR_DATA_ISO
DDR_A_DQS*	*	*	DDR_STROBE_ISO
DDR_B_DQ_BYTE*	*	*	DDR_DATA_ISO
DDR_B_DQS*	*	*	DDR_STROBE_ISO
DDR_*_DQ_BYTE*	=SAME	*	DDR_DQ2DQ
DDR_A_DQ_BYTE*	DDR_A_DQS*	*	DDR_DQ2DQS
DDR_A_DQ_BYTE*	DDR_A_DQ_BYTE*	*	DDR_BL2BL
DDR_B_DQ_BYTE*	DDR_B_DQS*	*	DDR_DQ2DQS
DDR_B_DQ_BYTE*	DDR_B_DQ_BYTE*	*	DDR_BL2BL
DDR_A_*	DDR_B_*	*	DDR_CH2CH

Note (1):

Deliberately set DQ to DQS spacing to 3:1 to avoid adding complexity to constraints, even though it can be less. Only one rule per channel is needed by trading off a little space.

Note (2):

Intel suggests 25 mil (0.65 mm) spacing for via to channel, and via to pad to two different channels. DDR3 draws about 20 mA per trace with edge rates in the 100s of ps. The main coupling mechanism is capacitive. A 0.65 mm spacing is used for power nets, which draw far more current (inductive coupling however). These rules are far too conservative. To meet these rules, the spacing must be applied to the net.

Note (3):

In order for the constraints DDR_*_DQ_BYTE* to =SAME to win out over DDR_{A,B}_DQ_BYTE* to DDR_{A,B}_DQ_BYTE* so that the small intra-bytelane spacing is used, the spacing rule DDR_DQ2DQ must have a weight greater than DDR_BL2BL.

DDR3

Electrical Constraint Set		Physical	Spacing		
Channel A					
E1170	DDR_A_CLK0	NDR_CLK_PHY	NDR_CLK	MEM_A_CLK P<0>	7 21 25
E1170	DDR_A_CLK0	NDR_CLK_PHY	NDR_CLK	MEM_A_CLK N<0>	7 21 25
E2250	DDR_A_CLK1	NDR_CLK_PHY	NDR_CLK	MEM_A_CLK P<1>	7 22 25
E2250	DDR_A_CLK1	NDR_CLK_PHY	NDR_CLK	MEM_A_CLK N<1>	7 22 25
E2720	DDR_A_CS0	NDR_CTRL_PHY	NDR_CTRL	MEM_A_CS I<0>	7 21 22 25
E1830	DDR_A_CS1	NDR_CTRL_PHY	NDR_CTRL	MEM_A_CS I<1>	7 21 22 25
E1180	DDR_A_ODT	NDR_CTRL_PHY	NDR_CTRL	MEM_A_ODT<0>	7 21 22 25
E1180	?	?	?	MEM_A_CKE<1..0>	7 21 25
E2400	?	?	?	MEM_A_CKE<3..2>	7 22 25
E1800	?	?	?	MEM_A_CAA<9..0>	7 21 25
E1800	?	?	?	MEM_A_CAB<9..0>	7 22 25
E1180	?	?	?	MEM_A_DQ<7..0>	7 19
E1180	?	?	?	MEM_A_DQ<15..8>	7 19
E1180	?	?	?	MEM_A_DQ<23..16>	7 19
E1180	?	?	?	MEM_A_DQ<31..24>	7 19
E1800	?	?	?	MEM_A_DQ<39..32>	7 19
E1180	?	?	?	MEM_A_DQ<47..40>	7 19
E1190	?	?	?	MEM_A_DQ<55..48>	7 19
E1190	?	?	?	MEM_A_DQ<63..56>	7 19
E1800	DDR_A_DQS0	NDR_DQS_PHY	NDR_A_DQS0	MEM_A_DQS P<0>	7 19
E1190	DDR_A_DQS0	NDR_DQS_PHY	NDR_A_DQS0	MEM_A_DQS N<0>	7 19
E1190	DDR_A_DQS1	NDR_DQS_PHY	NDR_A_DQS1	MEM_A_DQS P<1>	7 19
E1190	DDR_A_DQS1	NDR_DQS_PHY	NDR_A_DQS1	MEM_A_DQS N<1>	7 19
E2200	DDR_A_DQS2	NDR_DQS_PHY	NDR_A_DQS2	MEM_A_DQS P<2>	7 19
E1800	DDR_A_DQS2	NDR_DQS_PHY	NDR_A_DQS2	MEM_A_DQS N<2>	7 19
E2200	DDR_A_DQS3	NDR_DQS_PHY	NDR_A_DQS3	MEM_A_DQS P<3>	7 19
E1800	DDR_A_DQS3	NDR_DQS_PHY	NDR_A_DQS3	MEM_A_DQS N<3>	7 19
E1800	DDR_A_DQS4	NDR_DQS_PHY	NDR_A_DQS4	MEM_A_DQS P<4>	7 19
E2200	DDR_A_DQS4	NDR_DQS_PHY	NDR_A_DQS4	MEM_A_DQS N<4>	7 19
E2200	DDR_A_DQS5	NDR_DQS_PHY	NDR_A_DQS5	MEM_A_DQS P<5>	7 19
E2200	DDR_A_DQS5	NDR_DQS_PHY	NDR_A_DQS5	MEM_A_DQS N<5>	7 19
E2000	DDR_A_DQS6	NDR_DQS_PHY	NDR_A_DQS6	MEM_A_DQS P<6>	7 19
E2000	DDR_A_DQS6	NDR_DQS_PHY	NDR_A_DQS6	MEM_A_DQS N<6>	7 19
E2100	DDR_A_DQS7	NDR_DQS_PHY	NDR_A_DQS7	MEM_A_DQS P<7>	7 19
E2100	DDR_A_DQS7	NDR_DQS_PHY	NDR_A_DQS7	MEM_A_DQS N<7>	7 19
Channel B					
E2110	DDR_B_CLK0	NDR_CLK_PHY	NDR_CLK	MEM_B_CLK P<0>	7 23 25
E2110	DDR_B_CLK0	NDR_CLK_PHY	NDR_CLK	MEM_B_CLK N<0>	7 23 25
E2400	DDR_B_CLK1	NDR_CLK_PHY	NDR_CLK	MEM_B_CLK P<1>	7 24 25
E2400	DDR_B_CLK1	NDR_CLK_PHY	NDR_CLK	MEM_B_CLK N<1>	7 24 25
E2600	DDR_B_CS0	NDR_CTRL_PHY	NDR_CTRL	MEM_B_CS I<0>	7 23 24 25
E2600	DDR_B_CS1	NDR_CTRL_PHY	NDR_CTRL	MEM_B_CS I<1>	7 23 24 25
E2600	DDR_B_ODT	NDR_CTRL_PHY	NDR_CTRL	MEM_B_ODT<0>	7 23 24 25
E2110	?	?	?	MEM_B_CKE<1..0>	7 23 25
E2500	?	?	?	MEM_B_CKE<3..2>	7 24 25
E2600	?	?	?	MEM_B_CAA<9..0>	7 23 25
E2600	?	?	?	MEM_B_CAB<9..0>	7 24 25
E2220	?	?	?	MEM_B_DQ<7..0>	7 19
E2220	?	?	?	MEM_B_DQ<15..8>	7 19
E2220	?	?	?	MEM_B_DQ<23..16>	7 19
E2220	?	?	?	MEM_B_DQ<31..24>	7 19
E2220	?	?	?	MEM_B_DQ<39..32>	7 19
E2220	?	?	?	MEM_B_DQ<47..40>	7 19
E2220	?	?	?	MEM_B_DQ<55..48>	7 19
E2220	?	?	?	MEM_B_DQ<63..56>	7 19
E2300	DDR_B_DQS0	NDR_DQS_PHY	NDR_B_DQS0	MEM_B_DQS P<0>	7 19
E2300	DDR_B_DQS0	NDR_DQS_PHY	NDR_B_DQS0	MEM_B_DQS N<0>	7 19
E2300	DDR_B_DQS1	NDR_DQS_PHY	NDR_B_DQS1	MEM_B_DQS P<1>	7 19
E2300	DDR_B_DQS1	NDR_DQS_PHY	NDR_B_DQS1	MEM_B_DQS N<1>	7 19
E2300	DDR_B_DQS2	NDR_DQS_PHY	NDR_B_DQS2	MEM_B_DQS P<2>	7 19
E2300	DDR_B_DQS2	NDR_DQS_PHY	NDR_B_DQS2	MEM_B_DQS N<2>	7 19
E2300	DDR_B_DQS3	NDR_DQS_PHY	NDR_B_DQS3	MEM_B_DQS P<3>	7 19
E2300	DDR_B_DQS3	NDR_DQS_PHY	NDR_B_DQS3	MEM_B_DQS N<3>	7 19
E2300	DDR_B_DQS4	NDR_DQS_PHY	NDR_B_DQS4	MEM_B_DQS P<4>	7 19
E2300	DDR_B_DQS4	NDR_DQS_PHY	NDR_B_DQS4	MEM_B_DQS N<4>	7 19
E2300	DDR_B_DQS5	NDR_DQS_PHY	NDR_B_DQS5	MEM_B_DQS P<5>	7 19
E2300	DDR_B_DQS5	NDR_DQS_PHY	NDR_B_DQS5	MEM_B_DQS N<5>	7 19
E2200	DDR_B_DQS6	NDR_DQS_PHY	NDR_B_DQS6	MEM_B_DQS P<6>	7 19
E2200	DDR_B_DQS6	NDR_DQS_PHY	NDR_B_DQS6	MEM_B_DQS N<6>	7 19
E2400	DDR_B_DQS7	NDR_DQS_PHY	NDR_B_DQS7	MEM_B_DQS P<7>	7 19
E2400	DDR_B_DQS7	NDR_DQS_PHY	NDR_B_DQS7	MEM_B_DQS N<7>	7 19
SM COMP					
E2500	?	?		CPU_SM_RCOMP<0..2>	6

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DDR3 Constraints			
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PCH

PCH-specific Physical Rules

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
PCH_55S	*	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=STANDARD	=STANDARD
CLK_PCH_55S	*	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=STANDARD	=STANDARD

PCH-specific Spacing Definitions

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT	NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
CLK_PCH_ISO	*	=4:1_SPACING	?	CLK_PCH	*	*	CLK_PCH_ISO
COMP_PCH_ISO	*	=2:1_SPACING	?	COMP_PCH	*	*	COMP_PCH_ISO

LPC

LPC-specific Physical Rules

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
LPC_55S	*	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=STANDARD	=STANDARD
CLK_LPC_55S	*	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=STANDARD	=STANDARD

LPC-specific Spacing Definitions

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT	NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
LPC_ISO	*	=1.5:1_SPACING	?	LPC	*	*	LPC_ISO
CLK_LPC_ISO	*	=2:1_SPACING	?	CLK_LPC	*	*	CLK_LPC_ISO

HDA

HDA-specific Physical Rules

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
HDA_55S	*	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=STANDARD	=STANDARD

HDA-specific Spacing Definitions

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT	NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
HDA_ISO	*	=2X_DIELECTRIC	?	HDA	*	*	HDA_ISO

Crystal

Crystal-specific Physical Rules

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
CLK_XTAL	*	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF

Crystal-specific Spacing Definitions

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT	NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
XTAL_ISO	*	=4X_DIELECTRIC	?	XTAL	*	*	XTAL_ISO

SPI

SPI-specific Physical Rules

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
SPI_50S	*	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=STANDARD	=STANDARD
SPI_55S	*	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=STANDARD	=STANDARD

SPI-specific Spacing Definitions

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT	NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
SPI_ISO	*	=2:1_SPACING	?	SPI	*	*	SPI_ISO

LPC

Electrical Constraint Set	Physical	Spacing	
LPC			
	?	?	LPC AD<3..0> 14 43
	?	?	LPC AD_R<3..0> 14
	LPC_55S	LPC	LPC FRAME_L 14 43
	LPC_55S	LPC	LPC FRAME_R_L 14
LPC Clocks			
	CLK_LPC_55S	CLK_LPC	LPC CLK24M_SMC 17 43
	CLK_LPC_55S	CLK_LPC	LPC CLK24M_SMC_R 12 17

PCH Clocks

Electrical Constraint Set	Physical	Spacing	
PCH Reference Clock			
	CLK_XTAL	XTAL	PCH CLK24M_XTAL_IN 12 17
	CLK_XTAL	XTAL	PCH CLK24M_XTAL_OUT 12 17
	CLK_XTAL	XTAL	PCH CLK24M_XTAL_OUT_R 17
PCH RTC 32K			
	CLK_XTAL	XTAL	PCH CLK32K_RTCX1 12 17
	CLK_XTAL	XTAL	PCH CLK32K_RTCX2 12 17
	CLK_XTAL	XTAL	PCH CLK32K_RTCX2_R 17
SMC 32K			
	CLK_PCH_55S	CLK_PCH	PM CLK32K_SUSCLK_R 13 44
	CLK_PCH_55S	CLK_PCH	SMC CLK32K 43 44

25 MHz XTALS

Electrical Constraint Set	Physical	Spacing	
25M Reference Crystal			
	CLK_XTAL	XTAL	TBT CLK25M_IN 17 26
	CLK_XTAL	XTAL	TBT CLK25M_OUT 17 26
	CLK_XTAL	XTAL	TBT CLK25M_OUT_R 17
	CLK_XTAL	XTAL	ENET XTAL_IN 17 34
	CLK_XTAL	XTAL	ENET XTAL_OUT 17 34
	CLK_XTAL	XTAL	ENET XTAL_OUT_R 17

HDA

Electrical Constraint Set	Physical	Spacing	
HDA			
	HDA	HDA	HDA BIT_CLK 12 51
	HDA	HDA	HDA BIT_CLK_R 12
	HDA	HDA	HDA_RST_L 12 51
	HDA	HDA	HDA_RST_R_L 12
	HDA	HDA	HDA_SDOUT 12 51
	HDA	HDA	HDA_SDOUT_R 12 17
	HDA	HDA	HDA_SYNC 12 51
	HDA	HDA	HDA_SYNC_R 12
	HDA	HDA	HDA_SDINO 12 51
SPDIF			
	HDA	HDA	DP_INT_SPDIF_AUDIO 39 51 80
	HDA_55S	?	SPDIF_OUT_JACK 51 54
	HDA_55S	?	CS4208_SPDIF_IN 51
	HDA_55S	?	CS4208_SPDIF_OUT 51

SPI Bootrom

Electrical Constraint Set	Physical	Spacing	
SPI ROM			
	SPI_50S	SPI	SPI_CLK_R 14 45
	SPI_50S	SPI	SPI_CLK 45
	SPI_50S	SPI	SPI_ALT_CLK 45
	SPI_50S	SPI	SPI_SMC_CLK 43 45
	SPI_50S	SPI	SPI_MLB_CLK 45
	SPI_50S	SPI	SPI_CS0_R_L 14 45
	SPI_50S	SPI	SPI_CS0_L 45
	SPI_50S	SPI	SPI_ALT_CS_L 45
	SPI_50S	SPI	SPI_SMC_CS_L 43 45
	SPI_50S	SPI	SPI_MLB_CS_L 45
	SPI_50S	SPI	SPI_MOSI_R 14 45
	SPI_50S	SPI	SPI_MOSI 45
	SPI_50S	SPI	SPI_ALT_MOSI 45
	SPI_50S	SPI	SPI_SMC_MOSI 43 45
	SPI_50S	SPI	SPI_MLB_MOSI 45
	SPI_50S	SPI	SPI_MISO 45
	SPI_50S	SPI	SPI_ALT_MISO 45
	SPI_50S	SPI	SPI_SMC_MISO 43 45
	SPI_50S	SPI	SPI_MLB_MISO 45
	SPI_50S	SPI	SPIROM_USE_MLB 15 45

DC-DC

Power-specific Physical Rules

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
GND_P3MM	*	Y	0.300 MM	0.150 MM	12.7 MM	=STANDARD	=STANDARD
GND_P5MM	*	Y	0.500 MM	0.150 MM	12.7 MM	=STANDARD	=STANDARD
POWER_50S	*	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=STANDARD	=STANDARD
POWER_P3MM	*	Y	0.300 MM	0.150 MM	12.7 MM	=STANDARD	=STANDARD
POWER_P6MM	*	Y	0.600 MM	0.150 MM	12.7 MM	=STANDARD	=STANDARD

Physical Net Type to Rule Map

NET_PHYSICAL_TYPE	AREA_TYPE	PHYSICAL_RULE_SET
GND	*	GND_P5MM
GND	BGA	GND_P3MM
POWER	*	POWER_P6MM
POWER	BGA	POWER_P3MM
VR_CTL_PHY	*	POWER_P3MM
VR_CTL_PHY	BGA	STANDARD
VR_VID_PHY	*	POWER_50S

NET_PHYSICAL_TYPE	AREA_TYPE	PHYSICAL_RULE_SET
VR_DIDT_PHY	*	POWER_P6MM
VR_DIDT_PHY	BGA	STANDARD

Power-specific Spacing Definitions

Power and Common

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
POWER_ISO	*	=STANDARD	?
GND_ISO	*	=STANDARD	?

DC-DC Baddies

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
SWNODE_ISO	*	=8:1_SPACING	1000
SWNODE_SW2SW	*	=1:1_SPACING	?
SWNODE_SW2PWR	*	=2:1_SPACING	?
SWNODE_SW2GND	*	=2:1_SPACING	?

DC-DC Control

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
VR_CTL_ISO	*	=3:1_SPACING	?
VR_VID_ISO	*	=4X_DIELECTRIC	?

Constraints

Power and Common

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
POWER	*	*	POWER_ISO
GND	*	*	GND_ISO

DC-DC Baddies

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
VR_SWITCH	*	*	SWNODE_1SO
VR_SWITCH	*	BGA	BGA_P1MM
VR_SWITCH	VR_SWITCH	*	SWNODE_SW2SW
VR_SWITCH	POWER	*	SWNODE_SW2PWR
VR_SWITCH	GND	*	SWNODE_SW2GND

DC-DC Control

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
VR_CTL	*	*	VR_CTL_ISO
VR_VID	*	*	VR_VID_ISO

$$V_{DDQ} (1.2V) / V_{TT} (0.6V) \quad S3$$

Physical	Spacing	Voltage	DIDT	NO_TEST	
Input Bus					
U33 POWER	POWER	5V			PP5V_REG_P1V2_V5IN 59 77
Local Ground					
U14 GND	GND	0V			AGND_P1V2_S3 59 77
VDDQ S3					
U45 VR_D1DT_PHY	VR_SWITCH	12V	TRUE		REG_PHASE_P1V2_S3 59
U49 VR_D1DT_PHY	VR_SWITCH	12V	TRUE		REG_PHASE_P1V2_S3_L 59
U46 VR_D1DT_PHY	VR_SWITCH	12V	TRUE		REG_BOOT_P1V2_S3 59
U48 VR_D1DT_PHY	VR_SWITCH	12V	TRUE		REG_BOOT_P1V2_S3_RC 59
U49 VR_D1DT_PHY	VR_SWITCH	12V	TRUE		REG_UGATE_P1V2_S3 59
U52 VR_D1DT_PHY	VR_SWITCH		TRUE		REG_LGATE_P1V2_S3 59
U51 VR_D1DT_PHY	VR_SWITCH	12V	TRUE		REG_SNUBBER_P1V2_S3 59
U53 VR_CTT_PHY	VR_CTT				REG_P1V2_S3_VDDQSNS 59
U55 VR_CTT_PHY	VR_CTT				REG_P1V2_S3_VREF 59
U56 VR_CTT_PHY	VR_CTT				REG_P1V2_S3_REFIN 59
U57 VR_CTT_PHY	VR_CTT				REG_P1V2_S3_MODE 59
U58 VR_CTT_PHY	VR_CTT				REG_P1V2_S3_TRIP 59
U59 VR_CTT_PHY	VR_CTT				LDO_DDRVTT50_SNS 59
U61 VR_CTT_PHY	VR_CTT				REG_P1V2_S3_VTTREF 59
Output Bus					
U60 POWER	POWER	1.2V			PP1V2_S3 66
U62 POWER_DDR	POWER_DDR	0.6V			PP0V6_S3_DDRVTT 66
Sensed					
U64 POWER	POWER	1.2V			PP1V2_S3_DDR 66

PCH 1.05V S0

Electrical Constraint Set	Physical	Spacing	Voltage	DIDT	NO_TEST
Input Bus					
P1	POWER	POWER	5V		REG_VCC U7400
P2	POWER	POWER	5V		REG_PVCC U7400
Local Ground					
G1	GND	GND	0V		AGND_P1V05S0
1.05V S0					
B5	VR_D1DT_PHY	VR_SWITCH	12V	TRUE	REG_PHASE_P1V05S0
G6	VR_D1DT_PHY	VR_SWITCH	12V	TRUE	REG_PHASE_P1V05S0_L
G7	VR_D1DT_PHY	VR_SWITCH	12V	TRUE	REG_BOOT_P1V05S0
G8	VR_D1DT_PHY	VR_SWITCH	12V	TRUE	REG_BOOT_P1V05S0_RC
G9	VR_D1DT_PHY	VR_SWITCH	12V	TRUE	REG_UGATE_P1V05S0
G10	VR_D1DT_PHY	VR_SWITCH	12V	TRUE	REG_UGATE_P1V05S0_R
G11	VR_D1DT_PHY	VR_SWITCH	12V	TRUE	REG_LGATE_P1V05S0
G12	VR_D1DT_PHY	VR_SWITCH	12V	TRUE	REG_SNUBBER_P1V05S0
G13	VR_CTI_PHY	VR_CTI			REG_P1V05S0_OCSET
G14	VR_CTI_PHY	VR_CTI			REG_P1V05S0_VO
G15 SNS_CURRENT	SNS_DIFF_PHY	SENSE			REG_P1V05S0_PHASE_SNS_P
G16 SNS_CURRENT	SNS_DIFF_PHY	SENSE			REG_P1V05S0_PHASE_SNS_M
G17					
G18		SENSE			REG_P1V05S0_FB
G19		SENSE			REG_P1V05S0_RTN
G20	VR_CTI_PHY	VR_CTI			REG_P1V05S0_SREF
G21	VR_CTI_PHY	VR_CTI			REG_P1V05S0_FSEL
Output Bus					
P22	POWER	POWER	1.05V		FP1V05_S0

1.8V S3

Physical		Spacing	Voltage	DIDT	NO_TEST	
Input Bus						
	POWER	POWER	5V			PP5V_REG_P1V2_V5IN 59 77
Local Ground						
	GND	GND	0V			AGND_P1V2_S3 59 77
Switch						
	VB_D1DT_PHV	VB_SWITCH	3.3V	TRUE		REG_SW_P1V8_S3 59
	VB_D1DT_PHV	VB_SWITCH	3.3V	TRUE		REG_FB_P1V8_S3 59
Output Bus						
	POWER	POWER	1.8V			PP1V8_S3_REG_R 59
	POWER	POWER	1.8V			PP1V8_S3 66

1.05V S5

Physical	Spacing	Voltage	DIDD	NO_TEST
Input Bus				
Switch				
Output Bus				
102		1.05V		

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H1052	VR_CTL_PHY	VR_CTL				REG_CPUVCC_PROG3 57																																																																																																																																																																																																																																																
H1063	CPU_VIDSCCLK	VR_VID_PHY	VR_VID			CPU_VIDSCCLK 8 57																																																																																																																																																																																																																																																
H1062		VR_VID_PHY	VR_VID			CPU_VIDSCCLK_R 8																																																																																																																																																																																																																																																
H1063	CPU_VIDALERT_L	VR_VID_PHY	VR_VID			CPU_VIDALERT_L 8 57																																																																																																																																																																																																																																																
H1064		VR_VID_PHY	VR_VID			CPU_VIDALERT_R_L 8																																																																																																																																																																																																																																																
H1065	CPU_VIDSOUT	VR_VID_PHY	VR_VID			CPU_VIDSOUT 8 57																																																																																																																																																																																																																																																
H1064		VR_VID_PHY	VR_VID			CPU_VIDSOUT_R 8																																																																																																																																																																																																																																																
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H1065	POWER	POWER	1.8V			PPCPUVCC_S0_CPU 66																																																																																																																																																																																																																																																
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Backlight Controller

BLC-specific Physical Rules

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
BLC_P6MM	*	Y	0.600 MM	0.100 MM	3.0 MM	=STANDARD	=STANDARD
BLC_P3MM	*	Y	0.300 MM	0.100 MM	3.0 MM	=STANDARD	=STANDARD

Physical Net Type to Rule Map

NET_PHYSICAL_TYPE	AREA_TYPE	PHYSICAL_RULE_SET
POWER_BLC	*	BLC_P6MM
POWER_BLC_RET	*	BLC_P3MM
BLC_CTL_PHY	*	BLC_P3MM

BLC-specific Spacing Definitions

BLC High Voltage Output

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
BLC_HV_ISO	*	0.45mm	1000

BLC Baddies

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
PHASE_ISO	*	=8:1_SPACING	2000
PHASE_SW2SW	*	=1:1_SPACING	?
PHASE_SW2PWR	*	=2:1_SPACING	?
PHASE_SW2GND	*	=2:1_SPACING	?

BLC Control

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
BLC_CTL_ISO	*	=3:1_SPACING	?

Constraints

BLC High Voltage Output

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
BLC_HV	BLC_CTL	*	BLC_CTL_ISO
BLC_HV	BLC_HV	*	BLC_CTL_ISO
BLC_HV	*	*	BLC_HV_ISO

BLC Baddies

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
BLC_PHASE	*	*	PHASE_ISO
BLC_PHASE	BLC_PHASE	*	PHASE_SW2SW
BLC_PHASE	POWER	*	PHASE_SW2PWR
BLC_PHASE	GND	*	PHASE_SW2GND

BLC Control

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
BLC_CTL	*	*	BLC_CTL_ISO

Is it chel'oh or sel'oh?

Physical	Spacing	Voltage	DIDT	NO_TEST
Input Bus				
U602 POWER	POWER	12V		PP12V_BKLT_SNS
U605 POWER	POWER	12V		PP12V_BKLT_FUSED
U620 POWER	POWER	12V		PP12V_S0_BKLT_FILT
U622 POWER	POWER	12V		PP12V_S0_BKLT_PWR
U681 POWER	POWER	12V		PP12V_S0_BKLT_PWR_R
U695 POWER	POWER	5V		PP5V_S0_BKLT_R
U696 POWER	POWER	3.3V		PP3V3_S0_BKLT_VDDIO_R
Local Ground				
U681 BLC_CTL_PHY	BLC_PHASE	0V		PGND_BKLT
U683 BLC_CTL_PHY	BLC_PHASE	0V		DGND_BKLT
U690 BLC_CTL_PHY	BLC_PHASE	0V		LGND_BKLT
Backlight				
U695 POWER_BLC	BLC_PHASE	80V	TRUE	BKLT_PHASE
U693 BLC_CTL_PHY	BLC_PHASE	80V	TRUE	BKLT_GATE
U684 BLC_CTL_PHY	BLC_PHASE	80V	TRUE	BKLT_GATE_R
U655 BLC_CTL_PHY	BLC_PHASE	80V	TRUE	BKLT_SHUBBER
U697 BLC_CTL_PHY	BLC_PHASE	12V	TRUE	BKLT_SW_R
U695 BLC_CTL_PHY	BLC_CTL			BKLT_ISET
U693 BLC_CTL_PHY	BLC_CTL			BKLT_FLT
U665 BLC_CTL_PHY	BLC_CTL			BKLT_FLT_RC
U697 SNS_DIFP_PHY	SENSE			BKLT_SW_P
U698 SNS_DIFP_PHY	SENSE			BKLT_SW_M
U682	SENSE			BKLT_FB
U681	BLC_HV	67V		BKLT_FB_XW
U684	BLC_HV	67V		BKLT_FB_R
U676 POWER_BLC_RET	BLC_CTL			BKLT_ISEN1
U675 POWER_BLC_RET	BLC_CTL			BKLT_ISEN2
U678 POWER_BLC_RET	BLC_CTL			BKLT_ISEN3
U677 POWER_BLC_RET	BLC_CTL			BKLT_ISEN4
U680 POWER_BLC_RET	BLC_CTL			BKLT_ISEN5
U679 POWER_BLC_RET	BLC_CTL			BKLT_ISEN6
U689 POWER_BLC_RET	BLC_HV			BKLT_ISEN1_R
U688 POWER_BLC_RET	BLC_HV			BKLT_ISEN2_R
U690 POWER_BLC_RET	BLC_HV			BKLT_ISEN3_R
U691 POWER_BLC_RET	BLC_HV			BKLT_ISEN4_R
U692 POWER_BLC_RET	BLC_HV			BKLT_ISEN5_R
U693 POWER_BLC_RET	BLC_HV			BKLT_ISEN6_R
U681 POWER_BLC_RET	BLC_HV			LED_RETURN_1
U682 POWER_BLC_RET	BLC_HV			LED_RETURN_2
U684 POWER_BLC_RET	BLC_HV			LED_RETURN_3
U685 POWER_BLC_RET	BLC_HV			LED_RETURN_4
U686 POWER_BLC_RET	BLC_HV			LED_RETURN_5
U687 POWER_BLC_RET	BLC_HV			LED_RETURN_6
Output Bus				
U670 POWER_BLC	BLC_HV	67V		BKLT_BOOST
U693 POWER_BLC	BLC_HV	67V		BKLT_BOOST_1
U674 POWER_BLC	BLC_HV	67V		BKLT_BOOST_2

Cello Miscellaneous

Electrical Constraint Set	Physical	Spacing	
SPI			
U684	SMB_PHY	SMB	BKLT_SCL
U683	SMB_PHY	SMB	BKLT_SDA